Degree profiles of uniform polynomial consequences

Linear lower bounds for Nullstellensatz and Polynomial Calculus

Søren Riis

Abstract

This paper proves optimal linear degree lower bounds for Nullstellensatz and Polynomial Calculus, yielding a four-way polynomial-versus-exponential proof-size classification for uniform polynomial consequences over fields of positive characteristic. The size bounds use full, explicit coefficient lists, including zeros.

Apart from a fixed finite initial range, exactly one of four cases occurs: T_1, bounded degree; T_2, degree at least linear in the universe size n; T_3, constant degree on some residue classes and at least linear degree on the others; or T_4, infinitely many recurring finite degrees, with bounded-degree subsequences and finite $\Omega(n)$ peaks at arbitrarily large sizes. Thus the size patterns are polynomial bounds, exponential lower bounds, a periodic mixture, or polynomial subsequences with finite exponential peaks, respectively. Underivable targets have degree ∞, and both proof systems assign the same type.

The result strengthens my 2008–2009 refutation classification and extends it to uniform consequences. The proof uses degree-preserving lifting with onset and period bounds linear in the degree. A common p-adic hardness spectrum describes the four types topologically. Distinguishing any two types is undecidable already for refutations. The linear degree bounds, four-way classification and pairwise undecidability results have been formalised and verified in Lean. The Lean formalisation, verification records and an accompanying tutorial are available in the supporting materials [30].

1 Introduction

An algebraic consequence can be simple to state and difficult to derive. Given polynomial equations Γ_n on a universe of size n, we ask how much degree is needed to prove a prescribed equation $G^{(n)} = 0$ from them. Both the axioms and the target are generated by fixed rules as n varies. The answer has four mutually exclusive and exhaustive patterns: bounded degree, linear hardness, a periodic mixture of the two, and systematic fluctuation with finite linear peaks. This paper proves the classification for uniform polynomial consequences, with refutation as the special target $G^{(n)} = 1$. It strengthens the earlier logarithmic Nullstellensatz and double-logarithmic Polynomial Calculus lower bounds to linear bounds in the universe size.

The NS proof system was introduced by Beame, Impagliazzo, Krajíček, Pitassi and Pudlák [20]; PC was introduced by Clegg, Edmonds and Impagliazzo [21]. To fix the proof-theoretic meaning of a consequence, let $\Gamma = \{f_1, \ldots, f_m\}$ be a finite set of polynomials over a field K, interpreted as the equations $f_i = 0$. A *Nullstellensatz* (NS) derivation of G is an identity

$$G = \sum_{i=1}^{m} a_i f_i.$$

Its degree is the largest total degree of a nonzero product $a_i f_i$. *Polynomial Calculus* (PC) derives polynomials line by line: each line is an input axiom, a linear combination of earlier

lines, or an earlier line multiplied by a variable. A derivation of G ends with that actual polynomial, and its degree is the largest total degree of any line after collecting coefficients. Thus $x - y$ and y derive x at degree one, while deriving 1 certifies that the input equations have no common zero. Soundness makes every derivation a semantic consequence. The converse is not imposed as an inference rule: semantic entailment and the existence of an NS or PC derivation are distinct questions for a general polynomial system.

Degree measures the complexity of the algebraic calculation rather than its number of lines. A fixed degree bound confines all lines to a vector space with polynomially many monomials in the variable count. An NS certificate uses the span of the allowed monomial multiples of axioms. PC can also cancel terms and multiply the resulting polynomial again. Its bounded-degree consequences therefore form an iterated linear closure, and its minimum degree can be smaller than the NS minimum. Section 2 gives the exact characterisations used throughout the proof.

A *uniform family* $(\Gamma_n)_n$ is a family of systems of polynomial equations, generated by a fixed finite list of bounded-degree polynomial patterns using the same coefficient rules at every universe size. At each size n, all equations in Γ_n are imposed simultaneously. Each pattern can generate many equations as its indices range over the universe, so the number of equations can grow with n even though the list of patterns is fixed. Variable and axiom indices each carry a bounded number of labels, and relabelling the universe permutes the equations. Each axiom is fixed by permutations fixing its distinguished index labels, even when it contains sums over the entire universe. A *fully invariant uniform target* has no distinguished labels and is unchanged by every permutation; $G^{(n)} = \sum_{i<n} x_i$ is a simple example. Its logical antecedents include the Paris–Wilkie translation [19], treated systematically by Krajíček [33, Chapter 9], and my early work on independence and finitisation in bounded arithmetic [31, 32], followed by the complexity-gap results [1, 6]. My joint journal papers with Meera Sitharam develop the uniform translation and its permutation-module viewpoint [22, 23]. Here the hypotheses are stated entirely in algebraic terms. Definitions 2.1 and 2.2 make the coefficient rules and support conditions precise.

Examples of uniform families. For these examples label the universe by $1, \dots, n$. With three unary variable types x_i, y_i, z_i, consider the generating polynomial

$$P_{1,n} = x_1 + y_1 + \sum_{j,k=1}^{n} x_j x_k + \sum_{i,j,k=1}^{n} z_i z_j z_k.$$

Its total degree is at most three, independent of n, although the number of summands and variables grows with n. A permutation fixing 1 fixes this whole polynomial. Its distinguished index therefore has support $\{1\}$. Taking its permutation orbit gives the symmetric axiom system $P_{a,n} = 0$ for all $a \in \{1, \dots, n\}$, where only the first two terms become $x_a + y_a$: both global sums are unchanged. The degree bound concerns each monomial, not the number of terms or the union of labels mentioned by the sum. All sums are evaluated in the fixed coefficient field, including any coefficient cancellations.

Binary and higher-arity variable types are equally admissible. For binary variables u_{ij}, the row axioms

$$F_{i,n} = \sum_{j=1}^{n} u_{ij} - 1 = 0$$

have degree one and index support $\{i\}$, while each variable has support contained in $\{i, j\}$. A cubic polynomial such as

$$H_n = \sum_{i,j,k=1}^{n} u_{ij} u_{jk} u_{ki}$$

is fully invariant, with one empty-support axiom or target index. A single system can combine several such patterns. For example, it can impose all n row equations $F_{i,n} = 0$, all n^2 Boolean equations $u_{ij}^2 - u_{ij} = 0$, and the global equation $H_n = 0$ simultaneously. These three patterns define a system of $n + n^2 + 1$ indexed equations. There can be n^2 binary variables, or n^a variables of any fixed arity a, without violating the bounded-support hypothesis. These examples are instances of the rank-independent coefficient rules in Definition 2.1, rather than arbitrary symmetric polynomials chosen afresh at each rank.

Familiar examples from the literature. Algebraic encodings of pigeonhole and counting principles provide important examples of this uniform structure. Razborov studied Polynomial Calculus degree lower bounds for pigeonhole principles [34]; Beame and Riis studied Nullstellensatz bounds for onto-pigeonhole principles [26]. Fixed-block-size partition principles also underlie the Nullstellensatz lower bounds of Beame, Impagliazzo, Krajíček, Pitassi and Pudlák [20]. In their bounded-degree encodings, Boolean constraints, pairwise incompatibility constraints and sums expressing that each element is assigned exactly once are generated by fixed patterns invariant under relabelling. Fixed-offset pigeonhole families fit the present single-universe formulation by treating the extra elements as permanent parameters, as described below. Thus these bounded-degree, fixed-pattern encodings are instances of the uniformly generated polynomial systems considered here. The encoding matters: growing-degree axioms are outside the bounded-degree hypothesis used here.

Orbit generators and permanent parameters. A label in an orbit generator is moved when the orbit is taken. Thus the orbit of $x_1 y_2 = 0$, with $1 \neq 2$, consists of $x_a y_b = 0$ for all $a \neq b$. It does not include the diagonal equations. This differs from permanently naming two distinct domain elements: the single constraint involving those named elements need not impose the analogous constraint on every pair.

Permanent parameters can be included by using separate variable types. For one named element c, a binary family z_{ij} splits into the scalar variable z_{cc}, the unary families z_{ci} and z_{ic}, and the binary family z_{ij} for $i, j \neq c$. The letter c is part of the type name; only the other labels are permuted. Correspondingly,

$$\sum_{i,j \in \{c\} \cup U} z_{ij} = z_{cc} + \sum_{i \in U} z_{ci} + \sum_{i \in U} z_{ic} + \sum_{i,j \in U} z_{ij}.$$

The scalar type has empty support and is fixed by every permutation. Such zero-arity variable types are allowed by Definition 2.1: the positive parameter k is an upper bound on support size, not a lower bound on arity. More generally, a fixed set of ℓ distinct named elements splits each finite-arity variable family into finitely many types, according to which coordinates are named and which remain free. Sums split in the same way. In a universe of total size n, the remaining action is $S_{n-\ell}$; reindexing the free universe puts the system in the usual S_m form with $m = n - \ell$. This fixed shift preserves bounded degree and the asymptotic linear bounds; statements about residues and initial universes are read with the same shift. For example, with permanent parameters c_1, c_2, the equation $x_{c_1} y_{c_2} = 0$ is a single equation in two scalar variable types. Its orbit under permutations of the remaining labels contains only itself. Here the parameters are preselected distinct domain elements. An uninterpreted constant symbol, whose value is chosen as part of a model, is instead represented by its function-graph variables in the logical translation below.

The proof is organised around one question. If an invariant uniform target $G^{(n)}$ has a degree-at-most-d derivation from Γ_n, when can the corresponding target $G^{(n+q)}$ be derived within the same degree bound from Γ_{n+q}? Here $q = p^a$ is the number of additional elements

in the finite universe, whose size increases from n to $n+q$. Once such a lift is available and the universe is sufficiently large relative to the degree bound, it can be repeated:

$$\Gamma_{n+\ell q} \vdash_{\mathrm{S},\leq d} G^{(n+\ell q)} \qquad (\ell = 0, 1, 2, \ldots).$$

The selected numerical degree d is constant on this progression; the proof lines and their coefficients may change. The essential quantitative question is how large the initial universe must be relative to d. For indexed targets the same question is asked for the whole prescribed target family, using its coefficient map rather than a literal lift of each polynomial.

First-order logic and finite structures. The framework includes polynomial encodings of arbitrary fixed first-order sentences over a finite vocabulary of uninterpreted relations and functions. A sentence can first be Skolemised and normalised to universal clauses, with auxiliary relation or function symbols where needed. The new functions record existential witnesses: for example, $\forall u\, \exists v\, R(u, v)$ can be represented by $\forall u\, R(u, f(u))$. This preserves existence of models on each nonempty finite universe. A model may have several witness expansions, so it does not assert a one-to-one correspondence with models of the original sentence before those witnesses are chosen.

Relation variables record truth values, and function-graph variables record chosen outputs. Boolean equations, totality equations and pairwise functionality collisions give the translation of Section 2. Its polynomial system has a common root at size n if and only if the original sentence has an n-element model; roots correspond to models of the chosen normalised expansion. The vocabulary, arities, clause lengths and hence polynomial degrees are fixed while n grows. This is the translation setting used in LICS 2008 [6, Section 1.2, p. 273] and the expanded 2009 report [7, Section 1.2, pp. 3–4], following the joint work with Meera Sitharam [22]; a precise translation statement is Theorem 1 of its BRICS version [29, Section II]. The joint work gives the translation for existential second-order sentences, including these first-order existence problems, and its permutation-module setting is developed in [23].

Existential quantification is not simply replaced by a sum. For a witness graph, $\sum_j x_{ij} = 1$ is used together with Booleanity and $x_{ij}x_{ij'} = 0$ for $j \neq j'$. Only together do these require exactly one chosen output; in characteristic p, the sum equation alone also allows $p+1$ Boolean ones. The example of an injective function omitting one named point is obtained in this way and has no finite nonempty model, although it has an infinite model.

The domain symbols here are uninterpreted: order, arithmetic and other relations are themselves part of the structure being sought. One may axiomatise a linear order and then quantify over its possible interpretations. Fixing an external order or arithmetic on the labels in advance is a different translation and need not satisfy the full permutation and rank-independent coefficient hypotheses. The theorem does not automatically cover such built-in structures.

A fixed coefficient rule can likewise prescribe the polynomial expressions to be derived. At universe size n, write these targets as $G_b^{(n)}$, where b is the target index. The superscript records the universe size; it does not name a distinguished element. For a fully invariant target we omit b and write $G^{(n)}$. An indexed target family may request $G_b^{(n)} = 0$ for every b, or the representatives of one fixed index orbit. No arbitrary choice of a different orbit at each size is made.

The governing hypotheses are fixed bounds on axiom and target degrees, bounded supports of the variable and axiom indices, and coefficients prescribed consistently across universe sizes and under relabelling. An index supported on a few labels can specify a polynomial involving the entire universe: the totality equation $\sum_j x_{ij} - 1 = 0$ has index i, even though the sum uses every j. The same distinction applies to targets. Uniformity does not require all terms of a target polynomial to use one bounded set of labels. The parameter n is universe

size, not variable count; variables indexed by k-tuples may number on the order of n^k. The coefficient field has one fixed positive characteristic throughout.

There is a proof-size interpretation when each polynomial is represented by its *full, explicit coefficient list*. For a polynomial of total degree d, this lists all monomials of degree at most d in the entire ambient variable set, including zero coefficients. The convention applies to PC lines and NS multipliers as well as to the input. Bounded-degree derivations can then be written in polynomial size, whereas linear degree lower bounds force size at least $2^{\Omega(n)}$ on the relevant instances. These are lower bounds; no universal matching exponential upper bound is asserted. The size conclusions concern finite derivations of the prescribed targets. An underivable target has infinite degree and proof size. Over a fixed finite field, coefficient counts also give the corresponding bit-size bounds. If the input length is $L(n) = O(n^{c_0})$ for a fixed $c_0 > 0$, the exponential bound in n becomes a stretched-exponential bound $2^{\Omega(L(n)^{1/c_0})}$. Corollary 3.5 supplies the counting argument and the precise scope; sparse representations that omit zero coefficients use a different size measure.

Complexity gaps and the earlier classification. The restriction to a fixed uniform description belongs to a broader programme of complexity gap theorems. Resolution combines $A \vee x$ and $B \vee \neg x$ to derive $A \vee B$. In tree-like resolution, a derived clause cannot be shared between branches; general resolution permits this reuse. Riis's tree-resolution gap theorem [1] gives, for the standard translations of a first-order sentence with no finite models, either polynomial-size refutations or a $2^{\Omega(n)}$ size lower bound. The hard case occurs exactly when the sentence has an infinite model; if it has no model at all, the polynomial case holds. Distinguishing the two cases is undecidable.

Dantchev and Riis [2] proved a size gap between polynomial upper bounds and $2^{\Omega(n^\varepsilon)}$ lower bounds, for some fixed $\varepsilon > 0$, for *full resolution* on unary-relativised families. A unary predicate selects the domain on which the sentence is interpreted. When the original sentence has no finite models, its relativised translations have polynomial-size refutations, even tree-like ones, if it has no model at all. If it has an infinite model, every full-resolution refutation has size $2^{\Omega(n^\varepsilon)}$ for some $\varepsilon > 0$. That work also treats tree-like resolution with bounded-size conjunctions and shows that a built-in order destroys the gap phenomenon.

Dantchev, Martin and Szeider [3] refine the tree-resolution classification in the parameterised setting. Dantchev and Martin [4] obtain a constant-versus-polynomial rank gap for the Lovász–Schrijver and Sherali–Adams systems, again governed by infinite models. Beyersdorff, Clymo, Dantchev and Martin [5] establish tree-like QBF-resolution analogues, with explicit dependence on the translation. These results concern different resources—size, parameterised size or rank—and do not themselves yield an NS or PC degree bound.

Positive characteristic allows more than a dichotomy for algebraic degree. Easy and hard sizes can occupy different residue classes, and one family can have infinitely many recurring finite degree levels. My LICS 2008 paper [6, Theorems 1 and 10], with fuller treatment in the 53-page 2009 report [7], established four refutation types, their periodic mixed case and agreement of the NS and PC type labels. Its general degree lower bounds were $\Omega(\log n)$ for NS and $\Omega(\log \log n)$ for PC. Their periodicity and degree-dependent onsets came from Krajíček's Theorem 3.5 [8], building on Ajtai's analysis of symmetric linear systems [24]. Here the linear bounds are proved for the more general task of deriving uniform targets. The constant target 1 recovers the refutation classification and its strengthened bounds.

In the final remarks of the LICS paper and the longer report, I conjectured that the classification should admit a degree lower bound of the form $\Omega(n^\varepsilon)$, for sufficiently small fixed $\varepsilon > 0$, at least for NS and possibly for PC [6, Section 8][7, Section 9]. I recorded the considerable effort already spent on this question. Both the LICS abstract and the longer report explicitly asked about a linear degree lower bound [6, 7]. The present theorem proves

that bound for both systems, with the hard-residue and finite-peak qualifications below, and extends the setting from refutations to uniform polynomial consequences.

The journal work on uniformly generated submodules concerns fields of characteristic zero [23]. In a separate 1997 preprint with Meera Sitharam [28], we announced a nonconstant-to-linear degree gap for uniform NS and PC families. In my 2009 report, I explicitly acknowledged an unresolved gap in the inducing-down proof and stated that the main claim remained conjectured [7, p. 50, footnote 3; p. 53, reference 33]. That qualification concerns the proof; it does not refute the asserted gap theorem. The present argument gives independent finite algebraic constructions for the hypotheses and four profiles stated here.

The preliminary fresh-label construction—a construction introducing additional labels—in Section 6 already yields an $\Omega(\sqrt{n})$ degree gap in characteristic two, answering the positive-power question in that setting. The linear construction makes a structural change: corrections at successive tuple lengths reuse one retained pool of labels. The odd-characteristic argument supplies the additional nilpotent-chain structure needed to obtain the same linear degree lower bound in every positive characteristic.

The linear degree lower bound is asymptotically optimal in general. The injection example in Section 13 has minimum NS and PC degrees $\Theta(n)$, while the fluctuating example has finite $\Theta(n)$ peaks. These examples show that a superlinear degree lower bound cannot replace the linear bound throughout the classification.

The four degree profiles. For either proof system, let $h(n)$ be the least derivation degree of the uniform target; for an indexed target family, take the maximum of the least degrees over its finite rank-n index set. Infinity records an underivable target, and an empty index set has degree zero. Let $\mathrm{Rec}(h)$ be the set of finite degrees occurring at infinitely many sizes. The four alternatives and their coefficient-list size readings are as follows.

Type	Degree behaviour	Derivation-size consequence
T_1	Bounded degree beyond a fixed initial range	Polynomial upper bounds.
T_2	No recurring finite degree; linear lower bounds	Exponential lower bounds.
T_3	Finitely many recurring levels without eventual boundedness; fixed low degrees on a nonempty proper set of residues, linear lower bounds on the others	Periodic polynomial upper bounds and exponential lower bounds.
T_4	Infinitely many recurring levels; finite liminf and finite linear peaks	Polynomial subsequences and finite exponential peaks at arbitrarily large sizes.

Finite initial exceptions are allowed. In T_3, easy residues need not share the same degree. In T_4, a linear degree bound at every sufficiently large size is false: the least recurring finite degree gives arbitrarily late low-degree derivations. Its infinitely many recurring finite degrees distinguish it from T_3, which has only finitely many. Thus arbitrarily large sizes have finite linear peaks, while one fixed finite degree also occurs arbitrarily late. Section 3 proves exhaustiveness and exclusivity and states the exact quantitative bounds.

From unchanged-degree lifting to a linear bound. Fix a proof-degree budget d. The central statement says that derivability within this budget is periodic in universe size after an onset linear in d, with a period also linear in d. Consequently, a finite degree d attained beyond its own onset must recur on an infinite arithmetic progression. A degree which does not recur can only occur before that onset. Inverting this restriction gives the linear lower bound in T_2

and on the hard residues of T_3. In T_4, the bound on the period locates representatives of unbounded recurring degrees at sizes proportional to those degrees, producing finite linear peaks. Both the onset and period are needed.

The relevant budget counts labels. If variable indices have support at most k and axiom indices at most s, a degree-d monomial paired with an axiom index uses at most $kd+s$ labels. The fully invariant target uses this budget. For targets indexed by supports of size at most t, the general proof uses the conservative budget

$$R(d) = 2kd + s + t.$$

The extra factor of two comes from detecting the target image outside the consequence space by a pairing; both factors of that pairing must be transported. The fixed target degree affects the initial rank threshold and excludes budgets below the target's degree. Above that fixed threshold, a sufficient periodic onset is $n > 2R(d)$, and a period is the least power of the characteristic exceeding $R(d)$. Thus the extension to indexed targets retains the linear degree lower bounds.

There are three transport obligations. First, the operation must take the actual axiom and multiplication maps to those at the smaller rank. Second, it must preserve the sums, images and intersections which compute all bounded-degree consequences. This is the difficult PC requirement: a cancellation can create a new polynomial eligible for multiplication. Third, it must preserve and reflect whether the prescribed target belongs to the resulting space. Exact deletion and a common group-algebra projector meet the first two obligations; transport of the target coefficient map and nonzero-subquotient detection meet the third. An abstract isomorphism of spaces alone would supply none of these map compatibilities.

The construction is developed in stages. In characteristic two, a first lift uses separate fresh labels at successive tuple lengths; its quadratic label cost already explains a square-root degree gap. The linear construction reuses one pool. Its projector is constructed through tuple lengths $b \le R(d)$, repairing the current length while leaving the shorter lengths unchanged. The same retained labels serve all repairs, so the allocation is controlled by $\max_{b \le R(d)}(2b-1)$ rather than a sum of separate fresh-label allocations. The proof develops the square-zero construction in characteristic two, normalisation and the zero-marginal decomposition, the full rook-forcing induction, and the top-power construction in arbitrary positive characteristic. For odd p, square-zero identities are replaced by full length-p nilpotent chains and an ordered projector, with a separate transpose argument. These repairs retain the same linear label allocation. Section 4 states the exact map requirements and the resulting transfer, and Sections 5–9 establish its algebraic and combinatorial ingredients.

The comparison between the two systems is separate. Compressing invariant-subspace chains to a bounded universe gives $h_{\mathrm{PC}}(n) \le h_{\mathrm{NS}}(n) \le F(h_{\mathrm{PC}}(n))$ whenever the PC degree is finite, with a monotone function F independent of universe size. It applies to the entire consequence space and hence to each prescribed target. This establishes common type without asserting equality of exact degrees or of recurring degree sets. The linear PC bound is proved directly, rather than inferred by inverting the potentially large function F.

Examples, semantics and undecidability. The first-order examples illustrate the four refutation types. A fresh-variable construction also produces a nonconstant target whose finite derivation degree has linear peaks; this shows that the consequence theorem includes actual polynomial targets as well as the constant 1. The motivating Boolean translation uses totality sums and functionality collisions, with no automatic reduction modulo Boolean equations. Section 11 gives a p-adic hardness spectrum for all uniform requests: the four types correspond to an empty spectrum, the whole space, a nonempty proper clopen set, and a closed set that is not open. This makes explicit the residue-class structure already

implicit in my earlier report [7, Section 4], now in the setting of uniform consequences. The same section distinguishes linear witnesses from actual solutions and recalls Krajíček's Euler criterion. The topological argument needs no Boolean semantic hypotheses and does not assert a converse to that criterion.

The undecidability theorem already holds in the refutation special case. The no-finite-model promise and undecidability for every nonempty proper union of types were stated in LICS 2008 [6, Theorem 1]. Here explicit reductions separate any two types under that same promise, refining the published every-union formulation. On normalised first-order inputs, an explicit finite computation prefix gives a uniform degree bound when its machine halts, and a controlled disjunction supplies the pairwise reductions. The effective reduction and its finite-term certificate are included.

Algebraic antecedents. Krajíček's bounded-degree NS–PC comparison [8, Theorem 5.5] supplied an input to my earlier classification. The present proof retains the NS-span stabilisation argument and supplies its uniform chain bound independently. The behaviour of fixed-degree data as the number of labels grows is a central theme of representation stability and FI-modules, developed by Church, Ellenberg and Farb [14]. An FI-module assigns vector spaces and compatible linear maps to finite sets and injections; it organises comparisons as the label set grows. In positive characteristic, Harman [9] proves periodic equivalences for bounded-weight symmetric-group representations with a linear stable range. His equivalences already respect induction, restriction, tensor products and duality [9, Propositions 2.7 and 2.12]. The deletion functor $\ker z/(\ker z \cap \operatorname{im} z)$ used here is also one of the explicit functors in Hone and coauthors [10, Theorem A]. Its group-algebra operator comes from a generic shifted cyclic subgroup. Our argument proves exactness on its stated bounded-support category, without invoking a categorical equivalence. These antecedents place the linear-onset mechanism in an established setting. The present proof independently constructs exact deletion on the actual polynomial diagrams, preserves their consequence closures and detects the prescribed targets, with the quantitative bounds used here. The rook realisation step is related to the integral tensor-space centraliser results of Bowman, Doty and Martin [11]; the required realisation is obtained by the forcing and elimination argument in the paper. Uniform target membership already belongs to Krajíček's framework [12, 8]. The contributions here are the linear quantitative bounds for uniform consequences, the independent algebraic proof and explicit reductions for every pair of types under the earlier no-finite-model promise.

Notation and proof guide. The principal notation is collected here; local auxiliary symbols are defined in their respective constructions.

Notation	Meaning
n, $N = n + q$	Retained and larger universe sizes
d; D, D_G	Proof-degree budget; fixed axiom and target degree bounds
k, s, t	Variable, axiom-index and target-index support bounds
$R(d) = \kappa d + \sigma$	Support budget used in periodicity
$N(d)$, $Q(d)$	Degree-dependent onset and prime-power period
$I_d(n)$, $C_d(n)$	NS and ordinary-PC consequence spaces
$P_b(N)$, $\mathcal{C}_r(N)$	Injective-tuple modules and their subquotient category
$\mathcal{Z}_b(N)$, $J_r(N)$	Zero-marginal space and tuple annihilator ideal
z, w, e, $\mathcal{D}$	Block operator, common inner inverse, projector, deletion

The central dependency chain is

$$\text{rook realisation and layer splitting} \Longrightarrow \text{common inner inverse} \Longrightarrow \text{exact deletion},$$
$$\text{exact deletion and target detection} \Longrightarrow \text{periodicity with linear onset} \Longrightarrow \text{four profiles}.$$

The NS–PC comparison additionally uses a uniform subspace-chain bound. The matching obstruction supplies the concrete examples needed for pairwise undecidability. Standard background is in Lang [15], Alperin [16], the Stacks Project [18] and Cutland [17]; the specific algebraic and combinatorial arguments used here are proved below.

Outline of the paper. Section 2 defines the proof systems, uniform axiom and target templates, first-order encoding and bounded-degree consequence spaces. Section 3 derives the four profiles and their size consequences from quantitative transfer. Sections 4–9 construct the deletion and lifting operations, first in characteristic two and then in arbitrary positive characteristic. Section 10 proves uniform NS–PC type agreement. Section 11 gives the topological classification and discusses algebraic witnesses and Euler structures. Section 12 develops nonconstant targets and the fresh-variable construction. Sections 13–14 give the four first-order refutation examples and prove their matching obstruction. Section 15 completes the computation-prefix reduction. Section 16 closes with the mathematical consequences and acknowledgements.

2 Uniform systems and degree

Fix a field K of prime characteristic p. We study the degree needed to derive a specified polynomial from a system of equations. Refutation is the special case in which the specified polynomial is 1. All rings are ordinary commutative polynomial rings; Boolean equations, when present, are axioms rather than a quotient convention.

2.1 Consequences and their degree

Let $\Gamma \subseteq K[x_v : v \in V]$ be a finite system and let G be a polynomial in the same ring. We write $\Gamma \vdash_S G$, for $S \in \{\mathrm{NS}, \mathrm{PC}\}$, for an algebraic derivation of G from Γ. Since the input equations are $f = 0$, this means deriving the equation $G = 0$. In particular, deriving 1 gives a contradiction.

Following the NS proof system of Beame, Impagliazzo, Krajíček, Pitassi and Pudlák [20], an NS derivation of G is an identity

$$G = \sum_{f \in \Gamma} a_f f.$$

Its degree is the largest total degree of a nonzero product $a_f f$. Over a field this is $\deg a_f + \deg f$ for each such product. Zero summands impose no restriction and may be omitted; the empty sum is a certificate of $G = 0$ of degree zero. We use $\deg 0 = 0$ throughout.

In the system introduced by Clegg, Edmonds and Impagliazzo [21], an ordinary-PC derivation is a finite sequence $P_1, \dots, P_M$ whose last line is G. Each line is an input axiom, a linear combination $\alpha P_i + \beta P_j$ of earlier lines, or a variable multiple $x_v P_i$ of an earlier line. The zero polynomial is also allowed as the empty linear combination. The proof degree is the largest total degree of any line, computed after collecting coefficients. Allowing a linear combination of any finite number of earlier lines gives the same degree: it can be expanded into successive two-line combinations. Thus a cancellation may lower the degree of a derived polynomial and permit a later multiplication within the same degree budget. A Boolean simplification is not a further rule.

Write $h_S(\Gamma \vdash G) \in \mathbb{N} \cup \{\infty\}$ for the least such degree, assigning ∞ when there is no derivation. Every finite value is at least $\deg G$. For refutations we retain the notation $d_S(\Gamma) = h_S(\Gamma \vdash 1)$. Expanding an NS certificate into monomial multiples, deriving each product by successive variable multiplication, and adding gives

$$h_{\mathrm{PC}}(\Gamma \vdash G) \le h_{\mathrm{NS}}(\Gamma \vdash G), \qquad d_{\mathrm{PC}}(\Gamma) \le d_{\mathrm{NS}}(\Gamma). \tag{1}$$

The inequality concerns actual derivations of the specified target; the target is not added to the input axioms.

A short consequence proof. Over $\mathbb{F}_2$, take $f_1 = xy + x$, $f_2 = xy + 1$ and $f_3 = xz$. The following proof derives the nonconstant target $G = z$ in degree two:

$$f_1 + f_2 = x + 1, \qquad z(x+1) = xz + z, \qquad (xz + z) + f_3 = z.$$

We are keeping every line within the degree budget two. The quadratic terms cancel to give the degree-one line $x + 1$; multiplying that line by z is therefore a permitted step within this budget. Expanding these same steps into the NS identity $z = zf_1 + zf_2 + f_3$ gives product degree three. The input system itself is consistent: $x = y = 1$, $z = 0$ satisfies it. Thus consequences need not arise from refutations.

2.2 Uniform axioms and uniform targets

The running first-order example asks for an injective function whose image omits a distinguished point. If x_{ij} encodes the graph atom $g(i) = j$, totality and functionality contribute

$$f_{i,n} = \sum_{j<n} x_{ij} - 1, \qquad x_{ij}x_{ij'} \quad (j \neq j').$$

The same rules are used at every size. A permutation sends x_{ij} to $x_{\sigma(i),\sigma(j)}$ and sends $f_{i,n}$ to $f_{\sigma(i),n}$. Every permutation fixing i therefore fixes $f_{i,n}$, although the sum contains variables involving all labels. This distinguishes the *support of an axiom index* from the union of labels in its terms. The complete inconsistent system is given in (59).

We call a support assignment *equivariant* when relabelling an object also relabels its support. For linear maps the same word means commutation with relabelling: $f(\sigma v) = \sigma f(v)$. Uniformity below specifies both this symmetry and one coefficient rule used at every rank.

Definition 2.1 (Uniform template)**.** A bounded basis system consists of labelled objects, with an action of permutations of $\mathbb{N}$, and a finite support for each object, whose size has a fixed bound independent of rank. Support is equivariant, and a permutation fixing an object's support pointwise fixes that object. At rank n the basis consists exactly of the objects supported in $[n] = \{0, \dots, n-1\}$, and is finite.

A uniform polynomial template T has a variable basis $\mathcal{V}$ of support at most $k \ge 1$, an axiom-index basis $\mathcal{A}$ of support at most s, and coefficients $c(a, \mu) \in K$ indexed by an axiom index and a monomial in the labelled variables. The coefficients are invariant under simultaneous relabelling, independent of rank, and vanish when $\deg \mu > D$ for some fixed D. The rank-n instance is

$$\Gamma_n = \{f_{a,n} : a \in \mathcal{A}_n\}, \qquad f_{a,n} = \sum_{\mu \text{ supported in } [n]} c(a, \mu)\mu.$$

Choose $n_0 \ge \max\{1, kD + s\}$. The parameters k, s, D, n_0 belong to the fixed template.

Support zero is allowed for variable and index bases. A zero-arity variable type is a single scalar variable fixed by every permutation; a finite number of such types can be included. The parameter $k \geq 1$ is only an upper bound on variable support, so it does not exclude these types. This also explains the permanent-parameter encoding in the introduction without enlarging the template hypotheses.

Fixed finite types of tuples, with specified equality patterns, give familiar basis systems. A degree-d monomial uses at most kd labels, even if the polynomial containing it mentions every label in the universe. Products of bases add support bounds; restriction to an invariant subset does not increase them. A monomial–axiom pair has support at most

$$r = kd + s. \tag{2}$$

Overlaps can reduce this bound. For the injection system one may take $k = 2$ and $s = 3$: row totality has index support one, while an injectivity collision $x_{ij}x_{hj}$ has support at most three. The multiplication domain at degree d has basis pairs (x_v, μ) with $\deg \mu \leq d - 1$, of support at most $k + k(d - 1) = kd$.

Definition 2.2 (Uniform target family)**.** For a fixed axiom template, a uniform indexed target family consists of a bounded basis $\mathcal{B}$ of support at most t and coefficients $\gamma(b, \mu) \in K$, invariant under simultaneous relabelling of b and μ, independent of rank, and zero when $\deg \mu > D_G$. It requests the targets

$$G_b^{(n)} = \sum_{\mu \text{ supported in } [n]} \gamma(b, \mu)\mu, \qquad b \in \mathcal{B}_n.$$

Here n is the universe size and b is the target index; its support is a bounded set of elements of $[n]$. The degree of the request is the worst of the individual least degrees:

$$h_S(n) = \max_{b \in \mathcal{B}_n} h_S(\Gamma_n \vdash G_b^{(n)}), \qquad S \in \{\mathrm{NS}, \mathrm{PC}\}. \tag{3}$$

The empty maximum is zero; a single underivable target makes the maximum ∞. Thus $h_S(n) \leq d$ means that *every* requested target has a degree-d derivation, with no restriction on the total number of proof lines used.

There are three useful forms of this definition. A *fully invariant* target $G^{(n)}$ is the case of one target index with empty support, so $\sigma G^{(n)} = G^{(n)}$ for every permutation of $[n]$. For example, with variables x_i, the polynomial $\sum_{i<n} x_i$ is fully invariant: each monomial uses one label, although the full sum mentions all labels. An *indexed* family can instead request $G_i^{(n)} = x_i$ for each $i < n$. Its index has support one; its members are permuted rather than individually fixed.

One may restrict $\mathcal{B}$ to one fixed global orbit of indices. At every rank $n \geq t$ that orbit has a representative; all its representatives at that rank are related by a permutation of $[n]$ and have the same least degree. Hence one representative per rank gives the same profile. Indeed, restrict a global relabelling to the finite support of the index; its image is also in $[n]$. Extend this finite injection to a permutation of $[n]$, which has the same action on the supported index. There are only finitely many global orbits, since each meets the finite set $\mathcal{B}_t$. The aggregate request (3) takes their maximum. An arbitrary selection that switches between different orbits with n is a different problem and is not covered by the theorem.

Rank-independent coefficient rules are essential for both axioms and targets. Symmetry of each individual instance is insufficient: choosing $\{1 = 0\}$ at sizes in an arbitrary set and the empty system elsewhere produces an arbitrary degree-zero predicate, but is not a fixed template. Likewise, the target cannot be chosen by an arbitrary size-dependent rule.

Beyond n_0, every axiom-index orbit has a fixed polynomial degree. A monomial witnessing its formal degree uses at most kD labels; a relabelling fixing the index places these labels inside $[n]$ once $n \geq kD + s$. The same argument for targets gives the common stable starting rank

$$n_* = \max\{n_0, 1, kD_G + t\}. \tag{4}$$

Thus each target orbit has a fixed degree there, and their maximum is a degree floor for both profiles. The all-target degree-zero predicate is constant in this range: a nonconstant requested target fails it, whereas constant targets are consequences exactly when they are zero or the inputs span a nonzero constant. For general indexed requests the transport argument uses the larger support budget $2kd + s + t$; the extra factor of two detects a possible nonzero quotient, as explained in Section 4.

2.3 The normalised first-order input class

Any fixed first-order sentence in a finite uninterpreted vocabulary can be converted, by Skolemisation, flattening of terms and clausification, to a normalised expansion with the same existence of models at each nonempty finite size. Witnesses are retained as function symbols rather than translated as unrestricted sums. The resulting roots correspond to models of this expansion; several expansions may have the same reduct to the original vocabulary. This is the translation approach of [6, Section 1.2][7, Section 1.2] and [22]; see the explicit translation theorem in [29, Section II, Theorem 1]. We fix one such normalisation.

A normalised sentence is a finite collection of universally quantified clauses over relation symbols and function symbols. Its atoms are $R(x_{i_1}, \ldots, x_{i_a})$, flattened graph atoms $x_j = f(x_{i_1}, \ldots, x_{i_a})$, and equalities $x_i = x_j$. Constants are functions of arity zero. Symbols are uninterpreted; in particular there is no built-in order or arithmetic. Well-formedness requires the declared arities and variable references to match.

At size n introduce relation variables $r_{\vec{i}}^{R}$ and graph variables $x_{\vec{i},j}^{f}$. Include

$$z^2 - z = 0 \qquad \text{for every variable,} \tag{5}$$

$$\sum_{j<n} x_{\vec{i},j}^{f} - 1 = 0, \qquad x_{\vec{i},j}^{f} x_{\vec{i},j'}^{f} = 0 \qquad (j \neq j'), \tag{6}$$

$$\prod_{L \in C} (1 - \mathrm{val}(L)) = 0 \qquad \text{for every grounded clause } C. \tag{7}$$

An equality atom has value δ_{ij}; negation replaces a value by its complement. The collisions in (6) are indispensable in characteristic p: a sum of Boolean values equal to 1 need not contain exactly one 1 without them.

Proposition 2.3 (Compiler and semantics)**.** *The displayed translation is an actual uniform template. Its parameters can be computed from the normalised syntax. For every $n \geq 1$, its common roots over K correspond to models of the sentence on $[n]$.*

Proof. Use the atom tuples for variable indices and clause-grounding tuples for clause indices; totality and collision indices are also fixed tuples. Expansion of (7) has coefficients depending only on equalities among these labels. Repeated coordinates and coefficient cancellations are included in this expansion. Safe bounds, writing a_R, a_f for the maximum relation and function arities, are

$$k = a_R + a_f + 1, \quad s = k + 1 + \max_C \mathrm{nvars}(C), \quad D = 2 + \max_C |C|, \quad n_0 = kD + s + 1.$$

Here $\mathrm{nvars}(C)$ is one more than the largest variable index occurring in C, and is 1 when none occurs (the compiler reserves an unused variable slot). At a root every variable is 0 or 1.

Totality and collisions choose exactly one output of each function. Clause products vanish precisely when a literal is true. Conversely, a model supplies these 0/1 values and satisfies every equation. □

We use this explicit normalised input class for undecidability. There is no assertion that different choices of normalisation or Skolemisation preserve exact degree. Keeping the chosen encoding visible is especially important for the examples.

2.4 Nullstellensatz completeness and degree upper bounds

Let $I = (f_1, \dots, f_m) \subseteq K[x_1, \dots, x_v]$ be the ideal generated by a finite polynomial system. Hilbert's Nullstellensatz relates its common roots over an algebraic closure $\overline{K}$ to ideal membership: the system has no common root if and only if $1 \in I$. More generally, a polynomial G vanishes on every common root if and only if $G \in \sqrt{I}$, equivalently $G^r \in I$ for some $r \geq 1$; see [15]. The ideal-membership statements may be made over K itself, since solvability of the finite coefficient systems descends from field extensions (Lemma 2.5). An NS derivation of the actual target G requires $G \in I$, rather than merely $G \in \sqrt{I}$. Thus semantic consequence and algebraic derivability need not coincide for a general system.

For Boolean encodings, the equations $x_i^2 - x_i = 0$ force every variable to be 0 or 1, over K and over its algebraic closure. The quotient by these equations is a finite product of copies of K; hence every ideal containing them is radical. In this setting a target vanishes on all solutions if and only if it has an NS derivation. In particular, absence of a Boolean solution is equivalent to an NS refutation [20]. Without these equations, absence of roots in K alone is insufficient: $x^2 + 1$ has no root in $\mathbb{F}_3$ but has roots over its algebraic closure and admits no refutation.

The quantitative issue is the degree of a certificate. For an inconsistent Boolean system in v variables with input degree at most $D \geq 2$, the standard Boolean interpolation argument gives the safe bound

$$d_{\mathrm{PC}} \leq d_{\mathrm{NS}} \leq v + D. \tag{8}$$

Indeed, Boolean point indicators have degree at most v; multiplying each by an input nonzero at that point and reducing modulo the Boolean equations yields a certificate whose products have degree at most $v + D$. See [20] for the standard completeness argument.

For comparison, Razborov [34] gives modified pigeonhole systems with PC degree $\Omega(v)$, allowing logarithmic input degree. This does not establish sharpness for our uniform class, which requires a constant input-degree bound D independent of n.

Here v counts variables, whereas n denotes the universe size. For variables indexed by tuples of arity at most k, typically $v = O(n^k)$, so this argument does not give a general $O(n)$ bound. A bound linear in universe size requires additional structure: for the injection system, Section 13 gives an explicit certificate of degree at most $n + 1$. Together with its linear lower bound, that example establishes the asymptotic sharpness claimed in this paper.

All degree bounds use the ordinary polynomial ring. Multilinearisation must be justified by Boolean-equation multiples with their degrees counted: multiplying a squarefree degree-d monomial by a variable already in it first gives degree $d + 1$. The general uniform-family theorem does not require Boolean equations. In the first-order encoding, the pairwise collision equations are also needed for uniqueness: in characteristic p, a totality sum equal to one can otherwise have $p + 1$ nonzero summands.

2.5 The space of all bounded-degree consequences

Fix a rank-n instance, write $V = \mathcal{V}_n$ for its variable-index set, and choose a degree budget $d \geq 1$. Let $P_{\leq d}$ be the vector space with the monomials of degree at most d as basis. Zero

axioms may be omitted. For NS define

$$I_d = \operatorname{span}_K\{\mu f : f \in \Gamma,\ \mu \text{ a monomial}, \quad \deg \mu + \deg f \le d\} \subseteq P_{\le d}. \tag{9}$$

NS uses one span of permitted monomial–axiom products. PC requires an additional closure:

$$\begin{aligned} V_0 &= \operatorname{span}_K\{f \in \Gamma : \deg f \le d\}, \\ V_{j+1} &= V_j + \sum_{v \in V} x_v (V_j \cap P_{\le d-1}). \end{aligned} \tag{10}$$

For $d = 0$ omit the multiplication term, so $V_\infty = V_0$.

Proposition 2.4 (Exact consequence-space characterisation)**.** *For every polynomial G and every degree budget $d \ge 0$,*

$$\begin{aligned} h_{\mathrm{NS}}(\Gamma \vdash G) \le d \quad &\iff \quad G \in I_d, \\ h_{\mathrm{PC}}(\Gamma \vdash G) \le d \quad &\iff \quad G \in V_\infty := \bigcup_j V_j. \end{aligned}$$

The PC chain stabilises after finitely many stages and $I_d \subseteq V_\infty \subseteq P_{\le d}$. In particular, the least refutation degrees test membership of 1 in these spaces.

Proof. For NS, expand each multiplier into monomials. Every resulting product μf satisfies the degree bound and belongs to the spanning set in (9). Conversely, group a finite linear combination of these products by axiom. Its multiplier a_f has degree at most $d - \deg f$, giving the required identity with right-hand side G. The empty combination gives $G = 0$. Each permitted product μf is also a PC consequence: multiply f successively by the variable factors of μ, with every intermediate degree at most d, and add.

For PC, induct on a proof line. A usable axiom lies in V_0. A linear combination of vectors in V_a and V_b lies in $V_{\max(a,b)}$. If a nonzero line f is multiplied by a variable and the result has degree at most d, then $\deg f \le d - 1$. If $f \in V_j$, its multiple therefore lies in V_{j+1}. The zero line causes no difficulty.

Conversely, every vector of V_0 is derived by linear combinations of usable axioms. A vector of V_{j+1} is a finite sum of a vector of V_j and terms $x_v f_v$, with $f_v \in V_j \cap P_{\le d-1}$. Concatenate finite derivations of the f_v, append the permitted multiplication steps, and add until the last line is the requested vector. All partial linear combinations remain in $P_{\le d}$.

The ascending chain lies in the finite-dimensional space $P_{\le d}$. Every strict inclusion increases dimension, and equality of two successive stages makes all subsequent stages equal. Hence the chain stabilises. This argument places no length bound on candidate proofs; it shows that all their consequences are captured by a finite closure. □

The cancellation in our earlier example occurs in $V_0 \cap P_{\le 1}$: $f_1 + f_2 = x + 1$ becomes eligible for multiplication although neither of its two inputs was linear. A transport argument for ordinary PC must preserve the intersection in (10), including such vectors, rather than just transport a list of low-degree input axioms.

Extending scalars from K to a field E keeps the coordinate equations and allows coefficients in E. If $U \subseteq K^M$ has a K-basis $u_1, \dots, u_a$, its extension is $U_E = \operatorname{span}_E\{u_1, \dots, u_a\} \subseteq E^M$, also denoted $E \otimes_K U$. This is the finite-dimensional scalar-extension operation used in the next lemma; see [15] for its standard linear-algebra background.

Lemma 2.5 (Field extension)**.** *For a finite system Γ and target G over K, extending the coefficient field changes neither least NS nor least ordinary-PC degree of $\Gamma \vdash G$.*

Proof. At fixed degree, NS derivability is membership of G in a finite span; Gaussian elimination gives the same answer after extending scalars. For PC, scalar extension commutes with finite-dimensional sums, intersections and images, hence with every stage of (10). Membership of the original vector G in the stable space is therefore preserved and reflected. □

2.6 Explicit coefficient lists and proof size

Write $v(n) = |\mathcal{V}_n|$ for the number of polynomial variables, distinguishing it from the universe size n and the periodicity onset $N(d)$ used later. Variables, axiom indices and target indices all have polynomial counts in n: each bounded-support object is a relabelling of one of finitely many objects at its support bound.

Definition 2.6 (Size with explicit coefficient lists)**.** In an ambient ring with v variables, a nonzero polynomial of total degree d is stored as the full coefficient table of all monomials of total degree at most d, including zero coefficients. Its length is

$$B_v(d) = \binom{v+d}{d}.$$

The zero polynomial has a one-entry representation. A PC proof of G has size equal to the sum of these lengths over its lines, including the final line G. An NS identity $G = \sum_f a_f f$ has size equal to the sum over its nonzero multipliers; the input polynomials and target are supplied separately. The empty certificate for $G = 0$ has size zero.

Let $S_S^{\mathrm{coeff}}(\Gamma \vdash G)$ be the minimum certificate size in system S, with value ∞ if the target is underivable. For a uniform indexed request put

$$S_S^{\mathrm{coeff}}(n) = \max_{b \in \mathcal{B}_n} S_S^{\mathrm{coeff}}(\Gamma_n \vdash G_b^{(n)}),$$

with empty maximum zero. For the single target 1, this is the usual minimum refutation size.

A bundle of proofs for every requested target has total size between this maximum and $|\mathcal{B}_n|$ times the maximum, when the request is nonempty. Since $|\mathcal{B}_n|$ is polynomial in n, the two conventions give the same polynomial and exponential size cases. Size counts coefficient entries over K. For a fixed finite field, coefficients have bounded bit length; variable, axiom and line indices add only polynomial overhead to the bounds below. The uniformly bounded-degree inputs and targets themselves have polynomial-size tables. Every table uses *all ambient variables*, even those absent from its nonzero terms. Filling in missing powers in $x_1^d + 5$ alone would require only $d + 1$ entries, a different measure.

Lemma 2.7 (Degree and coefficient-list size)**.** *For v variables and input degree at most D, if G has a degree-d NS derivation, it has an NS certificate of size at most $B_v(d)^2$. If it has a degree-d ordinary-PC derivation, it has a PC proof of size $O(B_v(d)^3)$. Conversely, writing $h_S = h_S(\Gamma \vdash G)$, finite least degrees give*

$$S_{\mathrm{PC}}^{\mathrm{coeff}}(\Gamma \vdash G) \geq B_v(h_{\mathrm{PC}}), \qquad S_{\mathrm{NS}}^{\mathrm{coeff}}(\Gamma \vdash G) \geq B_v(\max\{0, h_{\mathrm{NS}} - D\}) \quad (G \neq 0).$$

Also $B_v(d) \geq 2^{\min\{v,d\}}$. For an indexed request the same upper bounds hold when $h_S(n) \leq d$. For a nonempty request, the lower bounds apply to a target attaining the finite maximum, with the NS zero-target exception just stated.

Proof. Put $M = \dim P_{\leq d} = B_v(d)$. For NS, membership of G in the span I_d has a witness using at most M monomial–axiom products. Grouping by axiom gives at most M nonzero multipliers, each with at most M coefficient entries.

For PC, retain a basis of the usable input axioms. Maintain a basis of the span W of the retained derivable polynomials. Unless W is closed under permitted multiplication, choose $f \in W \cap P_{\leq d-1}$ with $x_i f \notin W$. Derive f as a linear combination of at most M retained polynomials, using $O(M)$ two-line steps, then append $x_i f$ to the basis. At most M extensions occur. Once W is closed, it contains every degree-d consequence, including G. Append a

linear combination giving G as the last line. This constructs $O(M^2)$ lines, each with at most M entries; all intermediate linear combinations stay in $P_{\le d}$. If $G = 0$, the one-line proof 0 suffices.

Every PC proof has a line of degree at least its least possible proof degree. If $G \neq 0$, every NS certificate has a nonzero product of degree at least h_{NS}, whose multiplier consequently has degree at least $h_{\mathrm{NS}} - D$. Finally, the table contains every squarefree monomial in any $\min\{v, d\}$ variables, proving the exponential estimate. For a finite indexed request, a finite maximum is attained unless the request is empty, in which case its size is zero. □

This coefficient convention makes the size consequence elementary. It is a different resource from sparse monomial size: zero entries may dominate the table. We make no sparse-size lower-bound claim here.

3 Quantitative transfer and the four degree profiles

Fix a uniform axiom template and a stable uniform target family $G_b^{(n)}$ as in Section 2. The parameters k, s, t bound variable, axiom-index and target-index supports, respectively, and D_G bounds the target degree. Put

$$n_* = \max\{n_0, 1, kD_G + t\}.$$

For $\mathrm{S} \in \{\mathrm{NS}, \mathrm{PC}\}$, the aggregate profile $h_{\mathrm{S}}(n)$ is the maximum of the least degrees needed to derive the requested polynomials $G_b^{(n)}$, $b \in \mathcal{B}_n$. Zero targets have degree 0; an underivable target gives value ∞, and the maximum of an empty request is 0. We may instead restrict $\mathcal{B}$ to one fixed global orbit. At every $n \ge t$ that orbit has a representative, and equivariance makes its profile the least degree of any representative. These are two precise choices of a uniform request. Choosing different global orbits according to n is not covered.

Use the conservative parameters

$$\kappa = 2k, \qquad \sigma = s + t. \tag{11}$$

The general transfer budget is then $\kappa d + \sigma = 2kd + s + t$. For a fully invariant target, and in particular for refutation, a sharper version with $\kappa = k$ and $\sigma = s$ is given below. All the profile deductions are stated in the same notation so that both versions have a common presentation.

The next theorem is the quantitative lifting statement from which the classification follows. It fixes the degree budget and varies the universe size. Its two directions are important: an upward lift keeps a derivation available on one progression, while the equivalence also rules out a new derivation appearing later in that residue class.

Theorem 3.1 (Periodicity with linear onset)**.** *Over any field of prime characteristic p, let h be either the NS or ordinary-PC aggregate profile, or either profile of one fixed target orbit. For $d \ge 1$ put $R(d) = \kappa d + \sigma$. If*

$$n, n' \ge n_*, \qquad n, n' > 2R(d), \qquad n \equiv n' \pmod{q}, \qquad q = p^a > R(d),$$

then $h(n) \le d$ if and only if $h(n') \le d$. The degree-zero predicate is constant for $n \ge n_*$.

The proof in Section 4.4 transports the actual coefficient map of the targets and the actual consequence spaces. It does not require invariant proofs, and it does not infer the PC assertion from an NS lower bound. The same argument works whether or not the axioms are consistent.

Define

$$\begin{aligned} N(0) &= n_*, \qquad Q(0) = 1, \\ N(d) &= \max\{n_*, 2(\kappa d + \sigma) + 1\}, \qquad Q(d) = \min\{p^a : p^a > \kappa d + \sigma\} \quad (d \geq 1). \end{aligned} \tag{12}$$

Here $N(d)$ denotes the onset for a degree threshold; it is not the larger rank $N = n + q$ used in the deletion construction. $Q(d)$ is the chosen block size and period. Both are bounded linearly in d beyond the fixed template threshold. The onsets are nondecreasing, and $Q(e)$ divides $Q(d)$ for $e \leq d$. Applying Theorem 3.1 at every threshold up to d gives

$$n, m \geq N(d), \quad n \equiv m \pmod{Q(d)} \quad \Longrightarrow \quad \min\{h(n), d+1\} = \min\{h(m), d+1\}. \tag{13}$$

Here $\min\{\infty, d+1\} = d+1$. For $d \geq 1$,

$$Q(d) \leq p(\kappa d + \sigma). \tag{14}$$

Indeed, the preceding power of p is at most $\kappa d + \sigma$. The fixed initial threshold includes the stabilisation of target degrees. A budget below a target's stabilised degree cannot derive that target; thus such thresholds are not replaced by tests on a truncated target.

3.1 The classification theorem

Define the set of recurring finite degrees by

$$E(h) = \{d \in \mathbb{N} : \text{some } n \geq N(d) \text{ has } h(n) = d\}.$$

Equation (13) shows that every member d occurs along an infinite arithmetic progression:

$$h(n + \ell Q(d)) = d \qquad (\ell \geq 0).$$

The degree remains d throughout this progression: lifting does not lower the absolute degree. Its ratio to the universe size tends to zero. Nor does one such progression make the whole profile bounded; other residue classes may remain hard.

Conversely, a value occurring at arbitrarily large sizes occurs beyond $N(d)$. Thus $E(h) = \mathrm{Rec}(h)$, independently of the admissible onset. The key restriction is that a nonrecurring finite degree d can occur only before its own onset $N(d)$.

To include the small ranks in an exact formula, write $x_+ = \max\{x, 0\}$ and put

$$\beta(n) = \left\lfloor \frac{(n - 2\sigma - 1)_+}{2\kappa} \right\rfloor.$$

The positive part agrees with natural-number subtraction and is irrelevant once $n \geq 2\sigma + 1$.

Theorem 3.2 (Four-way classification)**.** *For each profile considered in Theorem 3.1, exactly one of the following four alternatives holds.*

T1. *The profile is bounded by a finite constant for all sufficiently large n.*
T2. *$E(h)$ is empty. For every $n \geq n_*$,*

$$h(n) > \beta(n), \tag{15}$$

including $h(n) = \infty$ in the extended order. In particular the finite values have an $\Omega(n)$ lower bound.

T3. *$E(h)$ is finite and nonempty, and h is not eventually bounded. Put $D = \max E(h)$. There is a nonempty proper set $U \subseteq \{0, \dots, Q(D) - 1\}$ such that, for each $c \in U$, one finite value $v_c \le D$ satisfies*

$$h(n) = v_c \quad \text{whenever } n \ge N(D), \quad n \bmod Q(D) = c.$$

For $n \ge N(D)$ with $n \bmod Q(D) \notin U$, the lower bound (15) holds.

T4. *$E(h)$ is infinite. For every bound B there are $n, d \ge B$, with $d \in E(h)$ and $h(n) = d < \infty$, such that*

$$n \le (p+2)(\kappa d + \sigma), \qquad h(n) \ge \frac{n}{(p+2)\kappa} - \frac{\sigma}{\kappa}. \tag{16}$$

Also $\liminf_{n\to\infty} h(n) = \min E(h) < \infty$.

In T1 and T3 the liminf likewise equals the least recurring finite value; in T2 it is ∞. All statements leave the fixed finite initial range unconstrained. NS and PC assign each fixed uniform request the same type, by Theorem 10.2.

Proof. For $n \ge n_*$, the inverse threshold satisfies $N(\beta(n)) \le n$. If $\beta(n) = 0$ this follows from $N(0) = n_*$. Otherwise $2\kappa\beta(n) \le n - 2\sigma - 1$, which gives the assertion from (12). Consequently, if $h(n) \le \beta(n)$, its finite value belongs to $E(h)$: its own onset is at most $N(\beta(n))$. If $E(h) = \varnothing$, this is impossible and proves T2, including the small ranks for which $\beta(n) = 0$.

Suppose $E(h)$ is finite and nonempty and let $D = \max E(h)$. By truncated periodicity, on each class modulo $Q(D)$ the profile has one fixed value at most D, or is always greater than D, once $n \ge N(D)$. The classes of the first kind form U. Recurrence puts some value from $E(h)$ beyond $N(D)$, so U is nonempty. If all classes belong to U, the profile is eventually bounded. Otherwise, at a size off U, the inequality $h(n) \le \beta(n)$ would give a recurring value, hence a value at most D, contrary to the definition of U. This proves the complete T3 split whenever the profile is not bounded.

If $E(h)$ is infinite, choose its elements d arbitrarily large so that $N(d) = 2(\kappa d + \sigma) + 1$. A recurring class for d has a representative in the interval

$$N(d) \le n < N(d) + Q(d), \qquad h(n) = d.$$

Using integrality and (14) gives $n \le (p+2)(\kappa d + \sigma)$, hence (16). The chosen d and n can both exceed any prescribed bound. These are finite degree values, rather than occurrences of ∞.

Whenever $E(h)$ is nonempty, let $e = \min E(h)$. Each of the finitely many values below e occurs only finitely often, so eventually $h(n) \ge e$. The value e recurs, proving the liminf formula. If $E(h)$ is empty, the same argument for every finite bound shows $h(n) \to \infty$. Finally, eventual boundedness forces $E(h)$ to be finite and nonempty: a recurring value cannot exceed an eventual bound, and at least one of the finitely many bounded values occurs infinitely often. This proves exhaustiveness and mutual exclusivity of the four cases. □

The finite liminf in T4 is one number, not an assertion that the profile is eventually constant. Its infinitely many recurring levels persist arbitrarily far out. The bound on $N(d)$ provides the degree gap; the bound on $Q(d)$ places finite representatives of large recurring levels at sizes linear in their degrees. Both estimates are needed for the peak statement. The easy classes in T3 may have different finite degrees. Neither the theorem nor the comparison identifies the NS and PC sets of recurring degrees.

Corollary 3.3 (Sharper invariant-target and refutation bounds)**.** *For one fully S_n-invariant target given by a stable coefficient rule, Theorems 3.1 and 3.2 hold with $\kappa = k$, $\sigma = s$ and $n_* = \max\{n_0, 1, kD_G\}$. For refutation, take the single target 1; then $D_G = 0$, $n_* = n_0$, and the transfer budget is $kd + s$.*

Proof. The target map has the trivial module K as source. Its image in a consequence-space quotient is zero or a copy of K, and deletion fixes K. Lemma 4.4 therefore detects membership without the doubled ambient support needed for indexed targets. The actual target matrix is still transported. The unchanged axiom and PC-closure budget $kd + s$ gives periodicity, and the preceding profile argument applies with the smaller parameters. □

Proposition 3.4 (The fluctuating case has no fixed periodic split)**.** *In T4 there is no period q_0, set of its residue classes U, finite bound B, and constant $c > 0$ such that, at every sufficiently large size, $h(n) \le B$ on U and $h(n) \ge cn$ off U.*

Proof. Choose a recurring finite value $e > B$. Its sufficiently late occurrences cannot lie in U, and for $n > e/c$ cannot lie off U. This contradicts recurrence. The argument uses unbounded recurring finite levels, not merely an unbounded profile. □

3.2 The polynomial–exponential consequence

With the full explicit coefficient lists of Definition 2.6, let $S(n)$ be the maximum of the minimum sizes of proofs of the requested targets. Use $S(n) = 0$ for an empty request and $S(n) = \infty$ if any target is underivable. A collection of proofs for every target has minimum total size between $S(n)$ and $|\mathcal{B}_n|S(n)$ when the request is nonempty. Since $|\mathcal{B}_n| = n^{O(1)}$, these two size conventions give the same conclusions below. For one fixed orbit, any representative has the same minimum size by relabelling.

Corollary 3.5 (Polynomial–exponential size bounds)**.** *For either NS identities or ordinary-PC derivations of a fixed uniform request, the type in Theorem 3.2 has the following size consequences.*

T1. *$S(n) \le n^C$ for all sufficiently large n, for some constant C.*
T2. *$S(n) \ge 2^{cn}$ for all sufficiently large n, for some $c > 0$.*
T3. *On the easy residue classes, $S(n) \le n^C$ for all sufficiently large n; on their complement, $S(n) \ge 2^{cn}$, for fixed $C, c > 0$.*
T4. *$S(n) \le n^C$ at infinitely many sizes for some fixed C, and $2^{cn} \le S(n) < \infty$ at arbitrarily large sizes for some fixed $c > 0$. Moreover, for every $e > 0$ there is b such that $n^e < S(n) \le n^b$ at infinitely many sizes. There is no eventual periodic partition with one uniformly polynomial part and one part satisfying an exponential lower bound.*

The exponential statements are lower bounds. They give no universal matching exponential upper bound. Infinite sizes are interpreted in the extended order; finiteness is asserted explicitly on the T4 peak subsequence.

Proof. Write D_A for a fixed upper bound on axiom degree. If the variable basis contains an object with support size $j > 0$, relabelling supplies a distinct object supported on each j-subset of $[n]$. Thus $v(n) \ge \binom{n}{j} \ge n$ for sufficiently large n, while $v(n) = n^{O(1)}$. The target and axiom index counts are polynomial as well. Lemma 2.7 gives polynomial-size proofs for every target whose derivation degree is bounded by a fixed constant, uniformly over the request.

If the aggregate degree is a finite value $d > D_A$, some target attains that maximum. Its minimum proof size is at least $B_{v(n)}(d - D_A)$ in either system: for NS, a degree-d identity must contain a multiplier of degree at least $d - D_A$; for PC, any proof must contain a line of degree at least d. Such a target is nonzero, so the empty NS certificate creates no exception. A linear degree lower bound therefore yields $2^{\Omega(n)}$. If a target is underivable, its infinite size satisfies the same lower inequality. This proves T1–T3 and the two subsequences in T4.

For the additional T4 assertion, choose a recurring degree d with $\ell = d - D_A > e$. At its infinitely many sufficiently large occurrences,

$$S(n) \geq B_{v(n)}(\ell) \geq \frac{n^\ell}{\ell!} > n^e,$$

while the fixed degree d gives $S(n) \leq n^b$ for some b. A periodic polynomial/exponential partition would be contradicted by choosing e above its polynomial exponent: these occurrences are too large for its polynomial part and too small for its exponential part.

If every variable support is empty, the entire variable basis is finite and every permutation fixes it. Each axiom-index orbit consequently gives one rank-independent polynomial; each target-index orbit does too. There are finitely many such orbits, and all appear at sufficiently large rank. The actual axiom and target polynomial sets therefore stabilise. The aggregate degree is then a finite constant, or is ∞, giving respectively T1 or T2 with infinite size. This covers the degenerate case without a growing variable count. □

The coefficient-list estimate is *at least* exponential in universe size. For instance, $v(n) = \Theta(n^2)$ and a line of degree $\Theta(n)$ require $2^{\Theta(n \log n)}$ coefficient entries. Over a fixed finite field, encoding the axiom and target polynomials by full lists has length $L(n) = n^{O(1)}$, since their degrees and index supports are fixed. If $L(n) = O(n^{c_0})$ for a fixed $c_0 > 0$, the hard-instance bound gives $2^{\Omega(L(n)^{1/c_0})}$. This is a stretched-exponential lower bound in input length, with an exponent depending on the template.

4 Deleting labels while preserving uniform consequences

The closure in Section 2.5 reduces a question about arbitrarily long proofs to one about subspaces and linear maps. The proof works with these spaces of consequences rather than with invariant certificates. Even an invariant target can belong to the image of an equivariant map without having an invariant preimage in characteristic p. Averaging over S_n would therefore miss the issue. The construction below transports the whole image and, for PC, its iterated closure.

A representation, or module, here is a vector space with a linear action of label permutations. An invariant subspace is preserved by every such permutation; a quotient by it inherits the action. A *subquotient* is a quotient of an invariant subspace. The trivial module is K with every permutation acting as the identity. In characteristic p, the usual averaging operator $(n!)^{-1} \sum_{g \in S_n} g$ is unavailable once $p \mid n!$. We therefore construct compatible maps directly, rather than obtain them by averaging.

There is also a reason not to delete variables by substitution. Setting all x_{ij} with a deleted row label i to zero turns the totality axiom $\sum_j x_{ij} - 1$ into -1. The resulting system is immediately refutable; it is not the intended smaller instance. Instead we need an operation which deletes labels on permutation bases while preserving the linear relations among their subspaces.

The proof separates three requirements: transport the actual coefficient and multiplication maps, preserve the linear operations forming the consequence spaces, and reflect target membership. We establish them below using one deletion operation. A common projector then realises the reverse direction as an explicit lifting operator. Sections 5–9 construct that projector; the present section explains the contracts it must satisfy and how they imply the desired transfer.

Here *exactness* means preservation of kernels and images, and hence of the operations in the PC recurrence. For example, $U + V$ is the image of $(u, v) \mapsto u + v$, while $U \cap V$ is

identified with the kernel of $(u, v) \mapsto u - v$ on $U \oplus V$. Preserving only dimensions, or only the spaces generated by the input axioms, would not suffice.

Remark 4.1 (Why prime-power blocks are natural). For $q = p^a$ and $0 \le j < q$, the images of the integers $\binom{n+q}{j}$ and $\binom{n}{j}$ agree in any field of characteristic p. Indeed, in $K[T]$ repeated use of the characteristic-p binomial identity gives

$$(1+T)^{n+q} = (1+T)^n(1+T^q).$$

Comparing coefficients below degree q proves the assertion. At $j = q$ an additional coefficient 1 appears, so the strict inequality matters. This explains the persistence of small counting expressions after a prime-power extension. It is only a counting observation: ordinary-PC transfer also has to preserve intersections and the restricted multiplication maps, which require the construction below.

4.1 The exact deletion lemma

For this representation-theoretic construction, K is algebraically closed of characteristic p. Lemma 2.5 will return the conclusions about proof degree to an arbitrary coefficient field. For a finite set X, $K[X]$ denotes the vector space with basis e_x for $x \in X$. A permutation action on X gives the permutation module $ge_x = e_{gx}$. The group algebra $K[S_N]$ consists of linear combinations of label permutations, with composition extended bilinearly as multiplication. Its elements act by the same expressions on every representation; an equivariant map commutes with these expressions. This is the standard module language used below [16].

Let $P_b(N) = K[\mathrm{Inj}([b], [N])]$ be the permutation module on injective tuples. Write $\mathcal{C}_r(N)$ for the finite-dimensional $K[S_N]$-modules annihilated by every group-algebra element which annihilates all $P_b(N)$, $0 \le b \le r$. Equivalently, it consists of subquotients of finite direct sums of the $P_b(N)$, $b \le r$. Indeed, their sum is a faithful module for the finite-dimensional quotient algebra; the regular module embeds into a finite sum of copies by evaluation on a basis. This class is closed under submodules, quotients and finite direct sums. It contains the permutation module of every bounded basis system of support at most r: each orbit is a quotient of a tuple orbit obtained by choosing an ordering of its support.

Thus $\mathcal{C}_r(N)$ imposes the operator relations shared by the tuple spaces up to length r. It does not require each vector to mention at most r labels: a vector in $P_1(N)$ may have nonzero coefficients at all N labels.

The regular action on the block means that exactly one translation takes any chosen block label to any other. Every retained label is fixed by those translations.

Take $q = p^a > r$, write $N = n + q$, and let $A \cong (\mathbb{Z}/p\mathbb{Z})^a$ act regularly on the last q labels. Its coordinate translations $\sigma_1, \dots, \sigma_a$ have order p and fix the retained labels $[n]$. Choose $c_1, \dots, c_a \in K$ linearly independent over $\mathbb{F}_p$ and put

$$z = \sum_{i=1}^{a} c_i(\sigma_i - 1), \qquad z^p = 0, \qquad \mathcal{D}(M) = \ker(z : M \to M)/(\ker z \cap \operatorname{im} z).$$

Here 1 is the identity permutation, and multiplication of operators means composition. The relation $z^p = 0$ expresses nilpotence of a linear operator; it does not assert that z itself is zero. Permutations of the first n labels commute with z, so $\mathcal{D}(M)$ is an S_n-module. Equivariant maps induce the maps of $\mathcal{D}$.

A nilpotent Jordan chain has basis $v_0, \dots, v_{\ell-1}$ with $zv_0 = 0$ and $zv_j = v_{j-1}$ for $j > 0$. Its kernel is Kv_0; for $\ell > 1$, this line is already in the image. A length-one chain has zero image instead. These are statements about the operator z; the chain decomposition need not respect the full label action.

On a Jordan decomposition for the nilpotent operator z, this quotient retains the one-dimensional blocks and discards the longer ones. That operation is generally not exact. The point of the following theorem is that the bounded-support representation class, with enough retained labels, has additional structure making it exact. On permutation bases it then has the concrete effect of deleting the last block of labels.

All tensor products below are over the working field. Concretely, $K[X] \otimes Y$ is a list of vectors $y_x \in Y$, written $\sum_x e_x \otimes y_x$. Labels act on both factors by $g(e_x \otimes y) = e_{gx} \otimes gy$. A natural identification is one which commutes with the applicable equivariant maps, rather than merely matching their dimensions.

Theorem 4.2 (Exact deletion)**.** *For $q > r$ and retained rank $n \geq \max\{0, 2r-1\}$ there is a choice of $c_1, \ldots, c_a$, linearly independent over $\mathbb{F}_p$ in the algebraically closed working field, for which $\mathcal{D}$ is exact on $\mathcal{C}_r(n+q)$. It fixes the trivial module. For a support-u basis system $\mathcal{U}$ with $u < q$, there are natural identifications*

$$\mathcal{D}(K[\mathcal{U}_{n+q}]) \cong K[\mathcal{U}_n], \tag{17}$$

$$\mathcal{D}(K[\mathcal{U}_{n+q}] \otimes Y) \cong K[\mathcal{U}_n] \otimes \mathcal{D}(Y) \tag{18}$$

for arbitrary Y. For rank-independent invariant coefficient matrices, the first identification transports the actual matrix to its realisation at rank n.

The algebraic certificate is a single group-algebra element w satisfying $z^{p-1} w z^{p-1} = z^{p-1}$ on all tuple modules of lengths at most r. Sections 5–9 construct the common witness and prove this theorem. The natural matrix and tensor identifications are part of that proof.

The witness is one shared inner-inverse expression; it is not an inverse to the nilpotent operator on the whole space. Exactness is asserted for the stated class and retained-rank range, with a suitable coefficient choice, and does not follow from $z^p = 0$ alone.

The tensor identification is required only with a permutation module in its first factor. When applying exactness to a tensor product with an arbitrary subspace, we place that product inside a known bounded-support ambient tensor product. Thus no unstated tensor closure of $\mathcal{C}_r$ is needed.

4.2 Transporting the entire consequence space

Fix a positive degree budget d and write $W_N = P_{N,\leq d}$ and $C_d(N) = V_\infty(N)$ for the stabilised PC consequence space. We first transport the spaces $I_d(N)$ and $C_d(N)$, without choosing any target. This separates the proof-system calculation from the question of whether a particular uniform expression belongs to it.

For NS the certificate map has source basis the pairs (μ, a) with

$$\deg \mu + \deg f_{a,N} \leq d, \qquad (\mu, a) \longmapsto \mu f_{a,N} \in W_N.$$

Its source has support at most $kd + s$, and W_N has support at most kd. The matrix entry at a target monomial ν is zero unless $\mu \mid \nu$, and otherwise is $c(a, \nu/\mu)$. These are the actual rank-independent invariant entries of the certificate map. The stability threshold n_0 makes each axiom degree independent of rank. Exactness and matrix naturality therefore identify the embedded image $\mathcal{D}(I_d(N))$ with $I_d(n)$.

For PC let $X_N = K[\mathcal{V}_N]$ and $L_N = P_{N,\leq d-1}$. The multiplication map

$$m_N : X_N \otimes L_N \longrightarrow W_N, \qquad x_v \otimes f \longmapsto x_v f$$

has stable entries. Its source is supported on at most $k + k(d-1) = kd$ labels. The usable axioms give the image $V_0(N)$ of their coefficient map. The recurrence is

$$V_{j+1}(N) = V_j(N) + \operatorname{im}\big(m_N|_{X_N \otimes (V_j(N) \cap L_N)}\big).$$

Every stage is a submodule of W_N, and every multiplication domain is a submodule of the fixed support-bounded space $X_N \otimes L_N$. The intersection is the kernel of $(v, l) \mapsto v - l$ on $V_j(N) \oplus L_N$. Exactness transports that kernel, the multiplication image and the sum. The natural tensor identification (18) supplies compatibility with the restricted multiplication map. Induction thus identifies the embedded $\mathcal{D}(V_j(N))$ with $V_j(n)$ for every j.

Both ascending chains stabilise because their ambient spaces are finite dimensional. Choose a stage beyond both stabilisation points. We obtain

$$\mathcal{D}(I_d(N)) = I_d(n), \qquad \mathcal{D}(C_d(N)) = C_d(n), \tag{19}$$

where equality means equality after the natural identification $\mathcal{D}(W_N) \simeq W_n$. This transports every cancellation in a PC proof, including a low-degree result subsequently multiplied by a variable. No bound on proof length and no symmetry of individual proof lines was used.

4.3 Why deletion also reflects target membership

Preserving an image is not by itself enough: an exact operation can annihilate a nonzero quotient. The next lemma supplies the reflection needed for a general target source.

The detector uses the bilinear coefficient pairing on a permutation basis, $\langle v, w\rangle = \sum_x v_x w_x$. Its orthogonal space $S^\perp$ consists of vectors pairing to zero with S. This pairing is nondegenerate over any field; it involves no conjugation or positivity. A vector represents the functional given by pairing against it.

Lemma 4.3 (Nonzero subquotients survive)**.** *Let $P = K[\mathcal{U}_N]$ have support at most u. Suppose deletion is exact on $\mathcal{C}_{2u}(N)$, its block size exceeds $2u$, and the natural identifications (17)–(18) hold. If $S \subsetneq S' \subseteq P$ are invariant subspaces, then $\mathcal{D}(S'/S) \neq 0$.*

Proof. The permutation basis gives a nondegenerate dot product on P, equivariant under S_N. It induces a pairing

$$S^\perp \otimes (S'/S) \longrightarrow K.$$

This pairing is surjective: for $v \in S' \setminus S$, a linear functional vanishing on S can take value one at v; the dot product represents that functional by an element of $S^\perp$. Let $Y = S'/S$. Tensoring over a field is exact, so the displayed source lies in $P \otimes Y$, itself a subquotient of $P \otimes P$. The last permutation module has support at most $2u$, justifying every use of exactness. If $\mathcal{D}(Y) = 0$, permutation-first-factor tensor compatibility gives $\mathcal{D}(P \otimes Y) = 0$. Exactness then takes the surjective pairing to a surjection from zero onto $\mathcal{D}(K) \simeq K \neq 0$, a contradiction.

We use this surjective pairing, not a trace or an averaged inner product. In characteristic p, composing a pairing with coevaluation can yield a vanishing dimension scalar and would not prove the lemma. □

The invariant-target case has a simpler detector: the trivial module K is one-dimensional, so a nonzero map from it has a one-dimensional image. It has no proper nonzero submodule, which is the meaning of *simple* in the following argument.

Lemma 4.4 (An invariant target needs no doubled budget)**.** *Let $j : K \to W$ and $H \subseteq W$ be equivariant and belong to the exactness class. Then $j(1) \in H$ if and only if its deleted image belongs to $\mathcal{D}(H) \subseteq \mathcal{D}(W)$.*

Proof. Membership is the vanishing of $K \xrightarrow{j} W \to W/H$. Its image is either zero or isomorphic to the simple trivial module K. Exactness preserves this image, and $\mathcal{D}(K) \simeq K \neq 0$. Thus the composite vanishes exactly when its deleted composite does. The vector $j(1)$ may be any invariant target, not just the constant polynomial one. □

4.4 The actual uniform target map

For indexed targets, let $\mathcal{B}$ be the fixed index basis and let $\gamma(b,\mu)$ be its stable coefficient rule. Write δ for the largest stabilised target degree, taking $\delta = 0$ for an empty request. At a rank where all target degrees are at most d, define

$$J_N : K[\mathcal{B}_N] \longrightarrow W_N, \qquad [b] \longmapsto G_b^{(N)}.$$

Its entry in row μ, column b is precisely $\gamma(b,\mu)$. Equivariance and rank independence are the hypotheses of the natural matrix identification in Theorem 4.2. Consequently $\mathcal{D}(J_N)$ is the actual smaller target map J_n. This statement concerns the prescribed full polynomials $G_b^{(n)}$; we do not replace a target by its degree-d truncation.

Let H_N be either $I_d(N)$ or $C_d(N)$. Every target belongs to H_N exactly when $\operatorname{im} J_N \subseteq H_N$. Put

$$S_N = H_N + \operatorname{im} J_N, \qquad Y_N = S_N/H_N.$$

The condition is $Y_N = 0$. Exactness, target-map naturality and (19) identify its deletion with

$$Y_n = (H_n + \operatorname{im} J_n)/H_n.$$

The polynomial permutation basis of W_N has support at most kd. Lemma 4.3, applied with this ORIGINAL support bound, reflects vanishing of Y_N once the budget covers $2kd$. The affine budget

$$R(d) = 2kd + s + t$$

also covers the certificate source $(kd + s)$, the target source (t) and every multiplication domain. It therefore proves membership in both directions. The target source need not be simple; reflection comes from a subquotient of the TARGET ambient polynomial space. Doubling $R(d)$ once again would be unnecessary: the pairing uses $W_N \otimes W_N$, whose support is $2kd$.

Proof of Theorem 3.1. First extend scalars to an algebraic closure, as permitted by Lemma 2.5. The degree of every indexed target stabilises beyond $n_* = \max\{n_0, 1, kD_G + t\}$. If d is smaller than the largest stable target degree, the all-target predicate is false at both ranks. Otherwise the full map J just described applies. Take $N = n + q$, $q = p^a > R(d)$ and retained rank $n \geq \max\{n_*, 2R(d) - 1\}$. Exact deletion transports H_N, J_N and their sum and quotient. The detection lemma yields

$$(\forall b \in \mathcal{B}_N,\ G_b^{(N)} \in H_N) \quad \Longleftrightarrow \quad (\forall b \in \mathcal{B}_n,\ G_b^{(n)} \in H_n).$$

Field descent holds for each actual target separately, so the equivalence holds over the original field. Repeated steps connect any eligible ranks in a common residue class and prove the stated periodicity. For one fixed global target orbit, replace $\mathcal{B}$ by that orbit; at eligible ranks, equivariance makes every representative equivalent.

At degree zero the stabilised targets are either all zero, or are scalars with at least one nonzero value. In the latter case every target is derivable in degree zero exactly when the original axioms include a nonzero constant. If a target has positive stable degree, degree-zero all-target membership fails. These are fixed predicates beyond n_* and give the separate zero-level assertion.

For a fully invariant target use $j_N : K \to W_N$, $j_N(1) = G^{(N)}$, in place of J_N. Lemma 4.4 reflects membership without the tensor pairing. The budget reduces to $kd + s$, with stable onset $\max\{n_0, 1, kD_G\}$. Taking $G^{(N)} = 1$ gives the sharper refutation special case. No target has been added to the axiom system. □

4.5 A lifting operator for bounded-degree consequences

Fix $d \geq 1$, write $W_n = P_{n,\leq d}$ and use a block $q = p^a > r = kd + s$ in the deletion range. Let $\iota : W_n \to W_N$ include retained monomials and $\rho : W_N \to W_n$ discard all others, where $N = n + q$. We seek a common projector lift satisfying

$$\mathsf{L} : W_n \longrightarrow W_N, \qquad \deg \mathsf{L}(f) \leq \deg f, \qquad \mathsf{L}(1) = 1, \qquad \mathsf{L}(V_j(n)) \subseteq V_j(N) \quad \text{for every } j. \tag{20}$$

It will also satisfy $\rho\mathsf{L} = \mathrm{id}$. The previously proved transfer gives

$$1 \in C_d(n) \quad \Longleftrightarrow \quad 1 \in C_d(N). \tag{21}$$

The operator below realises the upward direction while preserving the whole PC closure. Indexed targets still require the target-map argument above; their projector lifts need not equal the prescribed larger targets individually.

4.6 The lift from a common projector

We now give the map, leaving the construction of its certificate to the next sections. Temporarily allow a coefficient extension E/K and use the same notation for the extended polynomial spaces. Section 4.9 returns the lifting statement to K.

Identify the added block B with $A = \mathbb{F}_p^a$, acting on it by translations and fixing $[n]$. Let $\sigma_1, \ldots, \sigma_a$ be the coordinate translations. Choose $c_1, \ldots, c_a \in E$ linearly independent over $\mathbb{F}_p$ and set

$$z = \sum_{i=1}^{a} c_i(\sigma_i - 1) \in E[S_N], \qquad z^p = 0.$$

The group algebra $E[S_N]$ consists of linear combinations of label permutations; it acts on polynomials by the corresponding linear combinations of variable renamings. The permutations commute on the block, and $\sigma_i^p = 1$, which proves $z^p = 0$.

The finite algebraic target is one element $w \in E[S_N]$ for which

$$uwu = u \quad \text{on every } P_b(N) = E[\mathrm{Inj}([b], [N])], \quad b \leq r, \qquad u = z^{p-1}. \tag{22}$$

Thus the same expression must work on every tuple length up to the support budget. This inner-inverse identity does not make the nilpotent operator u invertible. For characteristic two put

$$e = (1 - zw)(1 - wz).$$

For general p, put $m = p - 1$ and

$$S_0 = 1, \qquad S_{j+1} = (1 - z^j w z^{m-j})S_j \ (0 \leq j < m), \qquad e = (1 - z^m w)S_m. \tag{23}$$

An idempotent, or projector, is an operator satisfying $e^2 = e$. For $v \in \ker z$, the constructed e will supply the representative ev of its deletion class. Its idempotence and this identification are proved below from the common witness and the block action.

The two formulas agree at $p = 2$. The general formula requires a further property of the block action: every nilpotent Jordan chain on a tuple module has length one or p. Nilpotence $z^p = 0$ alone would also allow intermediate lengths, for which this projector need not work. The block decomposition establishes this chain property; together with the common witness it gives the projector identities in Section 9. Once these inputs are proved, the formulas are explicit finite algebra expressions. The lifting operator is

$$\boxed{\mathsf{L}_{n,N,d}(f) = e\,\iota(f).} \tag{24}$$

Equivalently, writing $e = \sum_{g \in S_N} a_g g$,

$$\mathsf{L}(f) = \sum_{g \in S_N} a_g \, g(\iota(f)).$$

This explains why degree does not increase: inclusion and every renaming preserve degree. The difficult point is to choose the *common* coefficients a_g so that all consequence spaces are preserved. An arbitrary sum of renamings does not have that property.

4.7 Compatibility with the entire PC closure

The common projector constructed in Sections 5–9 satisfies $e^2 = e$, $ze = ez = 0$ and, on $\ker z$,

$$ev = v - z^{p-1}wv. \tag{25}$$

It identifies $\mathcal{D}(M)$ with eM naturally as vector spaces. Together with the retained-coordinate description of deletion, (19) and its stagewise proof give

$$\rho\big(V_j(N) \cap \ker z_{W_N}\big) = V_j(n) \quad \text{for every } j. \tag{26}$$

Thus the exactness and multiplication-map work need not be repeated.

Proposition 4.5 (The lifting formula preserves and reflects consequences)**.** *Assume the two facts in Section 4.6. Then* $\mathsf{L} = e\iota$ *preserves degree,* $\rho\mathsf{L} = \mathrm{id}$, $\mathsf{L}(1) = 1$, *and*

$$\mathsf{L}(f) \in V_j(N) \quad \Longleftrightarrow \quad f \in V_j(n).$$

The same equivalence holds for the stable spaces C_d.

Proof. Every retained monomial is fixed by the block action, so $z\iota(f) = 0$. Equation (25) gives

$$\mathsf{L}(f) = \iota(f) - z^{p-1}w\iota(f).$$

The image of z has no retained component: z kills retained basis vectors and preserves the span of moved ones. Hence $\rho\mathsf{L}(f) = f$. On constants, every permutation acts trivially, so the correction term vanishes and $\mathsf{L}(1) = 1$.

If $f \in V_j(n)$, equation (26) supplies $v \in V_j(N) \cap \ker z$ with $\rho v = f$. The permutation-space identification says that $v - \iota(f) \in \ker z \cap \operatorname{im} z$. Since $e \operatorname{im} z = 0$, we have $\mathsf{L}(f) = ev$. The space $V_j(N)$ is invariant under all label permutations, hence under e, and $ev \in V_j(N)$. Conversely, $\mathsf{L}(f) \in V_j(N)$ is killed by z, so (26) and $\rho\mathsf{L}(f) = f$ give $f \in V_j(n)$. Degree preservation follows from the renaming formula. Pass to the unions for the stable statement. □

This proves both the forward lifting implication and (21). The image of the lift need not be all of $C_d(N)$: the larger universe has many additional consequences. Every smaller consequence has this projector lift. In the transfer range, for $n \geq n_*$ and a fully invariant target G whose stabilised degree is at most d, naturality of its map from K gives $\mathsf{L}(G^{(n)}) = G^{(N)}$: the target map commutes with the common projector, which acts as the identity on K. More concretely, invariance gives $zG^{(N)} = 0$ and $eG^{(N)} = G^{(N)}$, while stable coefficients give $\rho G^{(N)} = G^{(n)}$. The retained-vector identification and the argument in Proposition 4.5 therefore yield $e\iota(G^{(n)}) = G^{(N)}$. Hence invariant targets, including 1, are exactly preserved. For indexed targets this equality need not hold. The image-subquotient argument above proves transfer of prescribed indexed targets without assuming it.

The common action also explains how individual inference steps are realised. For a retained axiom index a, naturality of its coefficient map F_N gives

$$\mathsf{L}(f_{a,n}) = F_N(e_A\iota_A(a)). \tag{27}$$

For $f \in V_j(n) \cap P_{n,\le d-1}$, act on the joint tensor input:

$$e_{X_N \otimes L_N}(\iota(x_v) \otimes \iota(f)) = \sum_y x_y \otimes g_y, \qquad X_N = K[\mathcal{V}_N], \quad L_N = P_{N,\le d-1}. \tag{28}$$

The tensor identification in (18) places each g_y in $V_j(N) \cap L_N$. Equivariance of multiplication then gives

$$\mathsf{L}(x_v f) = \sum_y x_y g_y. \tag{29}$$

The same permutation acts on both tensor factors; replacing this joint action by $e_X \otimes e_L$ would lose that compatibility. Storing closure-basis derivations and expressing the g_y in those bases recovers a literal PC proof, with no proof-length claim.

4.8 A one-label lifting example

Work in characteristic two. Retain the label a and add b, c. Let $z = 1 + (bc)$ and $w = (ab)$ act on the linear variables x_a, x_b, x_c. On their three-dimensional span,

$$z = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives $zwz = z$ and

$$e = (1 - zw)(1 - wz) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Consequently

$$\mathsf{L}(\alpha + \beta x_a) = \alpha + \beta(x_a + x_b + x_c). \tag{30}$$

Its retained coordinate is the original polynomial, and $\mathsf{L}(1) = 1$.

Take the uniform degree-one axiom $\sum_{i<m} x_i = 0$ at every universe size m. At size one the axiom is x_a; at size three it is $x_a + x_b + x_c$. Formula (30) sends the former exactly to the latter and lifts every degree-one consequence. Plain inclusion would leave x_a, which is not a consequence of the larger system: the assignment $(x_a, x_b, x_c) = (1, 1, 0)$ satisfies its sum but not $x_a = 0$.

Here the invariant target $G^{(m)} = \sum_{i<m} x_i$ is transported to the prescribed larger target. In contrast, the indexed expression $G_a^{(m)} = x_a$ has projector lift $x_a + x_b + x_c$, not x_a. The map is linear but is not a ring homomorphism and does not preserve every indexed expression literally. This calculation illustrates the distinction but lies outside the indexed transfer range: for $k = d = t = 1$ and $s = 0$, that theorem uses $R(d) = 3$, whereas here $q = 2$ and $n = 1$. It is therefore no counterexample to indexed periodicity.

4.9 Descent to the coefficient field

Lemma 2.5 returns every target-membership equivalence to the coefficient field K. A concrete forward lift can also be descended: all its coefficients lie in a finite extension E/K. Choose a K-linear map $\lambda : E \to K$ with $\lambda(1) = 1$ and apply it coefficientwise:

$$\widehat{\mathsf{L}}(f) = \lambda_*(\mathsf{L}_E(f)). \tag{31}$$

Since $V_j(N; E) = E \otimes_K V_j(N; K)$, this preserves membership in each closure stage, fixes 1, preserves degree and satisfies $\rho\widehat{\mathsf{L}} = \mathrm{id}$. Recover a proof from the K-closure bases; coefficientwise projection of individual proof steps need not preserve their scalar inferences. Reflection of prescribed target membership follows from Lemma 2.5, not from a claim that $\widehat{\mathsf{L}}$ reflects every individual consequence.

5 Nilpotent blocks and a common projector

The transfer proof reduces the whole theorem to Exact Deletion (Theorem 4.2). We now prove that finite-dimensional input, using a support budget r as an abstract parameter. For general indexed targets its application is $r = 2kd + s + t$; for invariant targets it is $r = kd + s$. The construction is independent of the equations and targets. We begin with the simpler square-zero operator in characteristic two, then build a witness with a quadratic retained-label cost, replace that cost by the linear threshold, and finally handle odd characteristic.

There are two obligations. On permutation bases, deletion must retain precisely the objects using the n retained labels and transport the actual coefficient maps. On their subspaces and quotients it must also preserve kernels and images, so that it transports PC intersections after cancellation. The orbit calculation supplies the first obligation; one common group-algebra witness will supply the second.

5.1 A block of labels and a square-zero operator

For this part $\operatorname{char} K = 2$. Choose $q = 2^a > r$, put $N = n + q$, and split the labels into retained labels $R = [n]$ and a block B with q labels. Identify B with the vector space $A = \mathbb{F}_2^a$. Translation by the ith unit vector gives a permutation σ_i of B, extended by the identity on R. These permutations commute and square to the identity.

Choose $c_1, \dots, c_a$ linearly independent over $\mathbb{F}_2$ in a suitable extension field, and put

$$z = \sum_{i=1}^{a} c_i(\sigma_i - 1). \tag{32}$$

Here z is a linear operator, not a scalar, and its products mean composition. Thus $z^2 = 0$ can hold although $z \neq 0$. Since the summands commute, cross terms cancel in characteristic two, and

$$z^2 = \sum_i c_i^2(\sigma_i - 1)^2 = 0.$$

For $q = 2$, z can be the swap of the two block labels plus the identity. For $q = 4$, work over $\mathbb{F}_4 = \mathbb{F}_2[\omega]/(\omega^2 + \omega + 1)$ and use $(c_1, c_2) = (1, \omega)$. During this finite-algebra construction, K denotes the chosen working extension field in all tuple spaces and group algebras. The extension field is used to construct the block operator and its witness; Lemma 2.5 preserves derivability and minimum degree over the original coefficient field.

The independence is essential. On a two-point orbit where σ_1 and σ_2 induce the same swap, choosing $c_1 = c_2 = 1$ makes $z = 0$. The orbit is moved, yet would survive deletion. Independent coefficients ensure that every nonzero binary combination of the c_i remains nonzero.

For a space M with label action define

$$\mathcal{D}(M) = \ker z / \operatorname{im} z. \tag{33}$$

This quotient is legal because $z^2 = 0$ implies $\operatorname{im} z \subseteq \ker z$. Two vectors killed by z represent the same class if their difference is zw for some w. A map commuting with label permutations commutes with z and induces a map between these quotients. Permutations of the retained labels commute with z, so they still act on the quotient.

5.1.1 The two-point orbit calculation

For the illustrative choice $z = \sigma - I$, on basis vectors e_β, e_γ swapped by σ,

$$ze_\beta = ze_\gamma = e_\beta + e_\gamma.$$

The kernel and image are both the one-dimensional span of $e_\beta + e_\gamma$, so the quotient is zero. On a fixed basis vector e_i, $z = 0$, and its line survives. This is the prototype of deletion: moved directions disappear, fixed directions remain.

5.1.2 The calculation for every moved orbit

An orbit comprises the objects reached from one object by the block translations. Its stabiliser consists of translations fixing that object. Since A is abelian, an orbit with stabiliser H can be coordinatised by the cosets A/H; the coordinate function below uses that description.

Let X be a finite A-set with no point fixed by all of A. On each orbit choose x_0. Its stabiliser H is a proper binary subspace. Choose a nonzero linear functional $\ell : A \to \mathbb{F}_2$ vanishing on H. The function

$$\pi(vx_0) = \ell(v)$$

is well defined: two choices of v differ by an element of H. Let D multiply e_x by $\pi(x)$, and put $e_i = \ell(\varepsilon_i)$. These are constants on that orbit. If S_i denotes the action of σ_i, direct evaluation at a basis vector gives

$$DS_i - S_iD = e_iS_i, \qquad Dz - zD = W := \sum_i c_ie_iS_i.$$

The difference $DS_i - S_iD$ is a *commutator*, measuring the failure of the two operators to commute. Here D is diagonal in the orbit basis, and $e_i \in \mathbb{F}_2$ is a scalar, distinct from a basis vector e_x.

Moreover $Wz = zW$ and

$$W^2 = \Big(\sum_i c_ie_i\Big)^2 I = \alpha^2 I, \qquad \alpha \neq 0.$$

Thus $\Theta = W^{-1}D$ satisfies $\Theta z - z\Theta = I$. For $zm = 0$ this gives $m = -z\Theta m$, hence $\ker z \subseteq \operatorname{im} z$. The reverse inclusion was automatic, so the quotient vanishes.

This calculation also works on $K[X] \otimes Y$ for *any* space Y carrying the label action. Let D inspect only the X coordinate and let S_i act on both factors. We still have $S_i^2 = I$ and the same commutator calculation. This stronger statement will justify multiplication, whose domain contains a variable factor tensored with a consequence subspace.

5.2 Natural deletion on permutation bases

Let $\mathcal{U}$ be a basis system whose objects have support less than q. An object fixed by all of A cannot have support meeting B. Otherwise invariance of its support under the regular action would force its support to contain all q block labels. Thus its fixed objects are precisely $\mathcal{U}_n$.

Split the basis into fixed and moved objects. The preceding calculation kills the moved summand and fixes the retained one. Hence

$$\mathcal{D}(K[\mathcal{U}_{n+q}]) \cong K[\mathcal{U}_n], \tag{34}$$

$$\mathcal{D}(K[\mathcal{U}_{n+q}] \otimes Y) \cong K[\mathcal{U}_n] \otimes \mathcal{D}(Y). \tag{35}$$

The second identification sends $e_u \otimes [y]$ to $[e_u \otimes y]$ for retained u. It is natural in Y: applying a map to y before or after taking its class gives the same result. For a product of two permutation bases, it simply keeps pairs of retained objects. There is no choice of unrelated isomorphisms here.

Suppose a matrix from one permutation basis to another has stable relabelling-invariant coefficients. On a retained input vector, its image is killed by z. Its retained component has exactly the smaller-rank coefficients; its moved component is in $\ker z = \operatorname{im} z$ on the moved summand, so vanishes in the quotient. Thus the *actual matrix*, not merely its rank, is transported correctly.

5.2.1 A small failure of exactness

Take the two-point permutation space $E = Ke_\beta \oplus Ke_\gamma$ and its invariant line $T = K(e_\beta + e_\gamma)$. Inclusion $T \hookrightarrow E$ is injective. But $\mathcal{D}(T) = T$ whereas $\mathcal{D}(E) = 0$. Deletion takes an injection to the zero map on a nonzero space. Therefore the basis calculation alone does not prove that deletion preserves kernels, images or intersections.

This failure is precisely why a proof cannot stop at "the monomials delete correctly." We must show that, with enough retained labels, every subspace produced by the PC calculation lies in a class where the quotient operation is exact.

5.3 A common group-algebra witness

The group algebra $K[S_N]$ consists of finite formal sums $\sum_g a_g g$ of label permutations. It acts on every space with an S_N action by the corresponding linear combination of operators. Its multiplication is composition. This provides a common language for certificates valid on many spaces at once.

For $b \geq 0$ let

$$P_b(N) = K[\mathrm{Inj}([b], [N])],$$

the space on ordered injective b-tuples. This is a tuple space, distinct from the polynomial ambient $P_{N,\leq d}$; b counts positions rather than polynomial degree. Label permutations act on entries; a second group S_b permutes tuple positions. These actions commute. Define

$$J_r(N) = \{a \in K[S_N] : aP_b(N) = 0 \text{ for every } b \leq r\}.$$

It is a two-sided ideal. Write $\mathcal{C}_r(N)$ for the finite-dimensional spaces with label action annihilated by this ideal. Subspaces invariant under labels, quotients and finite direct sums remain in this class. The ideal $J_r(N)$ is unrelated to the prescribed-target map J_N.

A two-sided ideal is closed under multiplication on either side by any group-algebra element. Here this follows because an operator annihilating every tuple space still does so after either composition. Its relations therefore hold on the subspaces and quotients in $\mathcal{C}_r(N)$.

Every bounded-support permutation basis of support at most r belongs to the class. For an orbit of an object, order its support of size $b \leq r$. Sending an injective tuple to the corresponding relabelling of that object gives a surjective equivariant map from $P_b(N)$ onto the orbit space. The support condition makes this well defined. Elements of J_r kill that quotient as well.

Definition 5.1. A support-r witness is one element $w \in K[S_N]$ satisfying

$$zwz - z \in J_r(N). \tag{36}$$

Such a w gives $zwz = z$ on *every* object of $\mathcal{C}_r(N)$. It is a inner inverse: for $v \in \operatorname{im} z$, applying w produces a preimage under z. It does not invert the nilpotent operator on the whole space. Choosing a separate matrix inverse on each space would not give this compatibility. A module map f, meaning a linear map commuting with label permutations, satisfies $f(wv) = wf(v)$ for this common group-algebra element.

5.3.1 The projector calculation, without an exactness assumption

Recall that a projector is an idempotent linear operator. Its image can model a quotient without being invariant under the original label action. What matters for exactness here is that the same formula for e commutes with every equivariant map between the spaces concerned.

Suppose $z^2 = 0$ and $zwz = z$. Set

$$e = (I - zw)(I - wz).$$

Then

$$ze = z(I - wz) = z - zwz = 0, \qquad ez = (I - zw)z = 0.$$

If $zv = 0$, then $ev = v - zwv$. Also

$$(zw)e = (zw - zwzw)(I - wz) = 0,$$

since $zwzw = zw$. Applying the formula on $\ker z$ to ev gives $e^2v = ev$. Thus e is a projector.

The map $[v] \mapsto ev$ is an isomorphism $\ker z / \operatorname{im} z \to eM$: it is well defined because $ez = 0$, its image lies in $\ker z$, and $ev - v = -zwv$ is in $\operatorname{im} z$ when $zv = 0$. Its inverse sends a vector of eM to its quotient class.

A short exact sequence $0 \to U \to V \to W \to 0$ consists of an injection, followed by a surjection whose kernel is the image of that injection. Preservation means that these three properties still hold after applying deletion. The common projector verifies each property directly.

Now let $0 \to U \to V \to W \to 0$ be an exact sequence in $\mathcal{C}_r(N)$. All maps commute with the same e. Restricting to eU, eV, eW preserves injectivity. A vector of eW has a lift $v \in V$; ev is a lift in eV, proving surjectivity. A vector ev mapping to zero lies in U and equals its own projection, so lies in eU. This proves exactness explicitly. By the natural isomorphism, $\mathcal{D}$ is exact.

The projector need not commute with permutations of the retained labels. We use its image as a vector-space model to prove exactness, and keep the original quotient for the residual label action. Nothing here asserts an unwarranted equivariant direct-sum decomposition for that residual group.

Example 5.2 (One retained label)**.** Let the retained label be a and the two block labels be b, c. Put $z = I + (bc)$ and $w = (ab)$. On $P_1(3)$,

$$zwz(e_b) = zw(e_b + e_c) = z(e_a + e_c) = e_b + e_c = ze_b,$$

and similarly for e_c; both sides vanish on e_a. The projector is the all-ones 3×3 matrix. Its image is the constant line, naturally representing the retained one-dimensional quotient. With no retained label, every element of $K[S_2]$ satisfies $zwz = 0$, so there is no such witness. The spare label makes a real difference.

6 A preliminary construction with fresh labels

The first construction exhibits all the algebraic steps with a generous label budget. In characteristic two, it repairs the common identity $zwz = z$ at tuple length b by allocating b new retained labels. The cumulative cost is quadratic in r; the next section improves this cost by using the residual's special shape on one pool.

6.1 Fresh-label construction

We now prove that a support-r witness exists in characteristic two when

$$n \geq T(r) := 1 + 2 + \cdots + r = \frac{r(r+1)}{2}, \qquad r < q.$$

This proof is independent of the rook theorem used later. Its ingredients are finite algebra, a position-compatible auxiliary inverse, and an explicit way to simulate that inverse using fresh labels.

Choose the generic coefficients in a finite field F of characteristic two, large enough to contain a independent elements. We may work inside the algebraic closure of the original field; the degree question descends by Lemma 2.5. The finiteness of F will be used only to normalise an element of the finite unital group algebra on the current label set. If that label set is Ω, the algebra $F[S_\Omega]$ has $|F|^{|S_\Omega|}$ elements, one coefficient in F for each permutation. This finite construction field imposes no finiteness restriction on the original coefficient field.

6.1.1 Normalisation of the residual

Suppose w_0 works on shorter tuples but not yet on length b. Put $y = z - zw_0z$. An auxiliary linear map h may satisfy $zhz = z$ on the larger tuple space. It is tempting to conclude $yhy = y$. In general the expansion contains an extra term zw_0zw_0z, and this need not collapse. Before correcting the error we arrange the additional identity $w_0zw_0 = w_0$ in the actual group algebra. This normalisation was a genuine mathematical step in the earlier construction.

Here normalisation replaces a ring element to impose an additional algebraic identity. A ring representation interprets ring elements as operators, respecting addition, multiplication and the identity. Thus a relation proved in the group algebra holds simultaneously in all its representations. This is why the next lemma preserves every previously successful tuple length at once.

Lemma 6.1 (Normalisation in a finite ring)**.** *Let a, z belong to a finite ring. There is an element b in the same ring with $bzb = b$, such that every ring representation in which $zaz = z$ also satisfies $zbz = z$.*

Proof. Put $x = az$. The sequence of powers of x eventually repeats, so $x^{i+t} = x^i$ for some $i \geq 0, t > 0$. Multiplication by further powers gives $x^{m+t} = x^m$ for $m \geq i$. Choose $k \geq i$ with $k + 1$ divisible by t. Then $2k + 1$ differs from k by a multiple of t, and $x^{2k+1} = x^k$.

Set $b = x^k a$. Direct multiplication gives

$$bzb = x^k(az)x^k a = x^{2k+1}a = x^k a = b.$$

In a representation where $zaz = z$, induction gives $z(az)^m = z$ for every $m \geq 0$. Hence $zbz = z(az)^{k+1} = z$. One exponent chosen in the original finite ring works simultaneously in all such representations. □

The notation b in this lemma is an element, not a tuple length; in the construction below we denote this normalised element by b_0.

Lemma 6.2 (Normalised residual)**.** *If $b_0zb_0 = b_0$ and $zhz = z$, then, for $y = z - zb_0z$,*

$$yhy = y.$$

Proof. Expand without commuting any factors:

$$\begin{aligned} yhy &= zhz - zhzb_0z - zb_0zhz + zb_0zhzb_0z \\ &= z - 2zb_0z + zb_0zb_0z \\ &= z - zb_0z = y. \end{aligned}$$

The last equality uses the normalisation. This identity works in any ring; characteristic two is not needed for this particular calculation. □

6.1.2 An auxiliary inverse compatible with tuple positions

Position permutations reorder the entries of a tuple; label permutations rename those entries. These actions commute. In this step we require compatibility with positions, since the subsequent simulation must work on all orderings of each underlying label set.

Let Ω consist of the block B and the retained labels already allocated. On $Q = P_b(\Omega)$, partition the tuples according to the A-orbit of their *underlying label set*. A set meeting B has proper stabiliser because it has size $b < q$. Choose the functional ℓ and coordinate π as in Section 5.1, now on these sets. Let D read π from the underlying set, and use the same W and nonzero α as before, defined separately on each orbit. Then on each moved orbit,

$$Dz - zD = W, \quad Wz = zW, \quad W^2 = \alpha^2 I, \quad z^2 = 0.$$

Multiplying the commutator by z shows $zDz = zW$. Thus

$$h = \alpha^{-2} WD \quad \text{satisfies} \quad zhz = z.$$

On tuples wholly outside B, put $h = 0$, since $z = 0$ there. Because D depends on the underlying set and translations act on labels, h commutes with every permutation of tuple positions. We have not claimed that h is a group-algebra element. It is just an explicitly constructed linear operator.

6.1.3 Fresh labels simulate any position-compatible operator

Adjoin b new labels $\kappa_1, \dots, \kappa_b$ to Ω. Let $i : P_b(\Omega) \to P_b(\Omega \cup C)$ include the old tuples and let p discard every tuple using a fresh label. We prove that any operator h commuting with position permutations has the form

$$h = pci \tag{37}$$

for some $c \in F[S_{\Omega \cup C}]$.

The inclusion and projection satisfy $pi = I$. The expression pci is the old-to-old part of the larger operator c: it makes no claim about inputs or outputs containing fresh labels. The residual localisation in the induction below is what makes this part sufficient.

Fix a b-element set $U = \{u_1, \dots, u_b\} \subseteq \Omega$, with one chosen ordering. Form

$$E_U = \prod_{j=1}^{b} (1 - (u_j \ \kappa_j)).$$

The transpositions are disjoint. On an old tuple missing some u_j, the corresponding factor kills the tuple, and the other factors cannot introduce u_j. On a tuple using all the u_j, the identity term leaves the tuple old, while every nonidentity term introduces a fresh label. Therefore $pE_U i$ is precisely the projector onto the orderings of U.

For an ordered output tuple v, choose a permutation $g \in S_\Omega$ taking $(u_1, \dots, u_b)$ to v. Extend g by fixing fresh labels. Then $pgE_U i$ takes that input ordering to v, takes its position permutations to the corresponding permutations of v, and kills all other underlying sets. The orderings of each U form a free orbit for S_b. Consequently an operator commuting with S_b is completely determined by its value on one ordering for each U. Expanding those values as linear combinations of output tuples and using the operators $pgE_U i$ proves (37). There is no division by $b!$.

A free position orbit means that no nonidentity position permutation fixes an injective tuple. Its $b!$ orderings are distinct basis vectors. Consequently position compatibility determines the remaining values from one ordering without averaging over S_b.

Example 6.3 (Two registers)**.** For $U = \{u_1, u_2\}$,

$$E_U = 1 - (u_1\kappa_1) - (u_2\kappa_2) + (u_1\kappa_1)(u_2\kappa_2).$$

On (u_1, u_2), projection p kills the last three terms. On (u_1, v) with $v \neq u_2$, the identity contribution is cancelled by the second swap factor, which fixes that tuple. The detector recognises the set U, while leaving the two possible orderings available for a position-compatible output rule.

6.1.4 The induction, including localisation of the error

For support zero take $w_0 = 0$: z is zero on constants. Suppose a witness exists on the block and $T(b-1)$ retained labels, collectively Ω. Normalise it in the finite ring $F[S_\Omega]$ to b_0. It still works on all tuple lengths below b and satisfies $b_0 z b_0 = b_0$ in the ring itself. Set $y = z - zb_0z$.

Adjoin b fresh labels. Decompose $P_b(\Omega \cup C)$ according to the positions occupied by fresh labels and their actual values. The group S_Ω fixes each fresh label, so every summand with at least one fresh label is a copy of $P_j(\Omega)$ for $j < b$. On it $y = 0$ by the old witness relation. The summand with no fresh labels is $Q = P_b(\Omega)$. Thus, as actual maps,

$$y = iy_Qp.$$

This localisation is what allows a compressed inverse to suffice. Construct the auxiliary h on Q above. Lemma 6.2 gives $y_Qhy_Q = y_Q$. Choose c with $pci = h$. Then

$$ycy = iy_Q(pci)y_Qp = iy_Qp = y.$$

Define the new group-algebra element

$$w_b = b_0 + (1 - b_0z)c(1 - zb_0).$$

Since $z(1 - b_0z) = (1 - zb_0)z = y$,

$$zw_bz = zb_0z + ycy = zb_0z + y = z$$

on length b. On every shorter tuple space the same fresh-label decomposition makes $y = 0$, so the relation remains valid there too. The new cost is exactly b retained labels, giving

$$T(b) = T(b-1) + b.$$

If still more retained labels are present, extend the constructed permutations by fixing them. Decomposing according to those outside labels reduces every tuple of length at most b to a shorter tuple on the constructed labels. Thus the same witness works for all $n \geq T(b)$.

Theorem 6.4 (Quadratic retained-label budget)**.** *For every $r < q = 2^a$ and $n \geq r(r+1)/2$, there is a common witness w with $zwz = z$ on all tuple modules of length at most r. Therefore deletion is exact on the support-r class. It transports the entire degree-d PC consequence space with $r = kd + s$, beyond the fixed template threshold. For prescribed indexed targets use the budget $r = 2kd + s + t$ to include the target map and the doubled-support detector.*

Every construction needed for this theorem has now been supplied. In particular, the proof is not an inference from witnesses found for small tuple lengths.

Remark 6.5 (The square-root degree lower bound). For either prescribed-target budget, the construction gives an onset $O(d^2)$ for degree-d transfer. Inverting that onset gives an $\Omega(\sqrt{n})$ lower bound when no finite degree recurs, and on the hard residues of the periodic case. The loss is precisely the cumulative allocation $T(r) = r(r+1)/2$. The following construction replaces this sum by the maximum $2r - 1$, using one retained pool at every layer.

7 A common witness with a linear label budget

Keep $N = n + q$ and the characteristic-two operator z, with $q > r$. The improvement is to repair only the old witness's residual: it vanishes on incoming shorter-tuple directions and has zero output marginals. We first construct a correction of this shape, then realise it by permutations annihilating all shorter tuples. Every stage uses the same retained pool. Its cost is therefore $\max_{b\le r}(2b-1) = 2r-1$ for $r \ge 1$, rather than a sum of fresh-label allocations.

7.1 The zero-marginal layer

We still work in characteristic two. Fix a tuple length $b \ge 1$, distinct from the proof-degree budget d. Let $V = P_b(N)$. For each position i there is a deletion map

$$\partial_i : P_b(N) \longrightarrow P_{[b]\setminus\{i\}}(N)$$

which forgets the entry in position i. On a vector of coefficients it sums over that entry, so we call it a marginal. The target is a copy of $P_{b-1}(N)$ with the surviving positions named. Give tuple spaces the dot product making their bases orthonormal, and put

$$V' = \sum_i \operatorname{im} \partial_i^{\mathsf{T}}, \qquad \mathcal{Z}_b(N) = \bigcap_i \ker \partial_i = (V')^{\perp}, \qquad T_b(N) = V/V'. \tag{38}$$

Here $\mathcal{Z}_b$ is the zero-marginal space; we also write $\mathrm{Soc} = \mathcal{Z}_b$ and $\mathrm{Top} = T_b$. This notation distinguishes the tuple zero-marginal space from the polynomial consequence space $C_d(n)$. All statements below use the displayed definitions.

The dot product is the bilinear coefficient pairing, not a positive inner product. Thus $U^{\perp}$ denotes the annihilator of U, not an assumed complement. The pairing identifies V/V' with $\mathcal{Z}_b^*$: a quotient class $[v]$ acts on $\zeta \in \mathcal{Z}_b$ by $\langle v, \zeta\rangle$, independently of its representative. We use only this elementary finite-dimensional duality; see [15] for background.

The incoming space V' contains directions lifted from shorter tuples. The zero-marginal space contains directions invisible after deleting any one position. The quotient T_b forgets the incoming directions. Label permutations commute with all marginal maps, and position permutations permute the marginal maps. Thus these spaces carry the two appropriate actions.

Example 7.1 (Lengths one and two)**.** For $b = 1$, ∂ sums coordinates, V' is the constant line and $\mathcal{Z}_1(N)$ is the zero-sum space. For $b = 2$, a coefficient vector is an off-diagonal array $f(i, j)$, $i \ne j$. Its marginals are its row and column sums. Thus $\mathcal{Z}_2(N)$ consists of off-diagonal arrays with every row and column sum zero. On three labels it is spanned by

$$e_{12} + e_{23} + e_{31} - e_{21} - e_{32} - e_{13}.$$

In characteristic two the six coefficients are all one, but every line still sums to zero because it contains two ones.

7.1.1 The old error already has a special shape

Suppose w_0 is a witness on all lengths below b, and set

$$\varepsilon = zw_0z - z \quad \text{on } V.$$

This is the action of an element of $J_{b-1}(N)$. For any $x \in J_{b-1}(N)$,

$$\partial_i x = x\partial_i = 0, \qquad x\partial_i^{\mathsf{T}} = \partial_i^{\mathsf{T}} x = 0.$$

The same element x acts on both tuple spaces: it commutes with the marginal maps and acts as zero at length $b-1$. Therefore x maps V into $\mathcal{Z}_b(N)$ and kills V'. In particular ε factors through

$$V \longrightarrow T_b(N) \longrightarrow \mathcal{Z}_b(N) \longrightarrow V.$$

It also commutes with position permutations. This is much more special than an arbitrary operator on V. The improved route exploits exactly this special form.

Factoring through the quotient means that $\bar{\varepsilon}([v]) = \varepsilon v$ is well defined because ε kills V'. Its output lies in the actual subspace $\mathcal{Z}_b \subseteq V$. The left and right correction maps must be defined on these two spaces.

There are two tasks. First, construct a position-compatible correction of this shape with $z\varphi z = -\varepsilon$. Second, realise that correction by an element of the actual label group algebra which kills all shorter tuples. The first task needs retained labels; the second will be solved by a rook-placement theorem without adding labels.

7.2 Partitioning a zero-marginal vector so that one label is avoided

For a finite position set S and label set L, let $P_S(L) = K[\mathrm{Inj}(S, L)]$ and $C_S(L) = \bigcap_{i \in S} \ker \partial_i$. This is the zero-marginal space also denoted $\mathcal{Z}_S(L)$. If $L' \subseteq L$, regard $C_S(L')$ as the vectors of $C_S(L)$ supported on tuples using only L'. Write $\mathcal{Z}_c$ when there are c positions.

Suppose a block label β occupies a new position and $y \in \mathcal{Z}_c(L)$ occupies the other positions. The vector $e_\beta \otimes y$ has zero marginals in those other positions, but its new-position marginal is y. We want a retained correction with the same marginal. If

$$y = \sum_{x \in L} \pi_x(y), \qquad \pi_x(y) \in \mathcal{Z}_c(L \setminus \{x\}),$$

then

$$e_\beta \otimes y - \sum_{x \in L} e_x \otimes \pi_x(y) \tag{39}$$

has every marginal zero. Avoiding x is essential: it makes each displayed tuple injective. The maps must also commute with permutations of the old positions, so they can be used consistently in all layers.

Example 7.2 (One old position)**.** Take $L = \{1, 2, 3\}$ and $y = ae_1 + be_2 - (a+b)e_3$. Define

$$\pi_1(y) = b(e_2 - e_3), \quad \pi_2(y) = a(e_1 - e_3), \quad \pi_3(y) = 0.$$

Each summand avoids its subscript and has coordinate sum zero; their sum is y. For $y = e_1 - e_2$, formula (39) is the six-term vector

$$e_{\beta 1} - e_{\beta 2} + e_{12} - e_{13} - e_{21} + e_{23}.$$

Check its two marginals by grouping terms with the same surviving label. With only labels $1, 2$, every $\mathcal{Z}_1(L \setminus \{x\})$ is zero, while $e_1 - e_2 \neq 0$. The third label is what permits the construction.

Here a partition of unity means a sum of linear maps equal to the identity. Its summands may overlap and need not be idempotent. Equivariance concerns tuple positions; the construction may choose distinguished labels.

Lemma 7.3 (Partition of unity)**.** *If $|L| \geq 2c+1$, there are position-equivariant linear maps*

$$\pi_x : \mathcal{Z}_c(L) \longrightarrow \mathcal{Z}_c(L \setminus \{x\}), \qquad \sum_{x \in L} \pi_x = I.$$

This statement and its proof work over any field.

The label called d in the following proof is a local distinguished label, not the polynomial-degree budget.

Proof. We induct on c. At $c = 0$ the space is K on the empty tuple; choose one label x, make $\pi_x = I$ and all others zero.

Let $c \geq 1$. Choose $d \in L$ and put $L' = L \setminus \{d\}$. For $y \in \mathcal{Z}_c(L)$ and position i, let $y^{(i)}$ be the coefficients of those tuples in which d appears in position i, with that position then removed. This lies in $\mathcal{Z}_{[c]\setminus\{i\}}(L')$. To see this, fix d in position i and sum another position: it is a marginal of y, hence zero. There cannot be a second occurrence of d in an injective tuple.

By induction, partition $y^{(i)}$ on L' as $\sum_{x\in L'} \pi'_x(y^{(i)})$, each term avoiding x. For a vector $v \in \mathcal{Z}_{[c]\setminus\{i\}}(L' \setminus \{x\})$, partition once more on $L' \setminus \{x\}$:

$$v = \sum_{x'\in L'\setminus\{x\}} \pi''_{x'}(v).$$

Both applications are justified: $|L'| \geq 2c$ and $|L' \setminus \{x\}| \geq 2c - 1 = 2(c-1) + 1$.

Construct $F_{i,x}(v)$ by inserting d into position i of v, then subtracting the sum obtained by inserting x' into position i of $\pi''_{x'}(v)$. Every term avoids x. All tuples are injective because the lower-dimensional vector avoids the inserted label. A marginal in a position other than i vanishes term by term. The marginal in position i is $v - \sum_{x'} \pi''_{x'} v = 0$. Thus

$$F_{i,x}(v) \in \mathcal{Z}_c(L \setminus \{x\}).$$

Its d-part in position i is v, and its d-parts in other positions are zero.

For $x \in L'$ set

$$\pi_x(y) = \sum_{i=1}^{c} F_{i,x}(\pi'_x(y^{(i)})), \qquad \pi_d(y) = y - \sum_{x\in L'} \pi_x(y).$$

The d-part of the last expression in every position is $y^{(i)} - \sum_x \pi'_x(y^{(i)}) = 0$. It therefore uses no label d, and belongs to $\mathcal{Z}_c(L')$. The sum identity is built into the definition.

For equivariance, use the same inductively constructed maps on every position set of size $c - 1$, transported along a bijection. The transport is independent of the chosen bijection because the inductive maps commute with position permutations. Permuting the c positions simply permutes the terms indexed by i in the displayed sum. No averaging over a symmetric group is used. □

The count $2c + 1$ is now visible in an actual induction: reducing one position removes two labels, the label d being cleared and the label x being avoided. This is a sufficient bound proved by the construction, not an appeal to a dimension formula.

7.3 Adding several positions without losing the zero marginals

A section of a surjection f is a linear map σ with $f\sigma = I$: it chooses preimages coherently with linear combinations. It gives a complement to $\ker f$, but compatibility with an operator or group action requires a further argument. Here the section adds positions and the surjection sums them out.

Let R_0 and J be disjoint position sets of sizes c and t. Here t counts new positions, not target-index support. We want to add the positions in J while keeping the marginals in R_0 zero. If $|L| \geq 2c + t$, the map deleting all positions in J has a position-R_0-equivariant section

$$\mathcal{Z}_{R_0}(L) \longrightarrow \{\Phi \in P_{R_0\cup J}(L) : \partial_i \Phi = 0 \ (i \in R_0)\}. \tag{40}$$

Here is the construction. Suppose some positions $J' \subsetneq J$ have already been added. Split the vector according to their injective labelling γ. On that component the remaining zero-marginal vector uses the label set $L \setminus \operatorname{im}\gamma$. Its size is at least

$$2c + t - |J'| \geq 2c + 1.$$

Partition that vector by Lemma 7.3. For a new position j, insert label x in the summand π_x. Summing the new position returns the original vector, and the old zero marginals stay zero. Repeat until all t positions have been added. Every step is linear and commutes with the R_0 position action, proving (40).

7.3.1 Sorting a zero-marginal vector by its block positions

Split the universe as retained labels $[n]$ and block labels B. Every vector $f \in P_b(n+q)$ has components f_D, for subsets $D \subseteq [b]$, consisting of tuples whose block positions are exactly D:

$$f_D \in K[\mathrm{Inj}(D, B)] \otimes P_{[b]\setminus D}(n).$$

Write ϵ_i for summing a block entry and $\partial_i^{\mathrm{ret}}$ for summing a retained entry. The condition of zero marginal in position $i \notin D$ is

$$\partial_i^{\mathrm{ret}} f_D + \epsilon_i f_{D\cup\{i\}} = 0. \tag{41}$$

It simply separates the possible value of the deleted entry into a retained label or a block label.

Let $\mathcal{Z}^{\leq t}$ be the zero-marginal vectors with no components using more than t block positions. The first space is $\mathcal{Z}^{\leq 0} = \mathcal{Z}_b(n)$. The top components of a vector in $\mathcal{Z}^{\leq t}$ have zero retained marginals by (41). We obtain a map onto the prospective layer

$$G_t = \bigoplus_{|D|=t} K[\mathrm{Inj}(D, B)] \otimes \mathcal{Z}_{[b]\setminus D}(n).$$

Its kernel is $\mathcal{Z}^{\leq t-1}$. We next prove surjectivity with a position-compatible section, so this is an actual quotient description rather than a count of dimensions.

The increasing spaces $\mathcal{Z}^{\leq t}$ form a filtration; its tth layer is the quotient by $\mathcal{Z}^{\leq t-1}$. In the displayed tensor coordinates, $e_\beta \otimes y$ places the retained vector y at a specified block pattern β. The direct sum separates the sets of block positions. This is a tuple layer, not a change in the polynomial-degree budget.

7.3.2 An explicit lift of one top component

Take $J = [t]$, an injective block tuple $\beta : J \to B$, and $y \in \mathcal{Z}_{R_0}(n)$ with $R_0 = [b] \setminus J$. If

$$n \geq 2(b-t) + t = 2b - t,$$

the section (40) gives Φ on the b retained positions such that $\partial_J \Phi = y$ and $\partial_i \Phi = 0$ for $i \in R_0$. For $J' \subseteq J$, set

$$\phi_{J'} = (-1)^{|J'|} \partial_{J\setminus J'} \Phi.$$

For $D \subseteq J$ define

$$f_D = e_{\beta|D} \otimes \phi_{J\setminus D},$$

and put the other components zero. At $D = J$ this is $e_\beta \otimes y$, the required top component. For $i \in R_0$, the retained marginal vanishes. For $i \in J \setminus D$,

$$\partial_i \phi_{J\setminus D} = -\phi_{J\setminus(D\cup\{i\})},$$

which cancels the block marginal of the adjacent component in (41). Thus the whole lift belongs to $\mathcal{Z}^{\leq t}$.

To lift all components equivariantly, choose an ordering of each t-element subset of block labels. Use it in the representative positions $[t]$. Its stabiliser among position permutations consists exactly of permutations of the remaining $b-t$ positions. The constructed lift commutes with this stabiliser, by the cone construction. Transport it along position permutations to every other placement of the same ordered block labels. If two permutations give the same placement, their difference is in the stabiliser, so the transported lift is the same. Extend linearly over all block subsets and retained vectors. This supplies the promised S_b-equivariant section without division by any group order.

For $1 \leq t \leq b$, the largest requirement $2b - t$ occurs at $t = 1$. Therefore the single hypothesis

$$n \geq 2b - 1 \tag{42}$$

supplies all these sections.

7.4 Making the splitting compatible with the block operator

The sections just constructed commute with position permutations, but need not commute with z. We now fix that, without adding labels.

A block pattern is an injective partial tuple with a specified nonempty position domain D of size t, and values in B. Its image is a nonempty subset of B of size $t < q$. Choose the orbit coordinate π from Section 5.1 on these *image subsets*. Let E be the space on these patterns, and give a pattern grade 0 or 1 according to π of its image. Let E_0, E_1 be the spans of the two classes. These spaces are stable under position permutations and split by the position domain, since neither operation changes the image subset.

The diagonal grade operator D (distinct from the position domain) and the invertible $W = [D, z]$ satisfy the same identities as before. For either $e = 0, 1$, suppose $v_0 + zv_1 = 0$ with $v_0, v_1 \in E_e$. Apply $D - eI$. The first term vanishes and the second gives Wv_1, so $v_1 = 0$, and then $v_0 = 0$. Thus $E_e \oplus zE_e$ embeds in the pattern space, giving $2 \dim E_e \leq \dim E$. Since $E = E_0 \oplus E_1$, both inequalities must be equalities. In particular

$$E = E_0 \oplus zE_0.$$

Tensor each domain component with its retained zero-marginal factor. Because z fixes retained labels, the layer has the analogous decomposition

$$G_t = Q_t \oplus zQ_t$$

with Q_t position-stable and $z : Q_t \to zQ_t$ injective by the preceding embedding. This injectivity makes the two components used below unique.

7.4.1 Turning a section into a commuting section

Let $f : E \to Q$ commute with both z and position permutations, suppose $Q = Q_0 \oplus zQ_0$ with $z|_{Q_0}$ injective, as just established for the layer, and let σ be a position-equivariant section. Define

$$\tau(q_0 + zq_1) = \sigma(q_0) + z\sigma(q_1).$$

The decomposition is unique. We have $f\tau = I$. Moreover

$$z\tau(q_0 + zq_1) = z\sigma(q_0) = \tau(zq_0),$$

using $z^2 = 0$. Thus τ commutes with z as well as with positions. Apply this to each quotient $\mathcal{Z}^{\leq t} \to G_t$ and split the filtration successively. We have proved

$$\mathcal{Z}_b(n+q) = \mathcal{Z}_b(n) \oplus C \oplus zC, \tag{43}$$

where C is position-stable, $z : C \to zC$ is an isomorphism, and z kills the retained summand. This is a concrete direct-sum decomposition, valid under $b < q$ and $n \geq 2b - 1$.

7.4.2 The four consequences used for correction

The witnesses below reverse the arrow $c \mapsto zc$ on the appropriate chain directions. They are inner inverses, not inverses of the nilpotent operator on all of V. The notation $z^{-1}(U)$ in the duality calculation denotes the inverse image $\{v : zv \in U\}$.

Write $V = P_b(N)$. The decomposition implies:

1. $zV \cap \mathcal{Z}_b(N) \subseteq z\mathcal{Z}_b(N)$.
2. $zV \cap V' \subseteq zV'$.
3. There is a position-equivariant w_C on $\mathcal{Z}_b(N)$ with $zw_Cz = z$ there.
4. There is a position-equivariant w_T on $T_b(N)$ with $zw_Tz = z$ there.

For the third assertion, let w_C send zc back to c and kill C and the retained summand. For the first, the kernel of z in (43) is $\mathcal{Z}_b(n) \oplus zC$. An element of $zV \cap \mathcal{Z}_b(N)$ is therefore $s + zc$. But zV consists of vectors on tuples meeting the block; so does zc. The remaining s is supported entirely on retained tuples, forcing $s = 0$.

For the other two assertions use transpose. In characteristic two each σ_i is an involution, so $z^{\mathsf{T}} = z$. The pairing identifies $T_b(N)$ with $\mathcal{Z}_b(N)^*$, giving w_T by duality. To see the second inclusion explicitly, take orthogonal complements. The first inclusion implies

$$z^{-1}(\mathcal{Z}_b) = \ker z + \mathcal{Z}_b.$$

Indeed, if $zv \in \mathcal{Z}_b$, it equals zc for some $c \in \mathcal{Z}_b$, so $v - c \in \ker z$. But

$$(zV')^{\perp} = z^{-1}(\mathcal{Z}_b), \qquad (zV \cap V')^{\perp} = \ker z + \mathcal{Z}_b.$$

The complements agree, hence so do the spaces. This proves the required inclusion, with no appeal to semisimplicity.

7.5 Correcting the error on one layer

The maps w_C and w_T are auxiliary inverses on $\mathcal{Z}_b$ and T_b, not yet common group-algebra elements. In products involving w_T, the residual is read as its induced map $T_b \to \mathcal{Z}_b$; composition with the quotient $V \to T_b$ and inclusion $\mathcal{Z}_b \hookrightarrow V$ returns the actual map on V.

Return to $\varepsilon = zw_0z - z$. It kills V' and has image in $\mathcal{Z}_b$. It also has image in zV, so the first consequence gives $\operatorname{im}\varepsilon \subseteq z\mathcal{Z}_b$. Hence

$$zw_C\varepsilon = \varepsilon.$$

If $zv \in V'$, the second consequence gives $zv = zv'$ for some $v' \in V'$. Since $\varepsilon = (zw_0 - I)z$,

$$\varepsilon v = (zw_0 - I)zv' = \varepsilon v' = 0.$$

On T_b, $zw_Tz = z$ says that $z(w_Tzv - v)$ belongs to V' when representatives in V are used. Applying the previous implication gives $\varepsilon w_Tz = \varepsilon$. This notation is unambiguous: ε kills V', so its value does not depend on the representative chosen after w_T.

Consequently the map

$$\varphi = -w_C \varepsilon w_T : T_b \longrightarrow \mathcal{Z}_b$$

satisfies $z\varphi z = -\varepsilon$, after composing with quotient and inclusion. It commutes with position permutations and kills V'. If this map is realised by an element $j \in J_{b-1}(N)$, then

$$z(w_0 + j)z - z = \varepsilon + zjz = 0$$

on length b, while j leaves all shorter tuple relations unchanged.

The remaining step is the realisation theorem. Section 8.1 proves it through a finite combinatorial argument. The realisation uses the existing labels, so the correction closes the witness induction on one retained pool for all $b \leq r$ once $n \geq 2r - 1$ and $r < q$.

8 Realising zero-marginal corrections by permutations

8.1 Rook coordinates and realisation

This section realises a position-equivariant correction with image in $\mathcal{Z}_b$ and kernel containing V' by one element of $J_{b-1}(N)$. It is independent of the block and works over any commutative ring. The proof translates the map into rook coordinates, proves that selected coordinates force the others, then solves for permutation coefficients by triangular elimination. Ordering the finite board is auxiliary bookkeeping; it adds neither labels nor a built-in order to the uniform input.

The centraliser of the position action is the algebra of linear operators commuting with every permutation of tuple positions. Label permutations lie in it because the two actions commute. The realisation problem here has further conditions: the required operator must have image in $\mathcal{Z}_b$ and kill V'. The rook calculation will realise operators of this specified shape.

8.1.1 From a map to a board of non-attacking rooks

A matrix entry of a position-equivariant map $\varphi : P_b(N) \to P_b(N)$ is indexed by an output tuple s and input tuple t. Permuting their positions together leaves the entry unchanged. Thus it depends only on

$$\rho = \{(s_i, t_i) : i = 1, \ldots, b\},$$

a set of b cells with distinct rows and distinct columns on an $N \times N$ board. Call this a b-rook. Board rows and columns are *labels*; they are different from the positions of the original tuples.

For instance, output $(3, 4)$ and input $(1, 2)$ give $\{(3, 1), (4, 2)\}$. Reordering both tuples preserves this rook; reordering only one generally changes it. These coordinates retain the input–output pairing while forgetting simultaneous position order.

The matrix therefore defines a function F on b-rooks. For a $(b-1)$-rook σ, choose an unused column y. A column line through σ is the family of rooks $\sigma \cup \{(x, y)\}$ as x ranges over unused rows. Define row lines analogously. The map has image in $\mathcal{Z}_b$ precisely when its column lines sum to zero. It kills V' precisely when its row lines sum to zero. Write Z_b for this zero-line-sum space of rook functions, distinct from the tuple-vector space $\mathcal{Z}_b$.

A label permutation g has graph $\{(g(y), y)\}$. Its matrix entry is one exactly when ρ is contained in that graph. Set

$$\chi_g(\rho) = [\rho \subseteq \operatorname{graph} g], \qquad L_b\Big(\sum_g a_g g\Big) = \sum_g a_g \chi_g$$

on b-rooks. For a fixed unused row or column, exactly one extension of a rook σ lies in the graph of g if σ itself does, and none otherwise. Therefore

$$\text{each line sum of } L_b(x) \text{ through } \sigma = L_{b-1}(x)(\sigma). \tag{44}$$

If $F \in Z_b$ is realised as $L_b(x)$, this identity forces $L_{b-1}(x) = 0$, then every smaller $L_j(x) = 0$. Thus $x \in J_{b-1}(N)$ automatically. It remains to prove that every $F \in Z_b$ is realisable.

8.1.2 A first example: matrices with zero row and column sums

At $b = 1$, Z_1 is the space of square matrices with zero row and column sums. If row and column N are designated as last, such a matrix is determined by its upper-left $(N-1)\times(N-1)$ entries and is

$$\sum_{i,j<N} F_{ij}(E_{ij} - E_{iN} - E_{Nj} + E_{NN}).$$

Each four-entry matrix is a difference of two permutation matrices: choose a permutation taking column j to row i and column N to row N, and subtract the permutation with these two outputs swapped. The other entries cancel. This proves realisation at length one over every ring. The general rook argument is a higher-length replacement for this elementary construction.

8.2 A forcing set of coordinates

Work on any two equally large finite sets of integer labels R, C, of size M, with their inherited orders. Here M is the board size and c the rook count; later $M = N$ and $c = b$. Numerical thresholds such as $y + 1$ denote the cut strictly above the integer label y. A permutation here is a bijection $g : C \to R$. An increasing chain is a subset of its graph increasing in both coordinates. Let $\mathrm{LIS}(g)$ be the largest chain length. Among the column sets of longest chains choose the lexicographically first one, I_g: at the first differing column, the set containing that column is earlier. Define

$$\rho_g = \{(g(y), y) : y \notin I_g\}.$$

If $\mathrm{LIS}(g) = M - c$, this is a c-rook. Let Ψ_c consist of these rooks.

A forcing set means that restriction to it is injective on the zero-line-sum functions. Two such functions agreeing there have a difference vanishing there, so the theorem below establishes uniqueness. Realisation by permutation coefficients is the separate existence step.

Theorem 8.1 (Forcing)**.** *A function in Z_c that vanishes on Ψ_c vanishes on every c-rook.*

Permutation strings use one-line notation: 312 means $g(1) = 3$, $g(2) = 1$, $g(3) = 2$.

Example 8.2 (The 3×3 board)**.** The permutations with longest increasing subsequence of length two are $312, 231, 213, 132$. Their canonical complement rooks are respectively

$$(3,1),\ (1,3),\ (1,2),\ (2,3).$$

If a zero-line-sum matrix is zero on these four cells, row one first forces $(1,1)$ to be zero. Column one then forces $(2,1)$; row two forces $(2,2)$; column two forces $(3,2)$; and row three forces $(3,3)$. This is the small model of the forcing argument.

8.2.1 Strengthened induction statement

To remove the bottom row and the first column, we must know which canonical longest chains survive contraction. Bounds on longest chains in upper parts of the board keep track of this. These are bookkeeping conditions for the induction, not new mathematical assumptions on F.

Let $L = M - c$. A condition is (s, θ, d), where s is "row" or "column", θ is a label threshold, and $d \geq 1$. This local d is a chain deficit, not polynomial degree. A bijection satisfies it if the longest increasing chain using only cells with that coordinate at least θ has length at most $L - d$. Bounds are interpreted as integer inequalities; a negative upper bound is impossible for a nonnegative chain length.

For a collection $\mathcal{E}$ of conditions let $\Psi(\mathcal{E})$ be the canonical complement rooks of permutations with $\mathrm{LIS} = L$ satisfying every condition. Let $Q(\mathcal{E})$ be the set of c-rooks for which, for at least one condition, at most $d - 1$ *unused* labels of the specified side lie below θ.

The two descriptions are paired deliberately. A bound on chains above a threshold will, through a line-sum identity, force a value at a rook with too few unused labels below it.

Lemma 8.3 (Strengthened forcing)**.** *If $F \in Z_c$ vanishes on $\Psi(\mathcal{E}) \cup Q(\mathcal{E})$, then $F = 0$.*

Theorem 8.1 is its case of no conditions.

8.2.2 Three facts about adding a corner cell

Let $r_0 = \min R$ and $y_0 = \min C$. The following statements describe the only possible new increasing chains.

1. Remove row r_0 and column y, and let $\widetilde{g}$ be a bijection on the smaller board with longest chain length L. Add the cell (r_0, y). Every new chain using it continues with a chain in columns strictly above y. If that region had longest chain at most $L - 1$, the new permutation has longest chain L. Its lexicographically first longest chain is still the old one: the old chain must use some column below y, whereas every new chain through the corner uses columns at least y. Therefore $\rho_g = \rho_{\widetilde{g}} \cup \{(r_0, y)\}$. For a column threshold $\theta \leq y$, the new regional maximum is
$$\max\{\mathrm{LIS}(\widetilde{g}|_{\mathrm{col} \geq \theta}),\ 1 + \mathrm{LIS}(\widetilde{g}|_{\mathrm{col} > y})\}.$$
2. Remove row x and column y_0, and add (x, y_0). If the old region of rows strictly above x has longest chain at most $L - 2$, every chain using the new corner has length at most $L - 1$. Thus all longest chains remain old, the canonical one remains unchanged, and $\rho_g = \rho_{\widetilde{g}} \cup \{(x, y_0)\}$. The analogous maximum formula holds for row thresholds at most x. The stronger deficit two eliminates a tie involving the smallest column.
3. Remove both minimal labels and add (r_0, y_0). This corner can be placed before every increasing chain of the smaller board. The longest length increases by one, every new longest chain contains the corner, and $I_g = \{y_0\} \cup I_{\widetilde{g}}$. Thus $\rho_g = \rho_{\widetilde{g}}$.

These facts follow just by splitting chains according to whether they use the added cell. They do not rely on a representation-theoretic result.

8.2.3 Base cases and degenerate conditions

The outer induction decreases the board size M; a decreasing corner sweep supplies the inner induction. Contracting a fixed cell changes (M, c) to $(M - 1, c - 1)$ and keeps $L = M - c$ fixed. Removing a row and column without fixing a cell leaves c fixed and decreases L. The threshold conditions record these distinct changes.

Prove Lemma 8.3 by induction on M. If $c > M$ there are no rooks. If $c = M > 0$, each line through an $(M-1)$-rook contains exactly one M-rook, so the zero-line-sum equation forces its value to zero.

A condition is degenerate if the entire board has at most $d-1$ labels below its threshold. Then every rook belongs to Q, and the conclusion is immediate. Assume no condition is degenerate. Every column threshold is therefore above y_0 and every row threshold above r_0.

If $c = 0$, consider the increasing bijection. It has longest length M. For a column condition, at least d columns lie below θ, so the part above θ has length at most $M-d$; the same holds for rows. Thus its empty complement rook belongs to $\Psi(\mathcal{E})$, and the sole value of F is zero. We may now assume $1 \le c < M$.

8.2.4 First sweep: eliminate every rook meeting the bottom row

For $y \in C$ in decreasing order, contract the cell (r_0, y) and define

$$G(\sigma) = F(\sigma \cup \{(r_0, y)\})$$

on $(c-1)$-rooks of the board with that row and column removed. Its line sums are line sums of F, so $G \in Z_{c-1}$. The smaller board has size $M-1$ and complement length $(M-1)-(c-1) = L$.

Retain the old conditions and add

$$\begin{aligned} &(\text{column}, y+1, 1),\\ &(\text{column}, y+1, d+1) \quad \text{for each old } (\text{column}, \theta, d) \text{ with } \theta \le y. \end{aligned}$$

The first new condition bounds the right region by $L-1$, giving corner fact 1. The second bounds it by $L-(d+1)$ whenever needed. An old column condition with $\theta \le y$ is preserved upon adding the corner because

$$\max\{L-d, 1+L-(d+1)\} = L-d.$$

Old column conditions above y and old row conditions are unaffected by the corner. Thus a canonical rook satisfying the new conditions extends to one in $\Psi(\mathcal{E})$, and G vanishes on its new Ψ set.

We must also verify every case of its new Q set. An old condition passes directly to the enlarged rook: the deleted row and column are occupied by the added cell, so the count of unused labels below an old threshold is unchanged.

For a new condition $(\text{column}, y+1, e)$, let

$$U = \{y' \in C : y' < y,\ y' \text{ unused by } \sigma\}.$$

On the contracted board this is exactly the unused set below $y+1$, so $|U| \le e-1$. If $e = 1$, there are no such columns. If the condition came from an old $(\text{column}, \theta, d)$, then $e = d+1$ and $|U| \le d$. For each $y' \in U$, the enlarged rook $\sigma \cup \{(r_0, y')\}$ has at most $d-1$ unused columns below θ. Indeed, if $y' < \theta$, its occupation removes one member from a subset of U; if $y' \ge \theta$, y' itself lies outside that subset, again leaving at most $|U|-1$. Thus every such rook lies in the old Q set.

Now use the row line through σ in row r_0. Its terms at columns greater than y are zero by the decreasing inner induction. Its terms below y are either absent ($e = 1$) or zero by the old Q condition. The only remaining term is $G(\sigma)$, so it is zero. This proves vanishing on the new Q set in all cases. The outer induction on M now gives $G = 0$.

Completing the decreasing sweep shows that F vanishes on every rook meeting row r_0.

8.2.5 Second sweep: eliminate every rook meeting the first column

Adjoin the condition $(\mathrm{row}, r_0+1, 1)$ to obtain $\mathcal{E}^+$. Its new Q rooks are exactly those using row r_0: there must be no unused row below r_0+1, and that row is the only candidate. They have just been proved zero. Its Ψ set is a subset of the old one. Thus the vanishing assumptions still hold for $\mathcal{E}^+$.

Sweep $x \in R \setminus \{r_0\}$ in decreasing order. Contract (x, y_0) and keep every old condition in $\mathcal{E}^+$. For each row condition $(\mathrm{row}, \theta, d)$ with $\theta \le x$, add $(\mathrm{row}, x+1, d+1)$. In particular the recently added condition produces $(\mathrm{row}, x+1, 2)$, because $x > r_0$. Therefore the region above x has maximum at most $L-2$, exactly the hypothesis of corner fact 2.

The verification for Ψ is now explicit: row thresholds at most x use the same maximum calculation as in the first sweep; row thresholds above x are unchanged; every column threshold exceeds y_0, so excludes the new corner. For Q, old conditions pass unchanged. A new condition from deficit d says that there are at most d unused rows below x. When any such row x' is occupied instead of x, the number unused below its source threshold is at most $d-1$, by the same two cases $x' < \theta$ and $x' \ge \theta$. These terms lie in $Q(\mathcal{E}^+)$. The column-line equation at y_0 then forces the remaining term to zero, since larger rows were handled earlier. The induction on M proves the contracted function zero.

The case $x = r_0$ was already covered by the first sweep. Thus F now vanishes on every rook meeting either row r_0 or column y_0.

8.2.6 The remaining smaller board

Restrict F to $(R \setminus \{r_0\}) \times (C \setminus \{y_0\})$ without changing c. A line in this board is an old line minus one term using a removed label; that term is zero, so the restriction has zero line sums.

For every condition in $\mathcal{E}^+$ with deficit $d \ge 2$, keep its side and threshold and replace its deficit by $d-1$. Discard the conditions with deficit one. The smaller complement length is $L-1$.

A permutation on the smaller board satisfying these conditions and having longest length $L-1$ extends by the minimal corner. Corner fact 3 gives longest length L and the same complement rook. Every old threshold excludes that corner. For $d \ge 2$ the regional bound is

$$(L-1) - (d-1) = L - d;$$

for $d = 1$ it follows from the whole smaller board having longest length $L-1$. So the restricted function vanishes on the new Ψ set.

If a smaller rook belongs to the new Q set, its unused count below a threshold is at most $d-2$. In the original board the removed minimum label is also unused by this rook and lies below that threshold, increasing the count by exactly one. The resulting count is at most $d-1$, so the rook lies in the old $Q(\mathcal{E}^+)$. Thus the new Q vanishes too. The induction on M finishes the proof of Lemma 8.3, and hence of the forcing theorem.

8.3 Triangular realisation and the common correction

This is back-substitution in a triangular system with diagonal one: each new coefficient is obtained by subtraction, with no division. The integral tensor-space centraliser context is discussed in [11]; the particular realisation required here follows from the forcing argument and the recursion below over any commutative ring.

The second ingredient is simpler. If ρ_g lies on the graph of $g' \ne g$, then g' agrees with g outside I_g and permutes the remaining values $g(I_g)$. On I_g the values of g are increasing. At the first column where g' differs, it must use a larger remaining value. Thus g' is lexicographically larger than g.

Fix $F \in Z_b(N)$ and let $\mathcal{T} = \{g : \mathrm{LIS}(g) \geq N - b\}$. Its canonical complement rooks have sizes at most b. Define coefficients a_g in decreasing lexicographic order by

$$a_g = \begin{cases} F(\rho_g) & |\rho_g| = b, \\ 0 & |\rho_g| < b \end{cases} - \sum_{\substack{g' \in \mathcal{T},\ g' \neq g \\ \rho_g \subseteq \mathrm{graph}\, g'}} a_{g'}. \tag{45}$$

Every coefficient on the right was already defined by the preceding observation. Put $x = \sum_{g \in \mathcal{T}} a_g g$, and let $H_k = L_k(x)$ for each $k \leq b$. The construction makes H_k zero on Ψ_k for $k < b$ and makes $H_b = F$ on Ψ_b.

At level zero the only coordinate is the empty rook, and it is prescribed zero when $b > 0$. Inductively suppose $H_{k-1} = 0$. Equation (44) shows that H_k has zero line sums. For $k < b$, it vanishes on Ψ_k, so forcing gives $H_k = 0$. At $k = b$, $H_b - F$ has zero line sums and vanishes on Ψ_b, so forcing gives $H_b = F$. The case $b = 0$ is immediate by a scalar multiple of any permutation.

We have proved that every zero-line-sum rook function is realised by a group-algebra element, and the realising element annihilates all smaller tuple modules. Translating back to matrices proves:

Theorem 8.4 (Realisation of the layer correction)**.** *For $b \geq 1$, every position-equivariant map $\varphi : P_b(N) \to P_b(N)$ with image in $\mathcal{Z}_b(N)$ and killing V' is the action of an element $j \in J_{b-1}(N)$. No additional labels are required.*

If $b > N$, the tuple space is zero and its only endomorphism is realised by $j = 0$. The witness construction uses $b \leq r < q \leq N$.

Example 8.5 (Actual permutation coefficients)**.** On a 3×3 board take

$$F = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}.$$

The recursion gives coefficients $-1, -1, 2, 1, -1$ for $312, 231, 213, 132, 123$, respectively. Therefore

$$F = -P_{312} - P_{231} + 2P_{213} + P_{132} - P_{123}.$$

Here P_g has its one in row $g(y)$ of column y. The coefficients sum to zero, so the action on the empty tuple is zero, as required for J_0. Reducing this integer identity modulo two or any odd prime gives the corresponding realisation in that characteristic.

9 Exact deletion in every positive characteristic

The characteristic-two construction used $z^2 = 0$ and $z^{\mathsf{T}} = z$. We retain its signed combinatorial sections and permutation realisation, but replace these operator properties by p-level chains, a top-power projector and conjugacy with the transpose. The goal remains exact deletion on $\mathcal{C}_r(n + q)$ using the same retained pool of size at least $2r - 1$ for $r \geq 1$.

9.1 The block operator in prime characteristic

Let now $\mathrm{char}\, K = p$ for any prime. Take a regular block $A = \mathbb{F}_p^a$ of size $q = p^a > r$, choose $c_1, \ldots, c_a$ independent over $\mathbb{F}_p$ in a suitable extension field, and put

$$z = \sum_i c_i(\sigma_i - 1).$$

The generators commute and have order p, so the binomial formula in characteristic p gives

$$z^p = \sum_i c_i^p(\sigma_i^p - 1) = 0.$$

As in Section 5.1, K denotes the working extension field during the construction; degree statements descend to the original field by Lemma 2.5. For $p > 2$ we cannot replace this by $z^2 = 0$. For example, for a three-cycle σ in characteristic three,

$$z = \sigma - I, \qquad z^2 = \sigma^2 + \sigma + I \neq 0$$

on its three-point permutation space.

9.1.1 The deletion quotient

Nilpotence means that a positive power of the operator is zero. Here $z^p = 0$ permits chain lengths from one to p; it does not exclude intermediate lengths. The quotient must therefore be read before any fixed/full-chain decomposition is established.

Since $\operatorname{im} z$ need not be contained in $\ker z$, the characteristic-two expression $\ker z/\operatorname{im} z$ is no longer defined. Use

$$\mathcal{D}(M) = \ker z/(\ker z \cap \operatorname{im} z). \tag{46}$$

A nilpotent chain has basis $v_0, \dots, v_{j-1}$ with $zv_0 = 0$ and $zv_i = v_{i-1}$ for $i \geq 1$. Its kernel is Kv_0. If $j = 1$, its image is zero and the chain survives in (46). If $j \geq 2$, v_0 is in the image and the chain disappears. Thus deletion keeps the one-dimensional chains and discards every longer chain.

One might instead try $\ker z/\operatorname{im} z^{p-1}$. It is a defined quotient, but it is the wrong one: a chain of length two in characteristic three would survive because $z^2 = 0$ there, even though it should be discarded by (46). Keeping the quotient precise is important.

9.2 The commutator argument in characteristic p

On a moved orbit A/H, choose a nonzero linear functional $\ell : A \to \mathbb{F}_p$ vanishing on the proper stabiliser H, and use the coordinate $\pi(vx_0) = \ell(v)$ from Section 5.1. Let D multiply a basis vector by its coordinate π, put $e_i = \ell(\varepsilon_i)$ and $\alpha = \sum_i c_i e_i \neq 0$. With S_i the coordinate-translation operators, defined orbit by orbit, we obtain

$$Dz - zD = W = \sum_i c_i e_i S_i, \quad Wz = zW, \quad W^p = \alpha^p I.$$

The commutator $[D, z] = Dz - zD$ measures the difference between the two orders of application. The displayed power identity gives the explicit inverse $W^{-1} = \alpha^{-p}W^{p-1}$; it is W, rather than the nilpotent z, that is inverted. Consequently the auxiliary operator $B = W^{-1}D$ satisfies

$$[B, z] = Bz - zB = I. \tag{47}$$

This proof works also on a moved permutation basis tensored with an arbitrary representation: D reads only the basis coordinate, while the S_i act diagonally and still have order p.

If $zm = 0$, commuting z past powers of B yields

$$zB^j m = -jB^{j-1}m, \qquad z^j B^j m = (-1)^j j!\, m.$$

The first identity follows by induction from $zB = Bz - I$; applying it repeatedly gives the second. At $j = p - 1$ the scalar $(p-1)!$ is nonzero in characteristic p, because every factor $1, \dots, p-1$ is nonzero. Therefore

$$\ker z = \operatorname{im} z^{p-1} \tag{48}$$

on the moved summand. The reverse inclusion uses $z^p = 0$.

This is the full replacement for the square-zero freeness calculation. It does not divide by p, by $|A|$, or by a symmetric-group order. It divides only by a nonzero orbit coefficient and by $(p-1)!$.

Here freeness concerns the action of z: a full chain is one copy of $K[T]/(T^p)$ with T acting as z, and full chains form a free module over that ring. It is not freeness for the full symmetric group.

On a bounded-support permutation basis the retained summand has $z = 0$ and the moved summand satisfies (48). Therefore the identifications with the retained basis and the tensor identifications from Section 5 still hold, with quotient (46). Stable matrices are transported by exactly the same retained-component argument.

9.3 The top-power witness

Set $u = z^{p-1}$. The common certificate becomes

$$uwu - u \in J_r(N), \qquad \text{or } uwu = u \text{ on every } P_b(N),\ b \le r. \tag{49}$$

This says that w takes the bottom of a full length-p chain back to a preimage at its top. It is enough to build a projector for the actual quotient, but more steps are needed than in characteristic two.

First note the property of the tuple modules supplied by the moved/fixed decomposition:

$$\ker z \cap \operatorname{im} z \subseteq \operatorname{im} z^{p-1}. \tag{50}$$

The image has no retained component, and a moved kernel vector is in the top-power image by (48). We will use (50) on the original tuple modules, then transfer polynomial identities through their common annihilator. We do not assume it for every arbitrary nilpotent operator.

Condition (50) is the purity condition needed here: a kernel vector already in $\operatorname{im} z$ must have a preimage under the top power. A nontrivial chain shorter than p violates it at its bottom. Fixed lines and full length-p chains satisfy it.

Example 9.1 (Why the distinction matters)**.** In characteristic three, a lone length-two chain has $u = z^2 = 0$. The relation $uwu = u$ is then vacuous for any w, but $\ker z \cap \operatorname{im} z$ is nonzero and $\operatorname{im} u = 0$. Such a chain does not satisfy (50). The argument must establish its projector identities on the actual tuple modules first; the top-power relation alone on an arbitrary vector space is insufficient.

9.3.1 A projector which removes one chain level at a time

Write $m = p - 1$, an exponent rather than a support parameter. Suppose $z^{m+1} = 0$, $z^m w z^m = z^m$ and (50) holds. Define

$$S_0 = I, \qquad S_{k+1} = (I - z^k w z^{m-k}) S_k \quad (0 \le k < m), \qquad e = (I - z^m w) S_m. \tag{51}$$

Operator products act from right to left. We verify every required property.

Eliminating high levels. Here the local integer d is a chain height, not a proof-degree budget. If $z^{d+1}R = 0$, then for $d \ge 1$ the vector $y = z^d R v$ is in $\ker z \cap \operatorname{im} z$, hence in $\operatorname{im} z^m$. The witness gives $z^m w y = y$. Thus

$$z^d (I - z^{m-d} w z^d) R v = y - z^m w y = 0.$$

Starting with $z^{m+1} S_0 = 0$, the successive factors of (51) give

$$z^{m+1-k} S_k = 0,$$

and finally $z S_m = 0$. It follows that $ze = 0$.

Action on a kernel vector. If $zv = 0$, every factor in S_m fixes v, because its rightmost power of z has exponent at least one. Therefore

$$ev = v - z^m wv \quad (zv = 0). \tag{52}$$

Killing the image. The first factor $I - wz^m$ fixes $\operatorname{im} z$, since $z^{m+1} = 0$. All later factors preserve $\operatorname{im} z$ because their correction terms have a left factor z^k with $k \geq 1$. Hence $S_m zv \in \ker z \cap \operatorname{im} z \subseteq \operatorname{im} z^m$. The final factor $I - z^m w$ kills this vector by the witness relation. Thus $ez = 0$.

Idempotence. The witness gives

$$z^m we = (z^m w - z^m wz^m w)S_m = 0.$$

Since $ev \in \ker z$, formula (52) now gives $e^2 v = ev$. We have proved $ze = ez = 0$ and $e^2 = e$.

9.3.2 Passing from tuples to every relevant space

Objects of $\mathcal{C}_r(N)$ are modules over $K[S_N]/J_r(N)$. An identity holding on every tuple test space of length at most r is an identity of this quotient algebra. Equivariant maps intertwine its operators; this does not make the projector central. Its image is used as a vector-space model of deletion, while the quotient retains the residual label action.

Construct e as the same expression in the group algebra. Once these identities hold on all tuple modules up to length r, the elements ze, ez and $e^2 - e$ belong to $J_r(N)$. They therefore act as zero on every object of $\mathcal{C}_r(N)$, whether or not we have a separate chain decomposition for that object. Formula (52) follows there directly from the displayed product formula.

Thus $[v] \mapsto ev$ identifies (46) with eM: e kills $\operatorname{im} z$, its image is in $\ker z$, and $v - ev = z^{p-1} wv \in \operatorname{im} z$ for $v \in \ker z$. The exactness proof using a common projector is now identical to Section 5. All maps commute with the common group-algebra expression.

Example 9.2 (A full chain in characteristic three)**.** Let $zv_2 = v_1$, $zv_1 = v_0$, $zv_0 = 0$, and let $wv_0 = v_2$, with w zero on the other chain vectors. Then $z^2 wz^2 = z^2$. The projector is

$$e = (I - z^2 w)(I - zwz)(I - wz^2).$$

The rightmost factor removes v_2, the middle factor removes v_1, and the leftmost removes v_0. If a separate fixed vector t is added and $wt = 0$, every factor fixes t. Thus the projector removes exactly the full chain and retains the fixed direction. For $p = 2$ the same formula has only the two factors $(I - zw)(I - wz)$ from Section 5.

9.4 The linear construction in odd characteristic

We now adapt the construction of Section 7. The partition proof in Section 7.2, cone and signed lifts in Section 7.3, and rook realisation in Section 8 use 1 and -1 without a characteristic-two cancellation. These combinatorial proofs remain valid over any field and supply the same label count $2b - 1$. The splitting and duality arguments require the replacements below.

At the current tuple length b, recall $V = P_b(N)$, $V' = \sum_i \operatorname{im} \partial_i^{\mathsf{T}}$, $\mathcal{Z}_b = \bigcap_i \ker \partial_i$ and $T_b = V/V'$. The layer G_t records exactly t block positions in the filtration $\mathcal{Z}^{\leq t}$. The tuple bases and marginal maps have the same definitions; their block action has a different chain structure.

9.4.1 The block layers have p levels instead of two

Let E be the space on the current block patterns with t positions, where $1 \le t \le b < q$, and give each pattern a grade $d \in \mathbb{F}_p$ using the orbit coordinate of its image set. Here d is a field-valued grade, not polynomial degree, and D is the diagonal operator reading that grade. Let E_d be the corresponding subspace. We claim

$$E = E_0 \oplus zE_0 \oplus \cdots \oplus z^{p-1}E_0.$$

For fixed d, suppose $\sum_{j=0}^{p-1} z^j v_j = 0$ with $v_j \in E_d$. The commutator identity and $Wz = zW$ imply

$$(D - dI)z^j v_j = jWz^{j-1}v_j.$$

Apply $D - dI$ to the relation and cancel the invertible W. Repeat this operation $p-1$ times. It gives $(p-1)!v_{p-1} = 0$, hence $v_{p-1} = 0$. Descending induction gives every $v_j = 0$. Thus $p \dim E_d \le \dim E$ for every $d \in \mathbb{F}_p$. Since the p spaces E_d partition the basis, their dimensions sum to $\dim E$, forcing equality in each inequality. The direct sum for E_0 is therefore all of E.

The grade depends only on the image set, so E_0 is position-stable and graded by the position domain. Tensoring with the retained zero-marginal factors gives

$$G_t = \bigoplus_{j=0}^{p-1} z^j Q_t$$

with Q_t position-stable and z^{p-1} injective on Q_t by the preceding directness argument. A position-equivariant section σ becomes a section commuting with z by

$$\tau\left(\sum_{j=0}^{p-1} z^j q_j\right) = \sum_{j=0}^{p-1} z^j \sigma(q_j).$$

The last term causes no problem because $z^p = 0$. Splitting the filtration gives

$$\mathcal{Z}_b(n+q) = \mathcal{Z}_b(n) \oplus \bigoplus_{j=0}^{p-1} z^j C \tag{53}$$

for a position-stable C, again under $n \ge 2b-1$ and $b < q$.

9.4.2 Top and zero-marginal spaces

The perfect pairing $T_b \times \mathcal{Z}_b \to K$ exchanges an operator on the quotient with its transpose on the zero-marginal space. Duality reverses composition order. Thus the odd-characteristic argument must use the actual transpose, not assume the characteristic-two equality $z^{\mathsf{T}} = z$.

With $u = z^{p-1}$, send uc back to c and kill the other levels in (53). This defines a position-equivariant w_C with $uw_C u = u$. Also

$$uV \cap \mathcal{Z}_b \subseteq u\mathcal{Z}_b.$$

Indeed a vector in this intersection is killed by z, so its decomposition in (53) is a retained vector plus an element of uC. Its membership in uV excludes the retained part, just as before.

For odd p, z need not be symmetric. Its transpose is

$$z^{\mathsf{T}} = \sum_i c_i(\sigma_i^{-1} - 1).$$

This is the ordinary bilinear matrix transpose; it does not conjugate the coefficients c_i. Let ι fix the retained labels and send a block label $v \in A$ to $-v$. Then $\iota\sigma_i\iota^{-1} = \sigma_i^{-1}$, so $z^{\mathsf{T}} = \iota z \iota^{-1}$. This is an actual relabelling, and preserves the tuple-layer constructions. The just-proved statements apply equally to u^{T}. Orthogonal complements give

$$uV \cap V' \subseteq uV',$$

and duality between T_b and $\mathcal{Z}_b$ gives a position-equivariant w_T with $uw_Tu = u$. This replaces the characteristic-two shortcut $z^{\mathsf{T}} = z$. The relabelling ι is the block inversion, distinct from the inclusion of a zero-marginal space into its tuple ambient.

9.4.3 The same correction, with the correct power

Suppose w_0 works on shorter tuples for the top-power relation. Its error is

$$\varepsilon = uw_0u - u.$$

It maps into $\mathcal{Z}_b$ and kills V'. The four properties just proved give, step by step as in Section 7.5,

$$uw_C\varepsilon = \varepsilon, \qquad \varepsilon w_T u = \varepsilon.$$

The correction $\varphi = -w_C\varepsilon w_T$ therefore satisfies $u\varphi u = -\varepsilon$. The characteristic-independent rook theorem realises it by $j \in J_{b-1}(N)$. Adding j repairs the witness at length b without changing any shorter length.

Inducting on b proves (49) on the same label pool whenever $n \geq 2r - 1$. The longer projector gives exactness, the moved-orbit calculation gives natural basis and tensor deletion, and the original PC closure transport gives the same period statement.

Completion of the proof of Theorem 4.2. Begin at tuple length zero with $w_0 = 0$, since z acts as zero on constants. At length b, the preceding correction and Theorem 8.4 replace w_{b-1} by w_b satisfying $z^{p-1}w_bz^{p-1} = z^{p-1}$ on every length at most b. The same retained labels are used at every step. The largest requirement is $\max_{1\leq b\leq r}(2b-1) = 2r-1$; length zero needs no label. The projector identities then imply exactness on $\mathcal{C}_r(N)$. The orbit commutator calculation supplies the natural retained-basis and tensor identifications, including the actual stable coefficient maps. These arguments prove every assertion of Theorem 4.2. □

The block construction is performed over its working extension field. Lemma 2.5 descends the transported polynomial consequence spaces to the original field without changing the degree budget.

10 An independent uniform NS–PC comparison

The linear PC lower bounds were proved by transporting the PC closure itself. We now establish a different result: every bounded-degree PC consequence has an NS identity of degree bounded independently of universe size. The comparison applies to arbitrary polynomials, not only invariant targets or refutations. Its qualitative consequence is that NS and PC assign every fixed uniform request the same type.

The NS-span plateau argument already occurs in Krajíček's uniform comparison [8, Theorem 5.5]. We give an independent proof of its required chain bound and carry the comparison through for the actual consequence spaces. A uniform bound on strict invariant-subspace chains forces the intersections of two successive NS certificate spaces with the fixed degree-d polynomial ambient to agree. Their common space is then closed under every

permitted PC operation. The plateau may depend on n, but the bound on its NS certificate degree does not.

At a degree cutoff, $I_e(n)$ is the linear space of polynomials with degree-at-most-e NS certificates. It is not an ideal closed under arbitrary multiplication. A *strict subspace chain* is a sequence in which each inclusion is proper. Its bound counts proper increases; consecutive equal spaces form a plateau. The proof uses deletion to preserve those increases, then uses a plateau to absorb variable multiplication. No decomposition into simple representations is required.

Lemma 10.1 (Uniform chain bound)**.** *For any bounded basis system $\mathcal{U}$, strict S_n-invariant subspace chains in $K[\mathcal{U}_n]$ have a length bound independent of n.*

Proof. Let r bound the basis support. Over an algebraic closure, choose a fixed $q = p^a > 2r$. At $N = n + q$ with $n \geq 4r + 1$, deletion is exact at the doubled budget $2r$. Lemma 4.3 shows that each nonzero subquotient B/A of $K[\mathcal{U}_N]$ survives. Thus $A \subsetneq B$ implies $\mathcal{D}(A) \subsetneq \mathcal{D}(B)$: the whole strict chain becomes a strict chain of the same length in $K[\mathcal{U}_n]$.

Repeatedly subtract q while the rank exceeds $M = 4r + q$. Every retained rank in such a step exceeds $4r$, so all steps are eligible. The final chain has at most $\dim K[\mathcal{U}_M]$ strict inclusions, because a smaller-rank basis injects into the rank-M basis. The same bound covers the initial ranks at most M. Each step may choose its own generic coefficients: the detector applies to arbitrary invariant subspaces, without a naturality assumption on the chain across ranks. Finally, scalar extension preserves strict inclusions, giving the result over the original coefficient field. □

Only a finite exactness threshold at each fixed support is needed for this argument. In characteristic two, the earlier quadratic construction also suffices. The bound may be large; its role is uniformity in n, not a sharp comparison of the two minimum degrees.

Theorem 10.2 (Uniform comparison for all consequences)**.** *For every uniform axiom template over a field of prime characteristic p, there is a nondecreasing $F : \mathbb{N} \to \mathbb{N}$, independent of n and of the target, with $F(d) \geq d$ and*

$$C_d(n) \subseteq I_{F(d)}(n) \qquad (n, d \geq 0). \tag{54}$$

Consequently, every polynomial with an ordinary-PC derivation of degree at most d from Γ_n has a total-product NS identity of degree at most $F(d)$ from the same axioms. For the aggregate profiles of any stable uniform target family, and for the profiles of any one fixed target orbit,

$$h_{\mathrm{PC}}(n) \leq h_{\mathrm{NS}}(n) \leq F(h_{\mathrm{PC}}(n)), \qquad F(\infty) = \infty. \tag{55}$$

Their NS and PC types in Theorem 3.2 therefore agree.

Proof. Fix d. Monomials of degree at most d form a permutation basis of support at most kd. Let L_d bound strict invariant-subspace chains in $P_{n,\leq d}$, uniformly in n, by Lemma 10.1. Consider the invariant spaces

$$W_j = I_{d+j}(n) \cap P_{n,\leq d}, \qquad 0 \leq j \leq L_d + 1.$$

They form an ascending chain. Some $W_j = W_{j+1}$ with $j \leq L_d$, otherwise there would be more than L_d strict inclusions.

This plateau contains every axiom usable within degree d and is closed under linear combinations. If $f \in W_j$ and $\deg f \leq d - 1$, then multiplication of an NS identity for f by a variable gives $x_v f \in I_{d+j+1}(n)$. Its degree is at most d, hence $x_v f \in W_{j+1} = W_j$. For $d = 0$ there is no permitted multiplication of a nonzero line; the zero line already lies in

W_j. Induction on *any* degree-d PC derivation now places every one of its lines in W_j. In particular,

$$C_d(n) \subseteq I_{d+L_d}(n).$$

This induction retains cancellations: a lower-degree linear combination can be multiplied because it lies in the plateau, even if its separate summands have degree d.

Choose

$$F(d) = d + \sum_{u=0}^{d} L_u.$$

It is nondecreasing, satisfies $F(d) \geq d$, and gives (54). No symmetry of a target was used. Conversely, expansion of NS identities into monomial–axiom products gives a same-degree PC derivation, so the first inequality in (55) holds. If the aggregate PC degree is finite, every requested target belongs to $C_d(n)$ at that budget; the inclusion puts every target in $I_{F(d)}(n)$. Empty requests have degree zero. Infinite PC degree forces infinite NS degree by the first inequality. These observations prove the complete extended-order comparison.

For type agreement write $g = h_{\mathrm{PC}}$ and $h = h_{\mathrm{NS}}$. The inequalities $g \leq h \leq F(g)$ transfer eventual boundedness in both directions. If g takes one finite value d infinitely often, then on those sizes h lies in the finite interval $[d, F(d)]$; some finite value of h recurs. A recurring finite value of h likewise forces a recurring finite value of g. Thus the recurring sets are empty together. If recurring values of g are unbounded, the same finite-interval argument yields unbounded recurring values of h. Conversely, choose a recurring value of h exceeding $F(B)$. On its infinite level set, $g > B$ and $g \leq h$, so some finite value above B recurs for g. Thus infinitude of the recurring sets agrees as well. Boundedness, emptiness and infinitude determine exactly the four alternatives. □

Corollary 10.3 (Common finite peaks)**.** *If a fixed uniform request has common type T4, there is a nondecreasing F such that, beyond every bound B, some size $n \geq B$ has*

$$h_{\mathrm{PC}}(n) = d \geq B, \qquad d \in \mathrm{Rec}(h_{\mathrm{PC}}), \qquad d \leq h_{\mathrm{NS}}(n) \leq F(d) < \infty,$$

with $n \leq (p+2)(\kappa d + \sigma)$. Both degrees satisfy the rational linear lower bound in (16) at these same sizes.

Proof. Take the finite PC peaks from Theorem 3.2. The pointwise lower comparison gives the NS lower bound, and the uniform upper comparison gives its finiteness at those sizes. □

The comparison does not identify exact degrees, easy-class values or recurring degree sets. Its potentially large function F cannot be inverted to obtain a linear PC lower bound from a linear NS lower bound. That quantitative result came directly from the PC transfer theorem.

11 Topological classification and semantic context

The four degree types have a purely topological description valid for all the uniform requests of Theorem 3.2. It uses only the eventual prime-power periodicity of bounded-degree sublevel sets. No Boolean encoding or model-existence assumption is needed. We then recall the distinction between algebraic witnesses and actual solutions, and the established Euler-characteristic criterion.

11.1 The p-adic hardness spectrum

The underlying structure is already implicit in the residue-class analysis of my 2009 report [7, Section 4, pp. 18–19]: once a degree threshold is fixed, sufficiently large sizes are classified by residues modulo a power of p. We express that earlier structure in p-adic language and apply it here to uniform consequences as well as refutations.

Fix any one of the aggregate or fixed-orbit profiles h in Theorem 3.1, over a field of characteristic p. A p-adic integer is a compatible choice of residues modulo all powers of p. The space $\mathbb{Z}_p$ is compact, and its basic residue neighbourhoods $a + p^L\mathbb{Z}_p$ are both open and closed (*clopen*). Using the onsets $N(d)$ and periods $Q(d)$ from (12), let $A_d(h) \subseteq \{0, \dots, Q(d) - 1\}$ consist of the residues a for which

$$h(n) \le d \quad \text{whenever } n \ge N(d), \qquad n \equiv a \pmod{Q(d)}.$$

Put

$$\mathcal{E}_d(h) = \bigcup_{a \in A_d(h)} (a + Q(d)\mathbb{Z}_p).$$

Thus $\mathcal{E}_d(h)$ records the eventually degree-d-easy residue classes. Each is clopen, and $\mathcal{E}_d(h) \subseteq \mathcal{E}_e(h)$ for $d \le e$: refine to the common period and use the periodicity on a sufficiently late tail. Set

$$\mathcal{H}_p(h) = \mathbb{Z}_p \setminus \bigcup_{d \in \mathbb{N}} \mathcal{E}_d(h). \tag{56}$$

We call this closed set the *hardness spectrum*. It does not depend on the choice of valid onsets or prime-power periods. Equivalently, $x \notin \mathcal{H}_p(h)$ exactly when some residue neighbourhood of x has a uniform finite degree bound at all sufficiently large integer sizes in that neighbourhood. Common residue refinements show the agreement of this description with (56).

Corollary 11.1 (The four topological shapes)**.** *For every uniform request considered in this paper, the NS and PC profiles have the following classification:*

Type	*Hardness spectrum* $\mathcal{H}_p(h)$
T_1	*Empty*
T_2	*All of* $\mathbb{Z}_p$
T_3	*Nonempty, proper and clopen*
T_4	*Closed but not open*

Moreover, for the same request the two systems have the same hardness spectrum, although their exact degrees and threshold sets can differ.

Proof. Write $E_d = \mathcal{E}_d(h)$ and $H = \mathcal{H}_p(h)$. If H is empty, compactness and the increasing open cover $(E_d)_d$ give $E_D = \mathbb{Z}_p$ for some D. This is eventual boundedness, or T1; the converse is immediate. Also $H = \mathbb{Z}_p$ exactly when every E_d is empty. This says that no finite degree recurs, or T2.

In T3, let D be the largest recurring finite degree. Theorem 3.2 gives a fixed nonempty proper union of easy residue classes. It is E_D, and no E_d meets a hard class: otherwise arbitrarily late integers in a common residue refinement would have both bounded and linearly growing degree. Hence $\bigcup_d E_d = E_D$, so H is nonempty, proper and clopen.

Conversely, suppose H is nonempty, proper and clopen. Its compact complement is covered by the increasing E_d, hence equals some E_D. If a finite degree e recurs, periodicity gives a residue neighbourhood on which that exact value occurs on a tail. This neighbourhood lies in $E_e \subseteq E_D$; comparing late integer representatives gives $e \le D$. There is a recurring finite degree because the complement of H is nonempty, but $H \neq \varnothing$ excludes T1. This is

T3. Since H is always closed and the four degree types are exhaustive, the remaining shape, closed but not open, is exactly T4.

Finally, the comparison theorem gives $h_{\mathrm{PC}} \leq h_{\mathrm{NS}} \leq F(h_{\mathrm{PC}})$, with F independent of size. Thus a residue neighbourhood is eventually bounded in one system if and only if it is eventually bounded in the other. The equivalent description above proves equality of their hardness spectra. □

In T3 a single finite congruence resolution separates the easy and hard regions. In T4 the closed spectrum has a nonempty boundary: every neighbourhood of a boundary point meets the easy region, but no such neighbourhood has one uniform finite degree bound. The spectrum records the qualitative organisation of the profile; the linear lower bounds and finite linear peaks still come from the quantitative transfer theorem. It is defined from the polynomial request and its degree profile, not from the ordinary model class of a first-order sentence alone.

11.2 Dual witnesses for failure to derive a target

A linear functional is a scalar-valued linear map on coefficient vectors. If it vanishes on every permitted consequence but not on the target, applying it to a proposed derivation gives a contradiction. The converse is finite-dimensional separation by a quotient basis; it does not require the functional to be evaluation at a solution.

For either system write $W_d(n)$ for its degree-d consequence space: $W_d(n) = C_d(n)$ for PC and $W_d(n) = I_d(n)$ for NS. Let $\delta = D_G$ be the fixed upper bound on target degree. We view $W_d(n)$ inside $P_{\leq \max\{d,\delta\}}(n)$, so the ambient space also contains every requested target even when its degree exceeds d.

Definition 11.2 (A degree-bounded target witness)**.** For a requested target $G_b^{(n)}$, a degree-d witness is a K-linear functional

$$\lambda : P_{\leq \max\{d,D_G\}}(n) \longrightarrow K, \qquad \lambda(G_b^{(n)}) = 1, \qquad \lambda(W_d(n)) = 0.$$

For an all-target request, a witness consists of an index b and such a functional. For a single fixed orbit, one may use any available representative. With target 1, this reduces to the usual normalisation $\lambda(1) = 1$ for a bounded-degree non-refutability witness.

Proposition 11.3 (Dual form of nonderivability)**.** *A degree-d target witness exists exactly when the target has no derivation of degree at most d in the chosen system. For an all-target request it exists exactly when $h(n) > d$.*

Proof. If $G_b^{(n)} \in W_d(n)$, a functional annihilating $W_d(n)$ cannot take value one on it. Conversely its class in the finite-dimensional quotient $P_{\leq \max\{d,D_G\}}(n)/W_d(n)$ is nonzero whenever it is not in $W_d(n)$. Extend that class to a basis, assign it value one and assign the other basis vectors zero. Composing with the quotient map gives the required functional. If the target degree is greater than d, it automatically lies outside $W_d(n)$; the enlarged ambient space also covers this degree-floor obstruction. Failure of all-target derivability is failure for at least one target. □

A K-valued solution a of Γ_n which violates $G_b^{(n)} = 0$ gives a witness by $\lambda(f) = f(a)/G_b^{(n)}(a)$. For target one this is ordinary evaluation, with its multiplicative property. A general witness need not be multiplicative, positive or normalised on one. It records a linear separation from all permitted bounded-degree consequences, not necessarily a truth assignment to the original variables.

Example 11.4 (Semantic entailment need not be derivation)**.** The equation $x^2 = 0$ over a field forces $x = 0$ at every solution. Nevertheless, neither NS nor PC derives x from this axiom: every consequence belongs to the ideal (x^2). For $d \geq 2$, the functional on $P_{\leq d}$ taking value one on x and zero on all other monomials annihilates the degree-d consequence space. This is a target witness although no actual solution violates $x = 0$. The example explains why our minimum degree is a syntactic ideal-membership resource.

Example 11.5 (Linear consistency without truth-value evaluation)**.** Over $\mathbb{F}_3$, take $x^2 - x = 0$, target one and degree budget two. The PC consequence space is $K(x^2 - x)$: no nonzero vector in that span has degree at most one, so no multiplication is usable. The functional

$$\lambda(1) = 1, \qquad \lambda(x) = \lambda(x^2) = 2$$

is a witness but is not evaluation: $\lambda(x^2) = 2$ whereas $\lambda(x)^2 = 1$. Boolean equations impose the intended truth-value semantics on actual solutions, without making every linear witness an actual evaluation.

For target 1, the NS witness is the design method for lower bounds surveyed and generalised by Buss [25]: a normalised linear functional annihilates every permitted axiom multiple. Annihilating only axiom multiples gives an NS witness. For PC the functional must annihilate the full closure, including polynomials obtained by cancellation followed by new multiplication. This is the dual form of the distinction between I_d and C_d.

11.3 What Krajíček's Euler criterion supplies

A strong Euler characteristic assigns a value $\chi(A)$ to each permitted parameter-definable set, possibly infinite. It gives a point value one, is additive on disjoint unions, is multiplicative on products, and is invariant under definable bijections. For a definable map $f : A \to B$ whose fibres all have characteristic r, it also satisfies $\chi(A) = r\chi(B)$. This last fibre rule is essential.

Such a rule replaces finite counting in a definable solution of a uniform polynomial system. In Krajíček's criterion, the values lie in $\mathbb{Z}/p^\nu\mathbb{Z}$, where the required precision ν depends on the degree budget and the indexing bounds. Reduction modulo p interprets a sum of a definable $\mathbb{F}_p$-valued function by

$$\sum_{x \in A}^{\chi} a(x) = \sum_{b \in \mathbb{F}_p} b\, \overline{\chi}(a^{-1}(b)).$$

With the definability and index-sort hypotheses of [12, Theorem 6.1], including the characteristic on the required index sorts M^*, a solution violating the target $g = 0$ excludes derivations within that fixed degree budget at sufficiently large sizes in the residue class specified by $\chi(M) \bmod p^\nu$. The refutation case is $g = 1$.

The mechanism is that bounded-degree calculations involve only bounded configurations of indices. The counting rules transport the relevant finite identities to the definable setting. A hypothetical derivation would make the target vanish there, contradicting the chosen solution. A characteristic with sufficient precision for each degree therefore provides a semantic obstruction to bounded-degree proofs. As the degree increases, the required precision may increase too.

Definable bijections, disjoint-union relations and products can be organised in the Grothendieck ring of definable sets [13]. The strong fibre rule is additional structure. Neither an arbitrary infinite model nor a merely mod-p counting rule supplies the full hypotheses of Krajíček's criterion. It is a sufficient criterion; no converse is used or asserted here.

Richer first-order counting theories can also describe bounded-degree algebraic reasoning under additional hypotheses [27]. Such model-theoretic representations are not needed for

Corollary 11.1; in particular, its scope is not restricted to a fixed Boolean compiler. A general converse to the Euler criterion remains a separate question.

12 Nonconstant targets with finite linear peaks

The degree of a target need not reflect the degree needed to derive it. A polynomial of degree one may require arbitrarily high proof degree. We first give an exact construction, then apply it to a uniform family whose hard values are finite.

A simple invariant target illustrates the distinction between finite hardness and failure of derivability. For $G^{(n)} = \sum_{i<n} x_i$, the input $\{1\}$ gives degree exactly one in both systems for $n \geq 1$. With no inputs the same target is underivable and has degree ∞. Both fit the uniform definition, but neither supplies a finite hard example. The next construction does.

Two elementary maps on the enlarged ring will be used. Substitution $y = 1$ is a ring homomorphism. Coefficient extraction $[y^1]$ is instead a linear map: it takes $\sum_j a_j(x)y^j$ to $a_1(x)$. It lowers the total degree by one on a nonzero extracted term and commutes with multiplication by a polynomial independent of y. This distinction accounts for the different exact NS and PC degrees.

Proposition 12.1 (Adjoining a fresh target)**.** *Let K be any field, let $\Gamma \subseteq K[x_v : v \in V]$ be any axiom set, and adjoin a fresh variable y. Each certificate uses only finitely many axioms. Let $\Gamma^{\uparrow}$ be the same axioms viewed in $K[x_v : v \in V, y]$, with no additional axiom involving y. Then*

$$\begin{aligned} h_{\mathrm{PC}}(\Gamma^{\uparrow} \vdash y) &= \max\{1, d_{\mathrm{PC}}(\Gamma)\}, \\ h_{\mathrm{NS}}(\Gamma^{\uparrow} \vdash y) &= d_{\mathrm{NS}}(\Gamma) + 1. \end{aligned} \tag{57}$$

These equalities include degree zero and infinity, with $\infty + 1 = \infty$.

Proof. For PC, embed a refutation of Γ in the larger ring and multiply its last line 1 by y. If the refutation had degree d, the resulting proof of y has degree $\max\{1, d\}$: only the new last line is added, and its degree is one.

Conversely, substitute $y = 1$ in every line of a PC proof of y. The old axioms are unchanged and linear combinations commute with the substitution. Multiplication by an old variable remains a multiplication step; multiplication by y becomes a repeated line, which the linear combination rule admits. No degree increases, and the last line becomes 1. Every proof of y also has degree at least one. These two lower bounds prove the first equality, including the case when no refutation exists.

For NS, multiplying the coefficients of an identity $1 = \sum_f a_f f$ by y gives an identity for y with degree one larger. For the reverse inequality, suppose

$$y = \sum_f a_f(x, y) f(x)$$

has total-product degree at most e. Take the coefficient of y^1. Since every f is independent of y, this gives the exact identity

$$1 = \sum_f [y^1] a_f(x, y)\, f(x).$$

Every nonzero product on the right has degree at most $e - 1$: it is the coefficient of y^1 in a product of total degree at most e. Zero products may be discarded. Hence $d_{\mathrm{NS}}(\Gamma) \leq e - 1$. In particular a finite proof of y exists exactly when a finite refutation exists. If the original refutation degree is zero, both new degrees are one, as asserted. □

Example 12.2 (Why the two exact identities differ)**.** Take $\Gamma = \{x, 1-x\}$. Both refutation degrees are one. PC derives $1 = x + (1-x)$ and then multiplies that last constant by y; every line still has degree at most one. An NS identity for the fresh target is

$$y = yx + y(1-x).$$

Its products have degree two. No degree-one NS identity can produce y, since its multipliers of these degree-one axioms would have to be constants. Thus the new PC degree is one and the NS degree is two. PC can finish the refutation before multiplying its last line; an NS identity carries the extra variable in its coefficient polynomials.

The absence of a y-axiom is part of the construction; in particular, we do not add $y^2 - y = 0$. For a uniform template, give the new variable empty support and keep all old axiom indices and coefficients. Set all coefficients of axiom monomials involving y to zero. The result is again a uniform template with the same bounds k, s, D, n_0, and $G^{(n)} = y$ is a fully invariant target of degree one at every rank. Thus (57) holds rank by rank.

This operation preserves each of the four types, separately for NS and PC. The NS recurring finite degrees are shifted by one. For PC, the map $d \mapsto \max\{1, d\}$ merges only degrees zero and one; all larger finite degrees and infinity are unchanged. Eventual boundedness, absence or unboundedness of recurring finite levels, and the mixed periodic case are consequently preserved.

Corollary 12.3 (A finite hard nonconstant consequence family)**.** *Let K have prime characteristic p, and let $\widehat{B}_n$ be the normalised weak-bijection system defined in (61), with all of its function and coverage witnesses retained. Adjoin a fresh variable y as above, and request the target $G^{(n)} = y$. Both target-degree profiles are of type T4.*

More precisely, write $n = p^a u$, where $p \nmid u$, and put $q = p^a$. Whenever $n \geq 2q$,

$$q \leq h_{\mathrm{PC}}(n) \leq q + 4, \qquad q + 1 \leq h_{\mathrm{NS}}(n) \leq q + 5,$$
$$h_{\mathrm{PC}}(n) \leq h_{\mathrm{NS}}(n).$$

In particular, at every size $n_a = (p+1)p^a$,

$$\frac{n_a}{p+1} \leq h_{\mathrm{PC}}(n_a) \leq h_{\mathrm{NS}}(n_a) \leq \frac{n_a}{p+1} + 5 < \infty. \tag{58}$$

Proof. The refutation bounds (62), proved in the next section, give $q \leq d_{\mathrm{PC}}(\widehat{B}_n) \leq d_{\mathrm{NS}}(\widehat{B}_n) \leq q + 4$. Apply (57); since $q \geq 1$, the maximum in the PC identity changes nothing here. At $n_a = (p+1)p^a$ we have $v_p(n_a) = a$ and $n_a \geq 2p^a$, yielding (58).

For each fixed a, infinitely many sizes $n = p^a(p(j+1)+1)$ have the same valuation and satisfy the displayed bounds. Some finite degree in each of the bounded windows must therefore recur infinitely often. The windows tend to infinity with a, proving T4 in each system. On the other hand, the old upper bound gives $h_{\mathrm{PC}}(n) \leq 5$ and $h_{\mathrm{NS}}(n) \leq 6$ whenever $p \nmid n$. The linear peaks thus coexist with infinitely many easy sizes, rather than giving an eventual linear lower bound at all sizes. □

This example has finite hard proofs of a nonconstant target, but its axioms are inconsistent: the difficulty is inherited from refuting $\widehat{B}_n$. It does not establish a family of satisfiable axioms with the same hard consequence profile. The small consistent example in Section 2 illustrates ordinary-PC consequences in a different setting; no growing lower bound is asserted for it.

13 The refutation special case: four first-order examples

We now take the constant target $G^{(n)} = 1$. This is the refutation special case of the uniform-consequence theorem; the sharper invariant-target budget applies. The four cases all occur among fixed first-order sentences with no finite nonempty models. We give examples over a field of fixed prime characteristic p. Each sentence is translated by the graph-and-sum convention: relation and function-graph variables are Boolean, and every function has a totality sum and pairwise functionality equations. The parameter $n \geq 1$ is the universe size. For the matrix notation in this section, we relabel the universe as $\{1, \dots, n\}$; this changes no equation or degree.

T1: a local contradiction. The sentence $\forall x\,(R(x) \wedge \neg R(x))$ contributes r_i and $1 - r_i$. Their sum refutes the system in degree one, independently of n. The contradiction is already visible at one element.

T2: an injection that misses a point. Let η say that a unary function g is injective and that a constant c is outside its image. Write x_{ij} for $g(i) = j$ and c_j for $c = j$. In addition to the Boolean equations, its system Γ_n^η consists of

$$\begin{gathered} \sum_j x_{ij} - 1 = 0, \qquad x_{ij}x_{ik} = 0 \ (j \neq k), \qquad x_{ij}x_{hj} = 0 \ (i \neq h), \\ \sum_j c_j - 1 = 0, \qquad c_j c_k = 0 \ (j \neq k), \qquad x_{ij}c_j = 0. \end{gathered} \tag{59}$$

A solution would inject n elements into the $n - 1$ elements other than c, so every instance is unsatisfiable. For either proof system, every degree-d refutation satisfies

$$n < 3pd \qquad (n \geq 3).$$

Here is how the internal matching bound gives this estimate. Choose the largest p-power $q \leq n/3$ and put $r = n - q$. Set $c_n = 1$, all other $c_j = 0$, and restrict the image of g to $[r]$. After transposition, every remaining axiom is an axiom of the bijection system $B_{r,n}$. Here $B_{r,m}$ is the Boolean r-by-m matrix system with every row and column sum equal to one and both row and column collisions equal to zero. If $d < q$, then $2d \leq r$ and $q \mid n - r$, contradicting the rectangle obstruction of Lemma 14.1. The substitutions do not increase ordinary-PC or NS degree. Thus $d \geq q$, while maximality of q gives $n < 3pq \leq 3pd$. This proves T2 without an external pigeonhole lower-bound theorem.

A matching linear upper bound. For the injection system, put

$$R_i = \sum_j x_{ij}, \qquad C = \sum_j c_j, \qquad P = \prod_{i=1}^{n} R_i.$$

Every monomial $c_j \prod_i x_{i,\lambda(i)}$ in the expansion of CP is divisible by an input equation from (59). If λ is not injective, it contains a column collision. If λ is injective, it is a permutation of the finite universe; for the row i with $\lambda(i) = j$, the monomial contains the omission factor $c_j x_{ij}$. Thus CP lies in the raw NS span at degree at most $n + 1$. The telescoping identity

$$CP - 1 = (C - 1) + C \sum_{i=1}^{n} (R_i - 1) \prod_{h<i} R_h$$

expresses $CP-1$ using the constant and row totality equations, with every product of degree at most $n+1$. Subtraction gives a refutation, so for every $n \geq 1$,

$$d_{\mathrm{PC}}(\Gamma_n^\eta) \leq d_{\mathrm{NS}}(\Gamma_n^\eta) \leq n+1.$$

This argument works over any field and uses no Boolean simplification. Together with the preceding lower bound, it shows that both degrees are $\Theta(n)$ for each fixed characteristic. The general linear gap is therefore sharp even among all-size inconsistent first-order families.

T3: hardness on one residue class. Let χ_p assert

$$\forall x \qquad f^p(x) = x \quad \text{and} \quad f(x) \neq x.$$

Its finite models are disjoint unions of directed p-cycles. In particular, it has a 0/1 graph assignment exactly when $p \mid n$. An explicit normalisation uses the function equations for f, the equations $f_{ii} = 0$, and

$$f_{i_0 i_1} f_{i_1 i_2} \cdots f_{i_{p-1} i_p} = 0 \qquad (i_p \neq i_0). \tag{60}$$

If $p \nmid n$, summing all length-p walks gives an NS refutation of degree at most p. Indeed, totality derives the sum of their monomials minus n within degree p. Nonclosed walks vanish by (60). A closed walk with a repeated vertex either contains a functionality collision or follows a shorter deterministic cycle; since p is prime, the latter is a forbidden fixed point. Each remaining cycle contributes its monomial p times, according to its starting vertex, and hence contributes zero. The resulting nonzero scalar n yields the refutation.

Now take $\psi_3 = \eta \wedge \chi_p$ in disjoint nonlogical symbols. For $p \mid n$, substitute a p-cycle assignment into the second block. This preserves proof degree and shows that the conjunction has exactly the degree of η, separately for NS and PC. For $p \nmid n$, the second block refutes in degree at most p. Consequently this all-size unsatisfiable family is bounded off the multiples of p and has linear degree on the multiples: it has type T3. No exact degree is needed on the bounded residue classes.

T4: the valuation of the universe size controls the difficulty. For a positive integer n, its p-adic valuation $v_p(n)$ is the largest a such that p^a divides n. Thus $n = p^a u$ with $p \nmid u$. We use this integer statistic, not a valuation on the coefficient field; the corresponding block size is p^a. The weak-bijection example from the earlier classification [6, Section 5] is more revealing; its algebraic complexity is closely related to the bijective pigeonhole principle studied by Beame and Riis [26]. Let unary functions f, g be injective with disjoint images, and require their images together to cover the universe. Use a unary witness h for coverage:

$$\forall z \qquad f(h(z)) = z \ \vee \ g(h(z)) = z.$$

Let a_{ij}, b_{ij}, w_{ji} represent $f(i) = j$, $g(i) = j$, and $h(j) = i$. The combined $2n$-by-n matrix $(a; b)$ has row sums equal to one and both row and column collisions equal to zero. The witness block consists of

$$\sum_i w_{ji} - 1 = 0, \qquad w_{ji} w_{ji'} = 0 \ (i \neq i'), \qquad w_{ji}(1 - a_{ij})(1 - b_{ij}) = 0, \tag{61}$$

together with the Boolean equations. Call this complete system $\widehat{B}_n$. Its witnesses are retained: it is not identified with the simpler matrix system $B_{n,2n}$.

Put $q = p^{v_p(n)}$. The proved bounds give

$$q \leq d_{\mathrm{PC}}(\widehat{B}_n) \leq d_{\mathrm{NS}}(\widehat{B}_n) \leq q+4 \qquad \text{whenever } n \geq 2q. \tag{62}$$

The upper bound holds at every positive size, without the displayed size condition. We give both degree-preserving reduction and certificate explicitly.

For the lower bound transpose the combined matrix and substitute $w_{ji} = a_{ij} + b_{ij}$ into every witness variable. Its totality becomes the corresponding totality of $B_{n,2n}$. The identities

$$(a+b)^2 - (a+b) = (a^2 - a) + (b^2 - b) + 2ab,$$
$$(a+b)(1-a)(1-b) = -(1-b)(a^2 - a) - (1-a)(b^2 - b)$$

put the substituted Boolean and coverage equations in the raw NS spans at degrees two and three respectively. A substituted witness collision expands into four column collisions of the raw combined matrix. Thus every substituted input has a derivation within its own degree. Substitution is affine: a multiplication by an old variable becomes a sum of permitted multiplications by new variables, with the preceding line still of degree at most $d-1$. A degree-d ordinary-PC refutation would therefore refute $B_{n,2n}$ within the same degree. The rectangle obstruction excludes this when $2d \le n$ and $d < q$, yielding the lower bound in (62).

For the upper bound fix a column j and abbreviate

$$u_i = a_{ij} + b_{ij}, \qquad T = \sum_i u_i, \qquad Z = \sum_i w_{ji} u_i.$$

The coverage axiom and disjointness collision give the exact identity

$$Z - 1 = (\sum_i w_{ji} - 1) - \sum_i w_{ji}(1 - a_{ij})(1 - b_{ij}) + \sum_i w_{ji} a_{ij} b_{ij}.$$

Consequently $Z - 1$ lies in the degree-three NS span. If x is one entry of this column, then

$$Tx - x = (x^2 - x) + \sum_{y \neq x} yx$$

lies in the degree-two span, by Booleanity and column collisions. Adding this for a_{ij} and b_{ij} gives $Tu_i - u_i$ in that span. Finally

$$T - 1 = \sum_i w_{ji}(Tu_i - u_i) - (T-1)(Z-1)$$

puts the missing column totality in the degree-four span. This retains the coverage witnesses throughout the derivation.

For completeness, consider any matrix with row sums R_i and column sums C_j, with their differences from one in the NS span at degree e, and with both collision families in degree two. For a fixed k-element row set S, telescoping the product of its row sums gives $\prod_{i \in S} R_i - 1$ in degree at most $e + k$. Summing over S gives

$$\sum_{|S|=k} \prod_{i \in S} R_i - \binom{\#\text{rows}}{k}$$

in that same span. Expand the products as partial functions from k rows to columns. A noninjective partial function contains a column collision, so its monomial is already in the degree-k span. The remaining monomials are exactly those of k-edge matchings. The column calculation gives the same matching sum, since inversion pairs partial injections with identical monomials. Subtracting derives

$$\binom{\#\text{rows}}{k} - \binom{\#\text{columns}}{k}$$

within degree $e + k$. Applied to the $2n$-by-n matrix with $e = 4$, this is the desired counting certificate.

Write $n = qu$ with $q = p^{v_p(n)}$ and $p \nmid u$. In characteristic p, $(1 + t)^n = (1 + t^q)^u$. The coefficient of t^q is therefore u; for $(1 + t)^{2n}$ it is $2u$. Thus $\binom{2n}{q} - \binom{n}{q} = u \neq 0$ in K. Rescaling the counting certificate gives an NS refutation of degree at most $q + 4$.

For each fixed a, the sizes $n = p^a(p(k + 1) + 1)$, $k \geq 0$, have $v_p(n) = a$ and satisfy (62). Hence some finite degree in $[p^a, p^a + 4]$ recurs infinitely often. These windows are unbounded as a varies, proving T4 for both proof systems. On the other hand, $p \nmid n$ implies degree at most five. There are explicit simultaneous linear peaks: at $n_a = (p + 1)p^a$,

$$\frac{n_a}{p+1} \leq d_{\mathrm{PC}}(\widehat{B}_{n_a}) \leq d_{\mathrm{NS}}(\widehat{B}_{n_a}) \leq \frac{n_a}{p+1} + 4.$$

For example, in characteristic two every odd n has degree at most five, whereas at $n = 3 \cdot 2^a$ both degrees lie between 2^a and $2^a + 4$. The peaks are linear in their universe size even though the profile returns to bounded values infinitely often. Thus linear peaks coexist with infinitely many easy sizes. This is the reason the T4 conclusion concerns peaks rather than an eventual linear lower bound at every size.

14 Matching designs and the concrete lower bounds

The examples use a lower bound for a concrete rectangular system. Its proof illustrates why a functional excluding a Nullstellensatz certificate does not by itself exclude a Polynomial Calculus derivation. To handle degree falls we construct a whole space of functionals, together with an extension operator between successive degrees. The concrete bijection system has its antecedent in Beame and Riis [26]; the use of designs as algebraic obstruction witnesses goes back to Buss [25]. The construction and the extension through ordinary-PC degree falls are proved explicitly below.

For positive integers $r \leq m$, let $B_{r,m}$ consist of the equations

$$\sum_{j=1}^{m} x_{ij} - 1 = 0 \quad (i \in [r]), \qquad \sum_{i=1}^{r} x_{ij} - 1 = 0 \quad (j \in [m]),$$

the Boolean equations $x_{ij}^2 - x_{ij} = 0$, and the row and column collision equations

$$x_{ij}x_{i\ell} = 0 \quad (j \neq \ell), \qquad x_{ij}x_{hj} = 0 \quad (i \neq h).$$

Here $[r] = \{1, \ldots, r\}$. A Boolean solution would describe a bijection between the two sides, so a proper rectangle has no solution. Nevertheless, in positive characteristic the two linear totality sums can agree because $m - r$ vanishes in the field. The following statement identifies a range in which this obstruction persists through ordinary PC.

Lemma 14.1 (Rectangle obstruction)**.** *Let K be a field of prime characteristic p, and let a, D be nonnegative integers. If*

$$1 \leq r \leq m, \qquad 2D \leq r, \qquad D < p^a, \qquad p^a \mid m - r,$$

then $B_{r,m}$ has no ordinary-PC refutation of degree at most D. Consequently it has no Nullstellensatz refutation of total-product degree at most D.

We give the full construction over the integers before passing to K. An *edge set* is a subset of $[N] \times [M]$. It is a matching if no two distinct edges share a row or a column. Write

rows P and cols P for its two projections. A degree-d design may take values in a coefficient ring: it is a function ψ on edge sets, zero on nonmatchings, for which

$$\sum_{j=1}^{M} \psi(P \cup \{(i,j)\}) = \psi(P), \qquad \sum_{i=1}^{N} \psi(P \cup \{(i,j)\}) = \psi(P) \quad (|P| < d). \tag{63}$$

The indices $i \in [N]$ and $j \in [M]$ are arbitrary. For K-valued designs, denote the resulting vector space by $E_d(N, M)$. The signed coordinate reductions below are constructed integrally and then interpreted in K. Normalisation, $\psi(\varnothing) = 1$, is a separate condition. We will construct a normalised member of $E_D(r, m)$ and prove that, for $D \geq 1$, *every* member of $E_{D-1}(r, m)$ extends to $E_D(r, m)$ while preserving its values through degree $D - 1$.

Associate an edge set P with the squarefree monomial $\prod_{e \in P} x_e$. A design assigns proposed values to these monomials and satisfies the displayed linear consistency tests. An actual solution evaluates multiplicatively; a design need not. Normalisation fixes the value of the constant monomial, without any probability or positivity assumption.

14.1 A terminating signed reduction

The reduction is a formal integer combination of matchings. Its coefficients may be negative, and are interpreted in K only after the integral identities have been proved. Termination is controlled by a decreasing integer rank, rather than by cancellation of coefficients.

A chain in a matching is a subset whose row coordinates and column coordinates both increase. Let $h(P)$ be the maximum chain length, and write $P_{\leq b}$ for the edges in rows at most b. A matching P is *terminal* if, for every positive integer b,

$$h(P_{\leq b}) \leq b - |\operatorname{rows}(P_{\leq b})|,$$

and the corresponding inequality holds for column prefixes. The right side counts unused labels in the prefix. This definition depends only on the labelled matching, not on the ambient rectangle. Prefixes beyond its last occupied row or column cannot introduce a new failure: their left side stays fixed and their right side increases.

For a nonterminal matching, use the least failed row prefix if one exists; otherwise use the least failed column prefix. In the selected prefix choose a longest chain, breaking ties by the lexicographically greatest list of its edges, with the edges themselves in row-first lexicographic order. In a column step use this rule after transposition. Thus all choices are deterministic.

Consider a row step with failed prefix b and chosen chain A. Put

$$P_0 = P \setminus A, \qquad S = \operatorname{cols} A, \qquad B = \{i \leq b : i \notin \operatorname{rows} P\},$$

$$U = \operatorname{rows} A \ \cup \ \{b < i \leq N : i \notin \operatorname{rows} P\}.$$

The sets B and U are disjoint and partition all rows unused by P_0. Moreover, $|B| < |A|$, because the selected prefix fails. A longest chain in the first failed prefix ends in row b. Indeed, if it ended earlier, the same chain would belong to the preceding prefix, whereas the number of unused labels cannot decrease when one adds the next row. That preceding prefix would already fail.

For sets of columns S and rows V, a full injection from S into V means a matching J with $\operatorname{cols} J = S$ and $\operatorname{rows} J \subseteq V$. A partial injection has only $\operatorname{cols} J \subseteq S$. Replace P by

$$\sum_{\substack{J \text{ partial injection} \\ S \to B}} (-1)^{|J|} [P_0 \cup J] \ - \sum_{\substack{J \text{ full injection} \\ S \to U \\ J \neq A}} [P_0 \cup J],$$

omitting colliding terms. Square brackets here denote formal basis elements, not polynomial equivalence classes. The transposed rule defines a column step.

To verify the identity for a degree-d design with $|P| \le d$, repeatedly apply its column equation to the columns of S. At each application the partial matching has fewer than $|P|$ edges, so the degree restriction is satisfied. An occupied row contributes zero by collision vanishing. For a still unassigned column the sum over available rows in U is therefore the previous value minus the sum over rows in B. Expanding these successive differences gives

$$\sum_{J:S\to U} \psi(P_0 \cup J) = \sum_{A'\subseteq S} (-1)^{|A'|} \sum_{J:A'\to B} \psi(P_0 \cup J).$$

Every surviving term is an injection, because collisions have value zero. The right side contains each partial injection exactly once, indexed by its column set. Isolating the term $J = A$ on the left proves the stated reduction. The column step follows by interchanging rows and columns in the design equations. No assertion that the deterministic reduction itself commutes with transposition is needed.

We next prove termination, including the equal-degree case. For the fixed grid $G = [N] \times [M]$, set $H = N + M$, $c = NM + 1$, and define

$$w(i,j) = c^{H-i-j}, \qquad W(P) = \sum_{(i,j)\in P} w(i,j), \qquad \rho(P) = |P|(W(G)+1) + W(P).$$

Every lower term has fewer edges, since an injection into B has at most $|B| < |A|$ edges. Thus its ρ is smaller irrespective of its weight. For a full injection $J \ne A$ into U, let e be the first edge of A not belonging to J, in increasing row order. Every earlier chain edge still belongs to J. If $x \in J \setminus A$ had row smaller than that of e, its row would be an earlier chain row already occupied by its retained edge; this contradicts the matching condition. A row from the other part of U is greater than b and hence cannot be smaller. Likewise, a column of x smaller than the column of e belongs to an earlier retained chain edge, again giving a collision. Consequently

$$e_1 \le x_1, \qquad e_2 \le x_2, \qquad e_1 + e_2 < x_1 + x_2.$$

At least one inequality is strict because $x \ne e$. Each newly added edge has weight at most $w(e)/c$, and there are at most $NM < c$ such edges. Their total weight is strictly smaller than $w(e)$. Cancelling the common edges of A and J shows $W(P_0 \cup J) < W(P)$. The identical argument applies to a column step, since $i + j$ is unchanged by transposition. Every child therefore has strictly smaller nonnegative integer rank ρ. The tree is finitely branching, and every path has length at most its initial rank. Hence the complete signed reduction tree is finite.

For terminal T, let $C_{N,M}(P,T) \in \mathbb{Z}$ be the sum, over paths from P to T, of the products of their step signs; set it to zero for colliding P. The empty path gives

$$C_{N,M}(S,T) = \delta_{S,T} \quad (S \text{ terminal}), \qquad C_{N,M}(P,T) = 0 \quad (|T| > |P|).$$

Induction on $\rho(P)$ using the reduction identity gives, for every degree-d design,

$$\psi(P) = \sum_{\substack{T \text{ terminal} \\ |T|\le d}} C_{N,M}(P,T)\psi(T) \qquad (|P| \le d). \tag{64}$$

Thus the terminal values determine every design in the relevant range. It remains to show that these coordinates themselves satisfy the required totality equations and can be restricted from a square to a rectangle.

14.2 Integral independence in a square

The next argument uses signed sums of evaluations at actual square permutations to isolate terminal coordinates. A diagonal value of 1 or -1 is invertible in every coefficient ring, unlike a factorial. This is why the resulting independence remains valid after reduction to any characteristic.

Let $T \subseteq [N] \times [N]$ be terminal. For an edge $e \in T$, let its height be the maximum chain length in the southwest downset $\{f \in T : f_1 \le e_1,\ f_2 \le e_2\}$. A longest such chain ends at e: otherwise e can be appended, because distinct edges of a matching have distinct row and column coordinates. If e is southwest of f with both coordinates strictly smaller, appending f raises the height. It follows that every equal-height level T_t is decreasing: as its row coordinate increases, its column coordinate decreases. The levels partition T.

Let $h = h(T)$. The terminal inequality at prefix N implies that there are at least h unused rows and h unused columns. Let a_t and b_t be the tth smallest unused row and column, respectively, for $1 \le t \le h$. These anchors precede every coordinate in T_t. For example, if $e \in T_t$, a chain of length t ending at e belongs to its row prefix. Terminality requires at least t unused rows in that prefix; row e_1 is occupied, so the tth unused row is strictly less than e_1. The column argument is identical. Form the paired blocks

$$A_t = \operatorname{rows} T_t \cup \{a_t\}, \qquad B_t = \operatorname{cols} T_t \cup \{b_t\}.$$

They are disjoint on each side, their paired sizes agree, and their minima are a_t, b_t. Their complements also have equal sizes. Fix the increasing bijection on the complements, and an increasing base bijection $A_t \to B_t$ in each block. Vary each base bijection by a permutation σ_t of A_t. The union of all these bijections is an actual permutation π_σ of $[N]$.

For any edge set P, define the integer signed moment

$$M_T(P) = \sum_{\sigma} \left(\prod_{t=1}^{h} \operatorname{sgn} \sigma_t \right) \mathbf{1}_{P \subseteq \pi_\sigma}.$$

If a block has two rows absent from rows P, interchange their images. This is a fixed-point-free involution on the choices, changes the sign, and preserves the containment condition. Its terms cancel over $\mathbb{Z}$. In particular, $M_T(P) = 0$ when $|P| < |T|$: otherwise every block would use at least $|A_t| - 1 = |T_t|$ rows, forcing $|P| \ge \sum_t |T_t| = |T|$.

The diagonal is a unit. In each block the edges of T_t prescribe the bijection on every row except a_t; its remaining value is necessarily b_t. Thus exactly one choice σ contains T, and

$$M_T(T) \in \{1, -1\}.$$

This is a single signed contribution; no factorial or positive integer multiplicity must be inverted.

For equal-degree triangularity, suppose $|P| = |T|$ and $M_T(P) \neq 0$. There is a choice containing P, so P is a matching. Cancellation shows that at most one row in each block is unused. Counting over the disjoint blocks forces exactly $|T_t|$ edges in block $A_t \times B_t$, one unused row in every block, and no edge in the complement. Write P_t for this block matching.

Among all matchings of size $|T_t|$ in $A_t \times B_t$, the decreasing matching T_t has uniquely least weight W. Here is a direct proof. Choose a weight minimiser Q from this finite set. If it uses the minimum row a_t, move that edge to the unique unused row. This is a legal matching, has the same cardinality, and strictly smaller weight, because its row coordinate has increased. Thus a_t is unused. The same argument shows that the minimum column b_t is unused. If Q contains an increasing pair $(i, j), (i', j')$ with $i < i'$ and $j < j'$, replace it by the crossed pair $(i, j'), (i', j)$. Both added edges have coordinate sum strictly larger than

that of (i, j), and their combined weight is smaller than its weight, since $c = N^2 + 1 > 2$ when such a pair exists. The unchanged context is disjoint from the crossed pair. This again contradicts minimality. Therefore Q is decreasing, uses precisely $A_t \setminus \{a_t\}$ and $B_t \setminus \{b_t\}$, and must equal T_t: on two fixed ordered sets there is exactly one order-reversing bijection. Any distinct matching with the same minimum weight would also be a minimiser and would give the same conclusion. Hence

$$W(P_t) \geq W(T_t), \quad \text{with equality only if } P_t = T_t.$$

Summing over the disjoint blocks proves that

$$|P| = |T|, \quad P \neq T, \quad M_T(P) \neq 0 \quad \Longrightarrow \quad W(P) > W(T).$$

Together with the lower-degree cancellation, this says that $M_T(P) = 0$ for every $P \neq T$ with $\rho(P) \leq \rho(T)$.

Now suppose a finite linear combination of terminal matching indicators vanishes at every permutation of $[N]$, with coefficients in any commutative ring R. Among its nonzero coefficients choose a terminal T of greatest rank ρ. Apply the integer signed moment M_T to this relation, then map the integers to R. All other nonzero terms vanish by triangularity; the result is the coefficient of T multiplied by 1 or -1. This contradicts its being nonzero. The terminal indicators are therefore independent over every commutative ring, including fields of characteristic two and rings with zero divisors.

Each square permutation satisfies the design equations in every degree. Equation (64), applied to its containment indicator, expresses every matching indicator in terminal coordinates. For a fixed row i, the permutation identity

$$\sum_{j=1}^{N} \mathbf{1}_{P \cup \{(i,j)\} \subseteq \pi} = \mathbf{1}_{P \subseteq \pi}$$

holds because π has exactly one edge in that row. Expand all terms in terminal coordinates and use the integral independence just proved. For each terminal T this yields

$$\sum_{j=1}^{N} C_{N,N}(P \cup \{(i,j)\}, T) = C_{N,N}(P, T).$$

The column identity follows in the same way from the unique edge in a column. Colliding sets have zero coordinates and zero permutation indicators, so they are included. Thus every square coordinate is an unnormalised design, in every degree, over the integers.

14.3 Whole-history compression and the marked binomial identity

The support recorded below belongs to an entire reduction path, not only its final matching. Grouping paths by this support will produce binomial coefficients in the number of appended rows. Keeping the initial mark ensures that the coefficient with support size zero is absent, even if its edge is later deleted.

Fix $P, T \subseteq [r] \times [m]$, with P a matching and T terminal, and a column $j \notin \operatorname{cols} P$. Append L rows, and consider all pairs consisting of a marked initial row $i \in \{r+1, \ldots, r+L\}$ and a signed reduction path from $P \cup \{(i, j)\}$ to T. The weight of a pair is the product of its path signs. Its *history support* is the set of appended rows appearing anywhere on the path, including the initial state and edges that are subsequently deleted.

If $|P| + 1 \leq D$ and $2D \leq r$, every such support is nonempty and has size at most $|P| + 1$. To prove the latter assertion, give one token to each initial edge. In a row rewrite, transport

a token by its column to the unique child edge in that column, or delete it if that column is absent. The child uses only columns of the parent, so this describes every child edge exactly once. In a column rewrite transport by row, with the analogous deletion rule. No deleted token reappears, and no two surviving tokens represent the same edge.

For any matching Q of at most D edges, failure at row prefix b implies

$$b < h(Q_{\le b}) + |\operatorname{rows}(Q_{\le b})| \le 2|Q| \le 2D \le r.$$

Thus every chosen row chain lies in old rows. An edge already in an appended row is part of the unchanged context of a row rewrite and is retained verbatim. A column rewrite preserves its token's row or deletes the token. It follows that a token can encounter at most one appended row throughout its lifetime. There are $|P| + 1$ initial tokens, so the full history has at most that many appended rows. The initial mark belongs to it even when its token later disappears.

We next verify that histories of the same size carry the same total integer weight. Let S be a prescribed k-element subset of the L appended rows, and let f be the increasing injection fixing $[r]$ and sending $r+1, \dots, r+k$ to the elements of S in order. For $t \ge 1$, set $f(r+k+t) = r+L+t$. This extends f increasingly and makes the inverse image of the large row interval $[r+L]$ exactly $[r+k]$. Rename the row coordinate of every edge by f and retain its column and step sign.

First, this renaming preserves the deterministic selector. Every failed row prefix lies below r, as just proved; those prefixes and their unused-row counts are unchanged. Their chains are unchanged as labelled edge sets. For a column prefix, the number of occupied columns and all chain lengths are preserved by the increasing row renaming. The first failed column prefix is therefore unchanged. The lexicographic tie choice is also preserved: a strictly increasing map preserves the lexicographic order of sorted edge lists, including the transposed lists used in a column step. Orientation, first failure and chosen chain all agree under renaming, and terminality is preserved.

Second, renaming gives a bijection on the branches whose child edges remain in $[r] \cup S$. In a row step, the inverse images of the sets B and U are exactly the corresponding sets in the compressed rectangle: order comparisons with the old endpoint and occupied-row membership are preserved. An image-supported replacement therefore has a unique inverse injection. Its column set, cardinality, exclusion of the original chain, collision status and sign are all preserved. For a column step the permissible column sets are unchanged; the same argument, using row membership, applies. A child cannot encode two different replacement injections, since its replacement edges are recovered by removing the fixed context: their columns are disjoint from the context columns in a row step, and their rows are disjoint in a column step. The lower and upper branch families are disjoint because their cardinalities differ. Thus no multiplicity is lost. There may be other branches using rows outside S, but these belong to different history supports and are not part of this correspondence.

Applying the branch bijection successively gives a two-way, sign-preserving bijection between marked paths with history exactly $\{r+1, \dots, r+k\}$ in the compressed rectangle and marked paths with history exactly S in the large rectangle. Its inverse simply renames every state and the initial mark back. The endpoint T is fixed because it uses only old rows. All intermediate states have at most D edges, so the selector argument applies at every step.

Define $c_k(P, T, j)$ to be the total weight of the marked paths in the rectangle $[r+k] \times [m]$ whose full history is all k appended rows. Partitioning the marked paths by their history support, and then by its cardinality, proves the integer identity

$$\sum_{i=r+1}^{r+L} C_{r+L,m}(P \cup \{(i,j)\}, T) = \sum_{k=1}^{D} \binom{L}{k} c_k(P, T, j). \tag{65}$$

The coefficients are independent of L. Its constant term is absent because each path retains its initial mark in the *history*, not necessarily in its final state. This distinction is essential.

14.4 Rectangular designs and their extensions

Assume now the hypotheses of Lemma 14.1, with $D \geq 1$, and put $q = p^a$ and $L = m - r$. For $1 \leq k \leq D < q$ the integer $\binom{L}{k}$ is zero in K. Indeed, $q \mid L$ and in characteristic p,

$$(1+z)^L = (1+z^q)^{L/q};$$

the right side has no coefficient in these degrees. Take the square of size m and an old terminal matching $T \subseteq [r] \times [m]$. The function

$$\theta_T(P) = C_{m,m}(P,T) \quad \text{in } K, \qquad P \subseteq [r] \times [m],$$

is a degree-D rectangular design. Its row equations are the square row equations, with the same column set. Its column equations follow from the square column equation after deleting the appended-row sum, which vanishes by (65) for $|P| < D$ and an unused column. If the column is occupied, every appended insertion collides, so the omitted sum is already zero. If P collides, every insertion still collides and both sides are zero. This covers all design equations.

On old terminal inputs, the square reduction is still terminal and

$$\theta_T(S) = \delta_{S,T}.$$

In particular, $\theta_\varnothing$ is normalised. For an arbitrary $\psi \in E_{D-1}(r,m)$ define explicitly

$$\widetilde{\psi}(P) = \sum_{\substack{T \text{ terminal in } [r]\times[m] \\ |T| \leq D-1}} \psi(T)\theta_T(P).$$

This is a member of $E_D(r,m)$ and agrees with ψ on all terminal inputs of size at most $D-1$. Apply (64) in the old rectangle to both designs, after restricting the new design to degree $D-1$. They agree on every edge set of size at most $D-1$. Thus the extension exists for every unnormalised design, with no choice of an unspecified lift.

14.5 The invariant through ordinary-PC degree falls

For $D \geq 1$, the extension property is stronger than existence of one normalised design. Multiplying a polynomial by an edge variable shifts its test functional to the preceding degree window. Extending that shifted design back to degree D lets the induction test every PC line, including lines whose leading terms have cancelled.

For a design ψ define a linear functional on the original polynomial ring by

$$\Lambda_\psi\left(\sum_\alpha b_\alpha x^\alpha\right) = \sum_\alpha b_\alpha \psi(\operatorname{supp}\alpha),$$

where $\operatorname{supp}\alpha$ is the set of edges with positive exponent. This formula is a functional on ordinary polynomials; it does not replace a PC proof by a proof in a Boolean quotient. Every degree-D design annihilates each axiom usable in a degree-D proof. Totalities use (63) at $P = \varnothing$; collisions use vanishing on nonmatchings; and the Boolean equations use the equality of the supports of x_e^2 and x_e.

For an edge e, put $(s_e\psi)(P) = \psi(P \cup \{e\})$. If $\psi \in E_D(r,m)$, then $s_e\psi \in E_{D-1}(r,m)$. Collision vanishing is immediate. In either totality equation at $|P| < D-1$, apply the

degree-D equation to $P \cup \{e\}$, whose cardinality is at most $D-1$. The insertion operations commute, including repeated insertion of the same edge. Also, directly from the monomial formula,

$$\Lambda_\psi(x_e f) = \Lambda_{s_e\psi}(f).$$

Induct through a degree-D ordinary-PC derivation, asserting that every line is annihilated by *every* member of $E_D(r,m)$. Input axioms were handled above, and linear combinations preserve the assertion even when their highest-degree terms cancel. Consider multiplication of an earlier line f by x_e. If $f=0$ there is nothing to prove. Otherwise ordinary polynomial degree gives $\deg(x_e f) = \deg f + 1$, so $\deg f \le D-1$. For a given $\psi \in E_D(r,m)$, extend $s_e\psi$ to a design $\nu \in E_D(r,m)$ using the preceding construction. The earlier line is annihilated by ν by the inductive assertion. Since the two designs agree on all edge sets of size at most $D-1$, and every monomial of f has support of that size or smaller,

$$\Lambda_\psi(x_e f) = \Lambda_{s_e\psi}(f) = \Lambda_\nu(f) = 0.$$

This proves the invariant for multiplication as well. Finally the normalised design $\theta_\varnothing$ sends the constant 1 to 1, so that constant cannot be the last line of the derivation.

For $D=0$, all displayed axioms have positive degree because both sides of the rectangle are nonempty. No nonzero constant axiom is available, and multiplication of a nonzero polynomial cannot produce a degree-zero line. Linear combinations of the available zero lines remain zero. Thus a degree-zero refutation is impossible directly. This completes the ordinary-PC proof in every case. The NS conclusion follows from the degree-preserving simulation of a total-product NS certificate by ordinary PC.

15 No algorithm can distinguish the types

Already in the special case of the fixed target $G^{(n)} = 1$, the classification cannot be turned into an algorithm which determines the case from a sentence. This remains true under the promise that every finite instance is inconsistent. Throughout this section, the input is a well-formed normalised sentence and the request is a refutation of its translated axioms, with the translation of Section 2; the characteristic p is fixed.

The promise is already part of the every-union undecidability theorem in LICS 2008 [6, Theorem 1]. Its Lemma 8 switches between T_1 and a prescribed other type. Here the controlled-join construction gives explicit reductions for every pair, including pairs wholly inside $\{\mathrm{T}_2, \mathrm{T}_3, \mathrm{T}_4\}$. This refines the published formulation and the inspected earlier construction; it is not a claim of exhaustive historical priority.

Theorem 15.1. *Let A, B be disjoint nonempty subsets of $\{\mathrm{T}_1, \mathrm{T}_2, \mathrm{T}_3, \mathrm{T}_4\}$. There is no partial computable algorithm which, on every normalised sentence having no finite nonempty model and having type in $A \cup B$, halts and correctly distinguishes membership in A from membership in B. This holds for NS and for ordinary PC, over every field of characteristic p.*

In particular, each of the six pairs is undecidable, and so is membership in any nonempty proper union of types. The last conclusion is the scope of the historical every-union theorem. The pairwise statement requires the additional construction below; it does not follow merely by listing all the undecidable union problems.

15.1 A sentence that records a computation prefix

A reduction is an effective transformation of finite input descriptions; it must not first decide whether its input computation halts. Register programs and the standard halting theorem are discussed in [17]. The normalised sentence and its degree bounds are constructed below.

A counter program has finitely many natural-number registers and instructions for incrementing a register, branching on zero and decrementing, and halting. All registers initially contain zero. The number of registers is finite but need not be two. Write c_M for one more than the largest register index occurring in M, so $c_M \geq 1$.

From a program M, form a sentence θ_M with a strict linear order, least and greatest elements o, e, a successor function S, counter-value functions C_r, and state predicates Q_ℓ. Its clauses have four roles:

1. *Time order.* Require $S(e) = e$ and require $S(t)$ to be the immediate successor of t when $t \neq e$.
2. *Initial and allowed configurations.* At o, all counters equal o and the state is the initial one. Exactly one state holds at each element, and halting states are forbidden.
3. *Increment transitions.* At $t \neq e$, the current state determines the instruction and the next state at $S(t)$. Increment replaces $C_r(t)$ by $S(C_r(t))$, with the guard $C_r(t) \neq e$.
4. *Zero tests and decrement transitions.* Decrement branches according as $C_r(t) = o$; on the positive branch, the new value v satisfies $v < C_r(t)$ and $S(v) = C_r(t)$.

In every transition, registers not changed by the instruction retain their values. These are finite universal clauses with flattened function graphs, and $M \mapsto \theta_M$ is primitive recursive.

Primitive recursion permits composition and loops with iteration counts specified by their inputs. It describes the finite procedure producing the sentence, not a bounded running time for the program encoded by it. The latter may perform unbounded search or never halt.

Lemma 15.2 (Computation switch)**.** *If M never halts, θ_M has a model of every size $n \geq 1$. If the run of M first reaches a halting state at time T, the translated system has an NS refutation of degree bounded by a constant C_M at every size $n \geq T + 1$.*

Proof. In the first case, use $[n] = \{0, \ldots, n-1\}$ in its natural order and record the first n configurations. Every counter value at time j is at most j, so no guarded increment overflows before the last element. This includes $n = 1$.

For the second case put $a_j = S^j(o)$. If the structure has $T+1$ distinct elements, then $a_j \neq e$ for $j < T$: otherwise immediacy of successor and linearity of the order leave only $a_0, \ldots, a_j$. Induction along this finite prefix forces the actual state at a_j and forces $C_r(a_j) = a_{c_r(j)}$, where $c_r(j)$ is the numerical counter value. At a_T this gives a forbidden halting state. The argument needs neither well-ordering nor an induction axiom in the structure.

This local contradiction gives a degree bound by the finite-term certificate in Lemma 15.3. More explicitly, with c_M counters one may take

$$m = T + 2 + 2c_M(T+1), \qquad C_M = m + \sum_R (m + T + 1)^{\mathrm{arity}(R)} + \max_{C \text{ a clause}} |C|.$$

The term nodes are $o, e, a_1, \ldots, a_T, C_r(a_j), S(C_r(a_j))$; $T + 1$ distinct additional labels supply the size condition. The constant depends on the program and its halting time, never on n. □

For completeness, the reduction needs effective universality, not a decision procedure for the halting time. Every primitive recursive function can be compiled into finite counter code: composition uses fresh registers, and primitive recursion uses a counted loop, preserving the input registers. Let $H(k, c)$ equal one when the program with code c, on its fixed encoded input, halts within k simulated steps, and zero otherwise. Bounded simulation computes H by a fixed number of primitive-recursive transition updates, so this predicate is primitive recursive. A counter program that successively tests $H(0, c), H(1, c), \ldots$ therefore halts exactly when the coded program does. Loading c into this fixed search program is primitive recursive. This reduces the usual undecidable halting problem to the programs just used. Only effective

compilation and the standard undecidability of halting are used; the number of counters is unrestricted but finite.

15.2 Finite local contradictions and their certificates

The halting switch uses a fixed finite computation prefix even though function totality is expressed by sums with n summands. The following elementary certificate construction separates these two sizes.

The degree bound counts a fixed list of term evaluations and relation atoms. The number of possible labellings, and hence the number of certificate summands, may still grow with n. Bounded degree here does not assert bounded certificate size.

Lemma 15.3 (Finite-term certificate)**.** *Let $n \geq 1$ and let a finite acyclic list of m function evaluations be given, with every argument supplied by an earlier node. Fix b distinct labels. Suppose that every assignment of labels to the nodes, and every Boolean assignment to the relation atoms on the named and fixed labels, either has inconsistent function values or falsifies a clause instance whose variables lie among the named and fixed labels and all of whose function evaluations occur in the list. If clauses have length at most L, the graph-and-sum translation has an NS refutation of degree at most*

$$m + \sum_{R} (m+b)^{\mathrm{arity}(R)} + L.$$

The hypothesis and conclusion concern sizes for which the b distinct labels exist.

Proof. For a labelling α of the nodes, let q_α be the product of the m graph variables asserting those function evaluations. Expanding the totality sum for each node, in the order of the acyclic list, gives

$$\sum_{\alpha} q_\alpha - 1$$

in the degree-m NS span. This is a telescoping identity: at step j, the product for the first $j-1$ evaluations multiplies the totality equation for evaluation j. Summing over the possible earlier values gives the next partial sum. Repeated graph variables cause no problem.

If α gives different values to the same function on the same input, q_α contains a functionality collision and already belongs to the span within degree m. Otherwise, list the distinct relation variables on the at most $m+b$ named labels. Write this actual set as $Y_\alpha = \{y_1, \ldots, y_{a_\alpha}\}$, where $a_\alpha \leq a_{\mathrm{rel}} = \sum_R (m+b)^{\mathrm{arity}(R)}$. Expand over these distinct variables:

$$1 = \prod_{i=1}^{a_\alpha} (y_i + (1 - y_i)) = \sum_{\beta} u_\beta.$$

Each u_β is the Boolean assignment monomial for one partial relational structure. By hypothesis that structure falsifies an available clause. Its clause polynomial is a product of at most L falsity indicators. Write $P = q_\alpha u_\beta$ and the clause polynomial as $F_1 \cdots F_\ell$, where each F_j is the falsity indicator of a literal false in the chosen partial structure. For each literal, $P(1 - F_j)$ has an explicit expression in the structural NS span:

- For a positive relation literal, $F_j = 1 - x$ and $u_\beta = (1-x)u'$. Thus $P(1-F_j) = q_\alpha u' x(1-x) = -q_\alpha u'(x^2 - x)$. For a negative relation literal, $F_j = x$ and $u_\beta = xu'$; the same identity with $x(1-x)$ applies.
- For a negative graph literal false in the partial structure, its graph variable x is one of the chosen evaluations, so $q_\alpha = xq'$. Its falsity indicator is $F_j = x$ and $P(1-F_j) = -q' u_\beta (x^2 - x)$.

- For a positive graph literal false there, the chosen evaluation has a different output variable x' on the same function input. Here $q_\alpha = x'q'$, $F_j = 1 - x$, and $P(1 - F_j) = q'u_\beta x'x$, an explicit functionality-axiom multiple.
- A false equality or disequality has the constant falsity indicator $F_j = 1$, so $P(1 - F_j) = 0$.

These are all cases of the normalised literals. Each identity has product degree at most $m + a_{\mathrm{rel}} + 1$. The exact telescoping identity

$$P(1 - F_1 \cdots F_\ell) = \sum_{j=1}^{\ell} P(1 - F_j) \prod_{i<j} F_i$$

therefore puts its left side in the NS span within degree $m + a_{\mathrm{rel}} + \ell$. Since $PF_1 \cdots F_\ell$ is a multiple of the falsified clause axiom, P is in that same span. Sum first over β, then over α, and use the totality identity to obtain one. □

A *partial structure* for θ_M has a domain of labels, all relations defined on it, and function values defined only on some arguments. It is *locally consistent* if every clause instance whose variables take values in the domain, and whose function atoms are all defined, has a true literal.

Lemma 15.4 (Local switch)**.** *Let the run of M first reach a halting state at time T, and let M have c_M counters. Suppose a locally consistent partial structure defines o, e, $a_{j+1} = S(a_j)$ for $j < T$ (with $a_0 = o$), $C_r(a_j)$ and $S(C_r(a_j))$ for $j \le T$, $r < c_M$, and that all these elements lie in its domain. Then its domain has at most T distinct elements.*

Proof. Suppose it contains $T + 1$ distinct elements. If $a_j = e$ for some $j < T$, take j least. Every domain element x satisfies $x = a_0$ or $a_0 < x$ (least element), and, if $a_i < x$ with $i < j$, then $x = a_{i+1}$ or $a_{i+1} < x$ (no element strictly between a_i and $S(a_i)$, and linearity). As $a_j = e$ is greatest, the domain is $\{a_0, \ldots, a_j\}$, too small. So $a_j \neq e$ for $j < T$, and the order clauses make $a_0 < a_1 < \cdots < a_T$. For the counter induction, the initial clauses give the initial state and $C_r(a_0) = a_0$ for every counter. A counter value at time j is at most j, so the notation a_v uses only this named prefix. Suppose its states and counters agree through time $j < T$. An increment of value v gives $S(a_v) = a_{v+1}$; because $v \le j < T$, this successor is among the defined evaluations and its guard is valid. The zero branch of a decrement is selected exactly when $v = 0$. On a positive decrement, the transition specifies a value $y < a_v$ with $S(y) = a_v$. The immediate-predecessor property forces $y = a_{v-1}$: if $y < a_{v-1}$ then a_{v-1} lies strictly between y and $S(y)$, and if $a_{v-1} < y < a_v$ then y lies strictly between a_{v-1} and $S(a_{v-1})$. Both alternatives contradict immediacy. Unchanged counters retain their named values. State uniqueness and the chosen transition identify the next state. Every clause used has its variables in the domain and its function atoms in the specified finite evaluation list. This proves the induction through time T. At $j = T$ the state is halting, which a clause forbids. □

Corollary 15.5. *In that case, for every $n \ge T + 1$ the translation of θ_M at size n has an NS refutation of degree at most*

$$C_M = m + \sum_R (m + T + 1)^{\mathrm{arity}(R)} + \max_C |C|, \qquad m = T + 2 + 2c_M(T + 1).$$

Proof. Apply Lemma 15.3 to the m evaluations o, e, $a_1, \ldots, a_T$, $C_r(a_j)$, $S(C_r(a_j))$ ($j \le T$, $r < c_M$), each argument supplied by an earlier node, with $b = T + 1$ fixed labels. A consistent labelling and a Boolean assignment give a partial structure of the kind in Lemma 15.4, whose domain contains the $T + 1$ fixed labels; so it is not locally consistent, which is the hypothesis of Lemma 15.3. □

Here c_M is one more than the largest counter index occurring in M (so $c_M \geq 1$), and T must be the first halting time. Sizes $n \leq T$ are genuine exceptions: θ_M can have models there.

15.3 A disjunction with controlled degree

Conjunction is easy to control when one block is satisfiable. In disjoint signatures, if β has a root at size n, then

$$d_*(\alpha \wedge \beta, n) = d_*(\alpha, n), \qquad * = \mathrm{NS}, \mathrm{PC}. \tag{66}$$

One inequality follows by using the refutation of α; the other follows by substituting the root of β into every proof line.

We also need a disjunction which does not accidentally simplify either branch. Introduce a fresh unary relation Z. Keep both branches' Boolean, totality and functionality equations. Replace every clause polynomial f of α by $z_t f$, and every clause polynomial g of β by $(1-z_t)g$, for a fresh universally quantified label t. Denote this normalised sentence by $\alpha \vee_* \beta$.

A guarded branch is simulated using the guard times each of its original proof lines; the unguarded line is not assumed derivable from the joined system. The same guard passes through linear combinations and variable multiplication. Structural axioms remain available, and can acquire the guard using one variable multiplication and linear combinations.

Lemma 15.6 (Controlled join)**.** *For $n \geq 1$, with ∞ allowed, each of NS and PC satisfies*

$$\max\{d_*(\alpha, n), d_*(\beta, n)\} \leq d_*(\alpha \vee_* \beta, n) \leq \max\{d_*(\alpha, n), d_*(\beta, n)\} + 1.$$

Proof. Set all selectors to one, and give the inactive block a structural root: all its functions take one fixed value and all its relations are false. Its clauses are guarded away, while its structural equations vanish. The resulting proof refutes α without increasing degree. Selectors zero give the other lower inequality.

For the upper bound fix a label t_0. Multiply a refutation of α line by line by z_{t_0} and a refutation of β by $1-z_{t_0}$. The guarded clauses are axioms; structural axioms can be multiplied by the guard at a cost of one degree. Linear combinations and variable multiplication remain valid, including after cancellations. The two derivations end in z_{t_0} and $1 - z_{t_0}$, whose sum is one. For NS the same construction multiplies the two certificates by the guards. □

15.4 The reduction

Let $\psi_1, \psi_2, \psi_3, \psi_4$ be the four sentences of Section 13. Each has no finite nonempty model. For these particular witnesses, the dominant type under the maximum operation is ordered by

$$\mathrm{T}_1 \prec \mathrm{T}_4 \prec \mathrm{T}_3 \prec \mathrm{T}_2.$$

This is a property of the chosen witnesses, not an ordering assertion about arbitrary profiles. A bounded profile has no effect on the type of a maximum; the all-size linear lower bound for ψ_2 makes it dominant; and the maximum of ψ_4 and ψ_3 is bounded off the multiples of p and linear on them. An additive error of at most one preserves these conclusions. More generally, adjoining an eventually bounded profile and allowing this error preserves each of the four intrinsic types: recurrence transfers through finite intervals of degrees, and unboundedness transfers directly.

Take $a \prec b$ and define, using disjoint signatures,

$$\Phi_M = \psi_a \vee_* (\psi_b \wedge \theta_M).$$

If M does not halt, (66) makes the second branch have exactly the profile of ψ_b. The join has type b. If M halts, the second branch is eventually bounded by Lemma 15.2, and the join has type a. Every Φ_M is within the all-finite-inconsistent promise, whether or not M halts. The map is primitive recursive and uses no information about that alternative. A separator of types a, b would decide halting. Choosing a and b on opposite sides of any disjoint nonempty A, B proves Theorem 15.1.

16 Conclusion

The linear bounds arise from a way of comparing bounded-degree algebraic calculations across universe sizes. For a fixed uniform request over a field of positive characteristic, the construction transports the actual axiom, multiplication and target maps. It preserves the linear-space operations defining the consequence spaces, including the cancellations which make ordinary PC stronger at a given degree. A separate detection argument reflects whether the prescribed target belongs to that space. Together these steps make the construction apply to uniform consequences, with refutation as the target-1 special case.

The quantitative gain is a linear onset and a linearly bounded prime-power period for each degree threshold. Repeated lifting keeps the numerical degree fixed while the universe grows. A nonrecurring degree can therefore occur only before its own onset; the period bound also locates recurring large degrees at proportionate universe sizes. This gives bounded degree in T_1, linear hardness at finite values in T_2, linear hardness on the hard residues in T_3, and finite linear peaks with a finite liminf in T_4. Underivable requests retain degree ∞, and the stated initial exceptions remain part of the results.

The linear degree lower bound is sharp in general. The injection example has matching $\Theta(n)$ degrees for both systems, and the weak-bijection example has finite $\Theta(n)$ peaks; its degree-one target extension has the same peak behaviour. These witnesses establish optimality of the general degree lower bound, while the theorem leaves the individual upper bounds to the particular uniform request.

The uniform NS–PC comparison shows that these patterns belong to the request itself: both systems assign it the same type, even when their minimum degrees differ. Their common hardness spectrum gives the same four alternatives as topological shapes in $\mathbb{Z}_p$, for all uniform requests. With full explicit coefficient lists, including zeros, the degree results give the stated polynomial upper bounds and exponential size lower bounds. The first-order refutation subcase connects this algebraic classification to logic: explicit reductions separate any two types under the no-finite-model promise already stated in LICS 2008 [6, Theorem 1], refining its every-union result. Thus the stronger quantitative bounds preserve the four-way structure of the earlier classification while extending its algebraic scope.

Proof sources and verification. The supporting archive [30] supplies the curated proof sources, pinned Lean 4.26.0 and mathlib dependency data, and the source, statement, axiom and import-closure audit commands and their recorded execution provenance. The core degree, comparison, classification and undecidability endpoints have Lean proofs; the archive identifies their exact types and assumptions. The injection upper certificate, the optimality deductions and the size presentation are written arguments. The topological corollary in Section 11 is likewise a written deduction from periodicity, comparison and compactness of $\mathbb{Z}_p$; it is not an addition to the Lean formalisation. Recorded fresh replay covers the named import closures, with separately compiled scope adapters distinguished in the evidence; it is not a fresh replay of every package module.

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References

[1] S. Riis, *A complexity gap for tree resolution*, Computational Complexity **10**, 179–209, 2001. doi:10.1007/s00037-001-8194-y.

[2] S. Dantchev and S. Riis, *On relativisation and complexity gap for resolution-based proof systems*, in Computer Science Logic (CSL 2003), Lecture Notes in Computer Science **2803**, 142–154, 2003. doi:10.1007/978-3-540-45220-1_14.

[3] S. Dantchev, B. Martin and S. Szeider, *Parameterized proof complexity*, in Proceedings of the 48th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2007), 150–160, 2007. doi:10.1109/FOCS.2007.53.

[4] S. Dantchev and B. Martin, *Rank complexity gap for Lovász–Schrijver and Sherali–Adams proof systems*, Computational Complexity **22**, 191–213, 2013. doi:10.1007/s00037-012-0049-1.

[5] O. Beyersdorff, J. Clymo, S. Dantchev and B. Martin, *The Riis complexity gap for QBF resolution*, Journal on Satisfiability, Boolean Modeling and Computation **15**(1), 9–25, 2024. doi:10.3233/SAT-231505.

[6] S. Riis, *On the asymptotic Nullstellensatz and Polynomial Calculus proof complexity*, in Proceedings of the 23rd Annual IEEE Symposium on Logic in Computer Science (LICS 2008), pp. 272–283, 2008. doi:10.1109/LICS.2008.30.

[7] S. Riis, *On the asymptotic Nullstellensatz and Polynomial Calculus proof complexity*, extended preprint, Electronic Colloquium on Computational Complexity, Report TR09-014, 2009. https://eccc.weizmann.ac.il/report/2009/014/.

[8] J. Krajíček, *On the degree of ideal membership proofs from uniform families of polynomials over a finite field*, Illinois Journal of Mathematics **45**(1), 41–73, 2001. doi:10.1215/ijm/1258138254.

[9] N. Harman, *Stability and periodicity in the modular representation theory of symmetric groups*, preprint, arXiv:1509.06414. https://arxiv.org/abs/1509.06414.

[10] C. Hone, F. Klein, B. Pauwels, A. Sherman, O. Yacobi and V. L. Zhang, *Semisimplifying categorical Heisenberg actions and periodic equivalences*, preprint, arXiv:2509.07377. https://arxiv.org/abs/2509.07377.

[11] C. Bowman, S. Doty and S. Martin, *Integral Schur–Weyl duality for partition algebras*, Algebraic Combinatorics **5**(2), 371–399, 2022. doi:10.5802/alco.214.

[12] J. Krajíček, *Uniform families of polynomial equations over a finite field and structures admitting an Euler characteristic of definable sets*, Proceedings of the London Mathematical Society (3) **81**(2), 257–284, 2000. doi:10.1112/S0024611500012557.

[13] J. Krajíček and T. Scanlon, *Combinatorics with definable sets: Euler characteristics and Grothendieck rings*, Bulletin of Symbolic Logic **6**(3), 311–330, 2000. doi:10.2307/421058.

[14] T. Church, J. S. Ellenberg and B. Farb, *FI-modules and stability for representations of symmetric groups*, Duke Mathematical Journal **164**(9), 1833–1910, 2015. doi:10.1215/00127094-3120274.

[15] S. Lang, *Algebra*, revised third edition, Graduate Texts in Mathematics **211**, Springer, 2002. doi:10.1007/978-1-4613-0041-0.

[16] J. L. Alperin, *Local Representation Theory: Modular Representations as an Introduction to the Local Representation Theory of Finite Groups*, Cambridge Studies in Advanced Mathematics **11**, Cambridge University Press, 1986. doi:10.1017/CBO9780511623592.

[17] N. Cutland, *Computability: An Introduction to Recursive Function Theory*, Cambridge University Press, 1980. doi:10.1017/CBO9781139171496.

[18] The Stacks Project Authors, Lemma 12.7.2 (Tag 010N) in *Additive functors*, and Remark 35.4.4 (Tag 08WK) in *Descent for universally injective morphisms*. https://stacks.math.columbia.edu/tag/010N; https://stacks.math.columbia.edu/tag/08WK.

[19] J. Paris and A. Wilkie, *Counting problems in bounded arithmetic*, in C. A. Di Prisco (ed.), *Methods in Mathematical Logic*, Lecture Notes in Mathematics **1130**, 317–340, Springer, 1985. doi:10.1007/BFb0075316.

[20] P. Beame, R. Impagliazzo, J. Krajíček, T. Pitassi and P. Pudlák, *Lower bounds on Hilbert's Nullstellensatz and propositional proofs*, Proceedings of the London Mathematical Society (3) **73**(1), 1–26, 1996. doi:10.1112/plms/s3-73.1.1. Preliminary version in FOCS 1994, 794–806.

[21] M. Clegg, J. Edmonds and R. Impagliazzo, *Using the Groebner basis algorithm to find proofs of unsatisfiability*, in Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC 1996), 174–183, 1996. doi:10.1145/237814.237860.

[22] S. Riis and M. Sitharam, *Generating hard tautologies using predicate logic and the symmetric group*, Logic Journal of the IGPL **8**(6), 787–795, 2000. doi:10.1093/jigpal/8.6.787.

[23] S. Riis and M. Sitharam, *Uniformly generated submodules of permutation modules: Over fields of characteristic 0*, Journal of Pure and Applied Algebra **160**(2–3), 285–318, 2001. doi:10.1016/S0022-4049(00)00069-4.

[24] M. Ajtai, *Symmetric systems of linear equations modulo p*, Electronic Colloquium on Computational Complexity, Report TR94-015, 1994. https://eccc.weizmann.ac.il/report/1994/015/.

[25] S. R. Buss, *Lower bounds on Nullstellensatz proofs via designs*, in P. Beame and S. R. Buss (eds.), *Proof Complexity and Feasible Arithmetics*, DIMACS Series in Discrete Mathematics and Theoretical Computer Science **39**, 59–71, American Mathematical Society, 1998. https://mathweb.ucsd.edu/~sbuss/ResearchWeb/designs/.

[26] P. Beame and S. Riis, *More on the relative strength of counting principles*, in P. Beame and S. R. Buss (eds.), *Proof Complexity and Feasible Arithmetics*, DIMACS Series in Discrete Mathematics and Theoretical Computer Science **39**, 13–35, American Mathematical Society, 1998. https://homes.cs.washington.edu/~beame/papers/riis.pdf.

[27] F. Part, N. Thapen and I. Tzameret, *First-order reasoning and efficient semi-algebraic proofs*, Annals of Pure and Applied Logic **176**(1), 103496, 2025. doi:10.1016/j.apal.2024.103496.

[28] S. Riis and M. Sitharam, *Non-constant degree lower bounds imply linear degree lower bounds*, Electronic Colloquium on Computational Complexity, Report TR97-048, 1997. https://eccc.weizmann.ac.il/report/1997/048/. The proof qualification is discussed in the introduction.

[29] S. Riis and M. Sitharam, *Generating hard tautologies using predicate logic and the symmetric group*, BRICS Report RS-98-19, 1998. Earlier version of [22]; Section II, Theorem 1 gives the Skolemised translation and model–solution correspondence. https://www.brics.dk/RS/98/19/BRICS-RS-98-19.pdf.

[30] S. Riis. *Degree profiles of uniform polynomial consequences: Linear lower bounds for Nullstellensatz and Polynomial Calculus—paper and supporting materials*. Version 1.0.0, Zenodo, 2026. Paper, tutorial, Lean formalisation, verification records and reproduction instructions. doi:10.5281/zenodo.23965838. Development repository: https://github.com/SR123/degree-profiles-linear-bounds.

[31] S. Riis, *Independence in Bounded Arithmetic*, Ph.D. thesis, University of Oxford, 1993.

[32] S. Riis, *Finitisation in Bounded Arithmetic*, BRICS Report Series RS-94-23, University of Aarhus, 1994. https://doi.org/10.7146/brics.v1i23.21644.

[33] J. Krajíček, *Bounded Arithmetic, Propositional Logic, and Complexity Theory*, Encyclopedia of Mathematics and its Applications, vol. 60, Cambridge University Press, 1995, Chapter 9.

[34] A. A. Razborov, *Lower bounds for the polynomial calculus*, Computational Complexity **7**(4), 291–324, 1998. https://doi.org/10.1007/s000370050013.